

STAT 27410 Final Project Proposal

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1. Introduction

Complex motor skills, which are precise and non-stereotypical movements, are essential for survival. For example, squirrels need to jump from branch to branch, and eagles need to catch small prey. Understanding brain activity during complex motor skills is essential in motor control and motor learning. Towards this goal, the MacLean Lab, a neuroscience lab at The University of Chicago, performed neuro-recording on the mouse primary motor (M1) cortex while the mouse was engaged in a reach-to-grasp movement. The M1 cortex can be seen as a recurrent neural network, receiving upstream signals on motor planning and body position, and outputting commands to downstream areas, such as the spinal cord and brainstem. While networks are very complex objects of study, we chose a narrow objective for this project: for any neuron in the network, we want to predict its activity using the activity of other neurons. We hope this can inspire future research on the dynamics and structure of neural networks.

1.1 Dataset

We use the data from the MacLean lab. During the experiment, the mouse reached for a rice grain in a head-fixed position. Before the experiment, the mouse underwent surgery so that its brain is attached to a microscope and that neurons would increase in fluorescence value when their activity increased. After the initial imaging, the MacLean Lab performed preprocessing to infer the estimated spike counts from the fluorescence data.

- Time series data of neural spikes: This is a time series of the inferred number of spikes at each time bin, inferred from the fluorescence values recorded by the microscope. Each time bin is 30 ms. We have 315 neurons and 148468 time bins.
- Inhibitory or excitatory neurons: This is an array that stores whether a neuron is inhibitory or excitatory, namely whether it increases or decreases the level of activity in downstream neurons.
- Spatial distribution: This is an array of 2-tuples that stores the x, y coordinate of each neuron.
- Mouse behavior: This is an array of 2-tuples that stores the start and end time of each bout of reaches.

1.2 Complications of Dataset

The dataset chosen is not without complications, however, some of these complications include:

- Incomplete network: Due to limitations of imaging techniques, the experiment records an incomplete network. We cannot record the complete M1 cortex, and not even all the neurons in the field of view. Hence, the recording is a small subset of neurons in the M1 cortex.
- Upstream signal: This entire brain region receives shared upstream signals. However, we cannot measure the upstream signals.

1.3 Scientific Question

For a target neuron, the goal is to predict its activity at each time bin using the activity of other neurons. In other words, the response variable is the activity of the target neuron. The predictor variables are the activity of the other neurons. The selected target neurons are Neuron 234, 168, and 281. The choice of selecting these three neurons comes from exploratory data analysis which showed that these three neurons had the potential to create relatively strong predictive models, making them neurons of interest and the three that we believe will provide the most interesting exploration and analysis of our data.

It is known that the activity of a target neuron can be explained by the activity of other neurons and their recurrent connections through linear models (Kotekal et al., 2020). This inspired us to construct linear regression models to capture the interaction between the predicted and predictor variables. In the frequentist analysis section, we hypothesize that different neurons integrate the activities of other neurons differently. This is a plausible hypothesis because the different neurons likely form different connections with other neurons, depending on factors such as their proximity. We seek to validate this hypothesis by comparing linear models that capture different integrations and linear models that require the same integration. This hypothesis, if true, would motivate hierarchical models in our Bayesian analysis.

Due to the high computational cost, we cannot perform the complete frequentist and Bayesian analysis on all 295 excitatory neurons. Instead, we pick three neurons. Because the experiment only records an incomplete network, the selection criterion is based on the information available regarding a neuron’s upstream neurons. (See further explanation in the frequentist analysis section.)

2. Exploratory Data Analysis

For the exploratory analysis, we are interested in visualizing the neuron activity and co-activity, neuron spatial distribution, type (inhibitory or excitatory), and mouse behavior.

2.1 Neuron Activity:

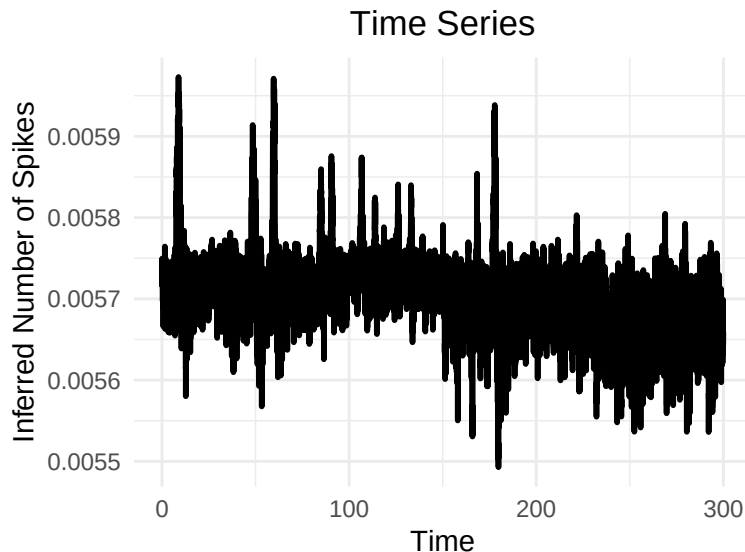


Figure 1: Time Series Plot of Neuron 111's Activity

As an example, we visualize the spikes time series of one of our target neurons, neuron 234 (Fig. 1). At each time bin, the y-axis displays the inferred number of spikes. A common question is why the quantity is continuous rather than discrete. Recall that spikes are not directly observable from the experiment. Instead, they are inferred from the observed fluorescence values of the neurons. Hence, the inferred number of spikes results from the spikes inference algorithm. We recommend interpreting the y-axis as an artificial measure of neuron activity level, and we refer to this quantity as inferred spikes. For more details, read the frequently asked questions section of the spike inference algorithm: <https://github.com/HelmchenLabSoftware/Cascade>.

2.2 Neuron Types:

Fig. 2 displays the counts of inhibitory and excitatory neurons in a barplot. Inhibitory and excitatory neurons are two types neurons, classified based on how they interact with downstream neurons. Each neuron has a membrane potential, namely the electric potential difference between the inside and outside of the cell. When a neuron's membrane potential reaches a certain threshold, it generates a spike, namely a sudden increase in action potential. This spike transmit signals to its downstream neurons and changes the action potential of the downstream neurons. Based on whether a neuron increases or decreases the action potential of its downstream neuron, the neuron is classified as excitatory or inhibitory respectively. The bar plot shows that there are many more excitatory neurons than inhibitory neurons.

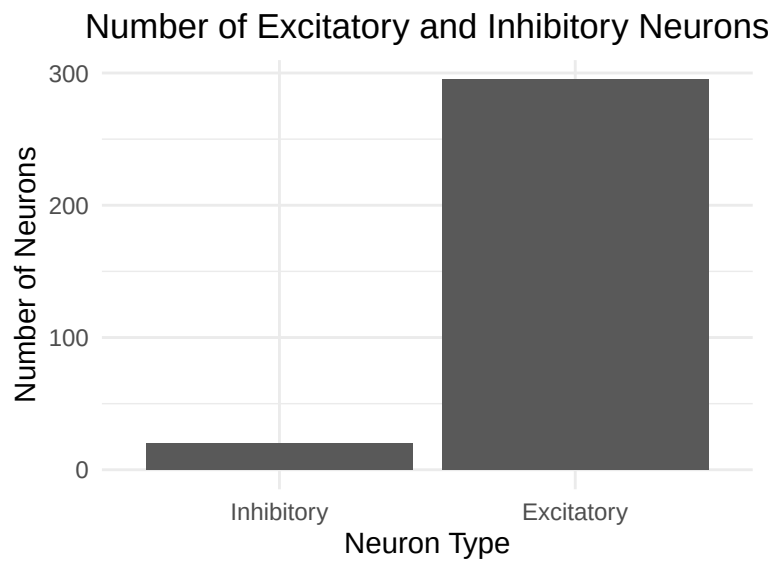


Figure 2: Distribution of Excitatory vs Inhibitory Neurons

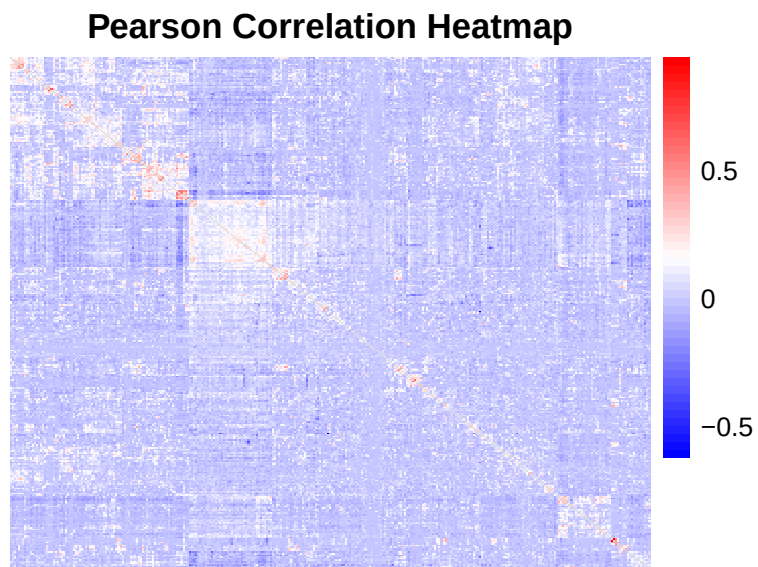


Figure 3: Heatmap of Pearson Correlation of Neuron Spikes

2.3 Correlation between Neuron Spikes:

In addition to single-neuron activity, we are interested in the collective dynamics of neurons. Here, we examine the co-activity between every pair of neurons, measured by Pearson correlation. The resulting matrix is a symmetric 315 by 315 matrix, where the ij -th entry is the Pearson correlation between the inferred spikes of the i th neuron and the j th neuron. Fig. 3 visualizes the Pearson correlation matrix using a heatmap. The heatmap shows that the correlation between neuron activity is low across the board. In addition, most correlations are positive, and negative correlations are small in magnitude.

2.4 Neuron Positions:

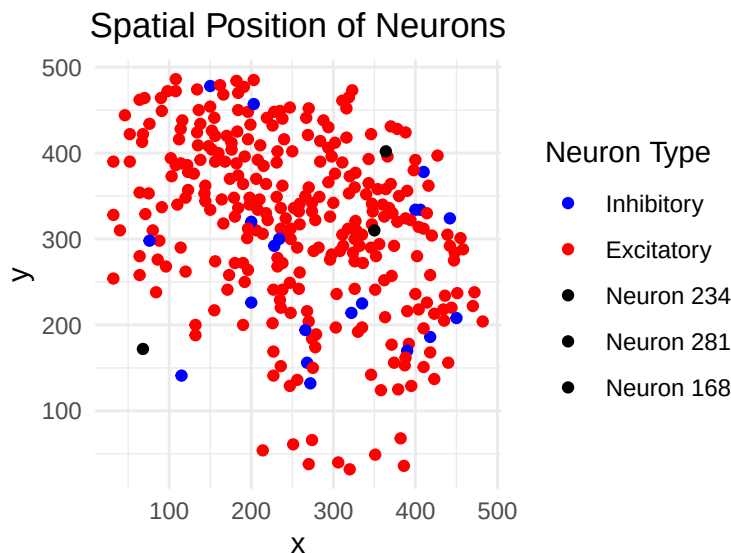


Figure 4: Spatial Positioning of Neurons

Fig 4. displays the spatial positions of the 315 excitatory and inhibitory neurons. x , y are the coordinates of the neurons in the image. Observe that some areas have more recorded neurons than others. In addition, because of the biophysics constraints, the neurons in proximity are more likely to be connected. Therefore, we expect the neurons in proximity to be more correlated.

2.5 Reach Times:

In the experiment, the mouse performs a reach-to-grasp task. Specifically, the mouse reaches for a tiny grain of rice. Given the task's difficulty, the mouse often makes several continuous attempts between rests. We call a set of continuous attempts "a bout of reaches". Fig. 5 displays a histogram of the lengths of the bouts, given by subtracting the start time of the bout from the end time of the bout. We see that most bouts are short, but there are a few long bouts of reaches.

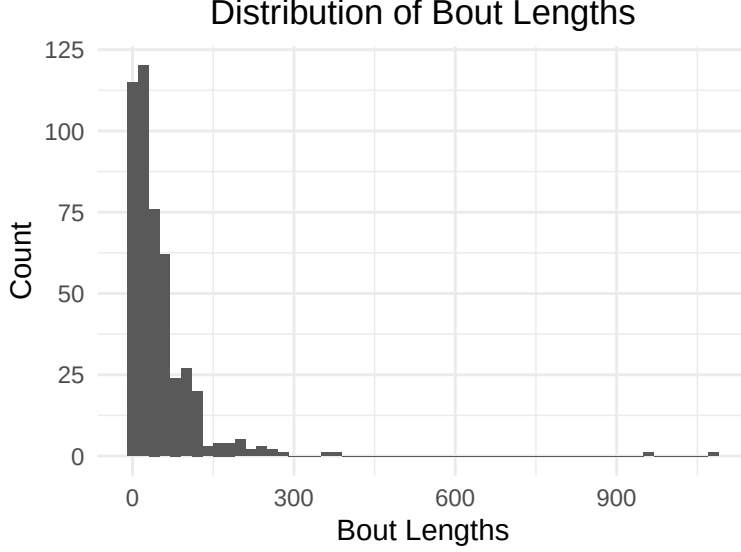


Figure 5: Distribution of Bout Lengths

3. Frequentist Analysis

3.1 Proposed Frequentist Model(s)

Our project aims to predict the inferred spikes of a target neuron using the activity of the other neurons. In the frequentist paradigm, our model of choice is linear regression. Linear models are a well-established method in computational neuroscience literature. Specifically, Kotekal et al. 2020 showed that recurrent interactions (i.e., activities of other neurons) can explain target neuron activity through a linear model.

In addition to past literature, our choice of linear regression holds several other advantages. Because of its relative simplicity, it has a closed-form solution and is computationally efficient. Given our model’s large number of predictors, computational complexity is an important consideration. Another advantage of simplicity is the interpretability of linear regression. We can understand how the predictors contribute to the target neuron activity and interpret the parameters as the strength of interaction between the two neurons. Given linear models’ computational efficiency and interpretability, it is our model of choice.

Using linear regression, we predict the inferred spikes of a target neuron using the inferred spikes of other neurons. To measure the performance of linear regression, we use the R^2 value, which is the percent of variance explained. In neuroscience, an R^2 around 0.1 is considered a good model.

Due to the computational cost of performing the complete frequentist and Bayesian analysis for all 295 excitatory neurons, we pick 3 neurons to analyze for this project. The criterion for neuron choice is as follows: Because our data represents an incomplete network, we want to pick neurons whose upstream neurons are observable (i.e., included in the data). Otherwise, we do not have information on the predictors, and the subsequent analysis wouldn’t be meaningful. To determine the neurons with observable upstream neurons, we first perform

a linear regression on all 295 neurons. We take the 3 neurons with the highest R^2 value. Fig. 6 displays a histogram of the R^2 of all the neurons.

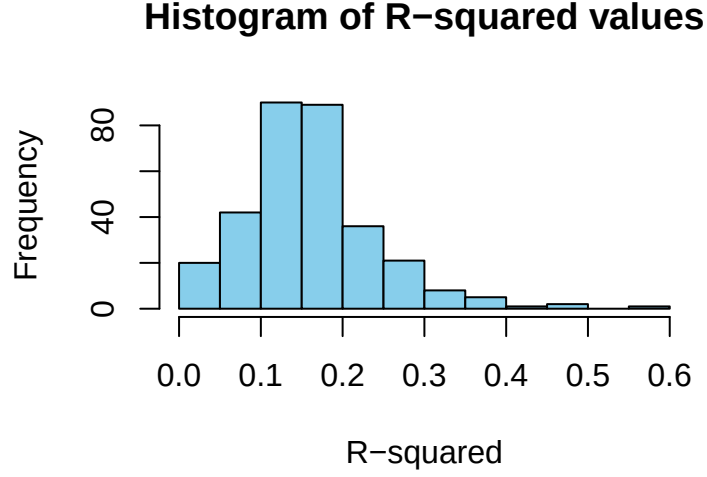


Figure 6: Histogram of R^2 Values for Regression Models

Using this method, we pick neurons 234, 168, and 281.

For these three neurons, we consider two models for comparison.

Model 1:

$$Y_{it} = \alpha_i + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt} \beta_{ij} + \varepsilon_{it}$$

Where:

- Y_{it} is the inferred spikes of neuron i at time t .
- X_{jt} is the inferred spikes of neuron j at time t .
- α_i and β_{ij} are parameters to be fitted.
- ε_{it} is the error term for the model, $\varepsilon_{it} \sim \mathcal{N}(0, \sigma_i^2)$.

Model 2:

$$Y = \alpha + X\beta + \varepsilon$$

Where:

- Y is the time series of neurons 234, 168, and 281, concatenated vertically. It is an array of dimension $(3 \times \text{number of time bins})$ by 1

- X_t is the time series of all neurons other than neurons 234, 168, and 281, concatenated three times vertically. It is a matrix of dimension ($3 \times \text{number of time bins}$) by ($\text{number of other neurons}$).
- α and $\tilde{\beta}$ are parameters to be fitted.
- ε is the error term for the model, where each entry follows $\mathcal{N}(0, \sigma^2)$ distribution.

We chose these two models to compare because we want to understand if neurons integrate the activities from other neurons similarly or differently. In model 1, the coefficient for the predictors could be different across neurons 234, 168, and 281, representing different integrations of the activity of other neurons. On the other hand, model 2 requires the coefficient and integration to be the same. Based on neuroscience knowledge, we hypothesize that model 1 will have better performance, as different neurons have different connectivity with other neurons and integrate signals differently.

In reality, we do not use all the neurons as predictors. This is because later we fit a linear mixed effect model and we have to discard the columns with high colinearity. Hence, for later comparison, we also discard all the predictors with high colinearity here.

3.2 Fitting the Frequentist Model(s)

The two frequentist models were fit by compiling the predictor neurons into matrices and solving for Beta using the following equation:

$$\hat{\beta} = (X^T X)^{-1} (X^T y)$$

The resulting Beta is a column vector containing the intercept value and the 312 coefficients for the predictors. The models for Neurons 234, 168, and 281 are representative of our first frequentist model, which creates 3 separate models for each Neuron that we are analyzing and trying to predict for. The concatenated model is representative of our second frequentist model, which compiles all three response variables into a single matrix and tries to predict for all of them together.

Here, we display the value of the coefficients for the first few predictors in Table 1. (See full set of coefficients in Appendix.)

3.3 Comparing the Frequentist Models

After fitting the models, we conclude that model 1 is better than model 2. In other words, considering each neuron separately and predicting for each one individually is better than trying to compile the predictions into a single model. We reached this conclusion after examining the residual plot, R^2 values, and MSE from five-fold cross validation of each model.

Predictor	Neuron 234	Neuron 168	Neuron 281	Concatenated Model
Intercept	(0.0019820749, 0.0042973961)	(-0.0077705987, -0.0005312599)	(-0.0009465221, 0.0015422220)	(-0.0025887456, 0.0021131830)
Neuron_3	(-0.0000477226, -0.0000121633)	(0.0001155287, 0.0002267124)	(-0.0000541075, -0.0000158847)	(-0.0000007129, 0.0000715006)
Neuron_4	(-0.0000392193, 0.0000027763)	(0.0000499259, 0.0001812339)	(-0.0000493106, -0.0000041694)	(-0.0000191026, 0.0000661815)
Neuron_5	(-0.0000642615, 0.0000006114)	(-0.0005035074, -0.0003006687)	(-0.0000723183, -0.0000025862)	(-0.0002229934, -0.0000912502)
Neuron_6	(-0.0000022124, 0.0000247042)	(-0.0000079717, 0.0000761888)	(-0.0000462356, -0.0000173029)	(-0.0000228026, 0.0000318594)
Neuron_7	(-0.0000574232, -0.0000265640)	(-0.0000763770, 0.0000201110)	(-0.0000013821, 0.0000317886)	(-0.0000496421, 0.0000130265)
Neuron_8	(0.0000010607, 0.0000354907)	(0.0000456376, 0.0001532904)	(-0.0000032178, 0.0000337912)	(0.0000093820, 0.0000793022)
Neuron_9	(-0.0000066706, 0.0000666413)	(0.0001317571, 0.0003609820)	(-0.0000770307, 0.0000017723)	(0.0000051348, 0.0001540157)
Neuron_11	(-0.0000917205, -0.0000370458)	(0.0001899531, 0.0003609050)	(-0.0000242702, 0.0000344997)	(0.0000165372, 0.0001275699)
Neuron_12	(-0.0000337814, -0.0000135145)	(0.0000392893, 0.0001026580)	(0.0000186658, 0.0000404507)	(0.0000050491, 0.0000462069)

Table 1: 95% Confidence Intervals for Predictor Coefficients for Models without Interaction Terms

Residual plots: Residual plots are displayed in Figure 7. None of the residual plots appear to be traditionally desirable, with all of them lacking a random spread across the x-axis and instead making similar patterns to one another. However, this does not mean the models are all bad or useless. Due to lack of certainty when making predictions in the neuroscience field, what might appear as a bad or weak model in a different statistical context is considered an acceptable model in a neuroscience context. The residual plots for our first frequentist model (the residual plots belonging to Neurons 234, 168, and 281) are slightly more desirable than the residual plot for our second frequentist model (the residual plot belonging to the Concatenated Model) as the spread across the x-axis seems to be more balanced and evenly split in our residual plots for our first frequentist models than our second.

To motivate further analysis with an added interaction term between behavior and neuron activity level, we color coded the residual plot. The red circles correspond to when the mouse is reaching and the blue dots correspond to the mouse resting. We observe that neurons tend to spike more during reaches, as the red dots are denser at higher fitted values. In addition, residual is smaller at higher fitted values and bigger at lower fitted values. This motivates us to fit separate models for different behaviors, reach or nonreach, by adding an interaction term (see section 3.4).

R^2 values and MSE: The adjusted R^2 values of our first frequentist model (0.491, 0.41, and 0.357) are better than the R^2 value of 0.0796 for our second frequentist model. However, comparing a model consisting of three separate models using R^2 can be difficult, so Figure 8 shows the results from performing five-fold cross validation on the models to find the MSE. The results in Figure 8 highlight that our first frequentist model is better than our second frequentist model.

This conclusion aligns with our intuition from neuroscience: each neuron connects to the overall neural network differently and so they will each be affected uniquely by every predictor neuron. This result is further supported by comparing the coefficients between the models: the coefficients between the concatenated model and the separated models tend to differ. The numerical values of the differences between these coefficients are displayed in Figure 9.

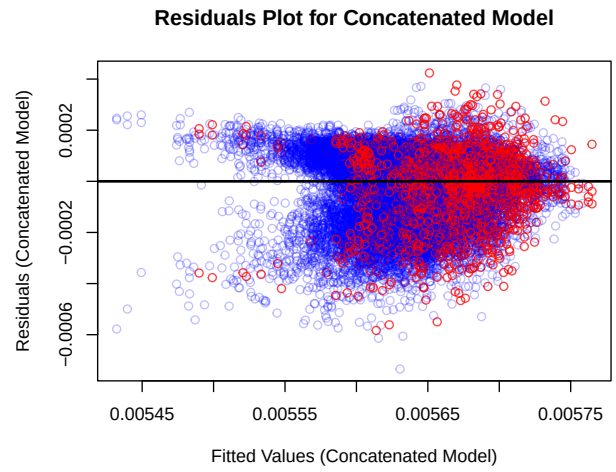
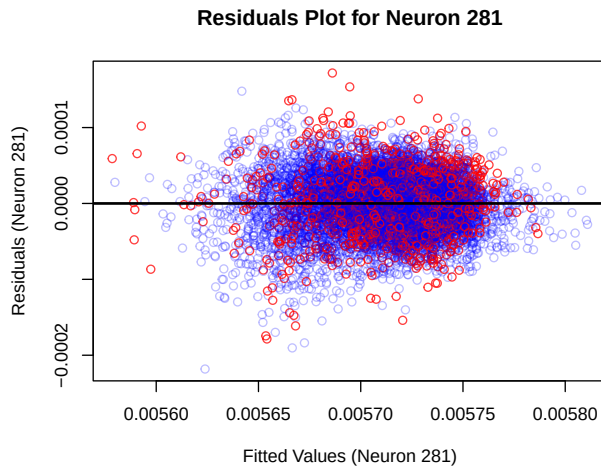
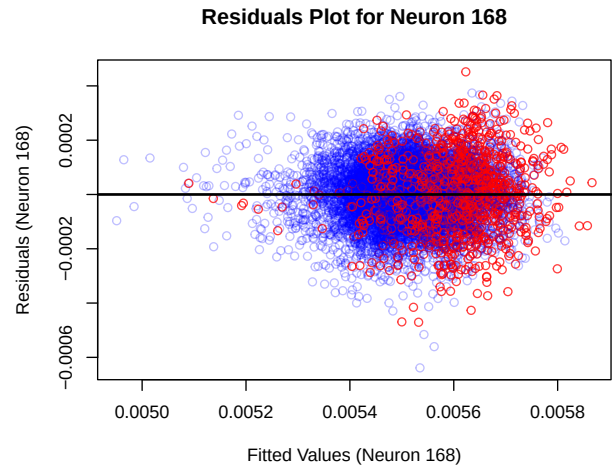
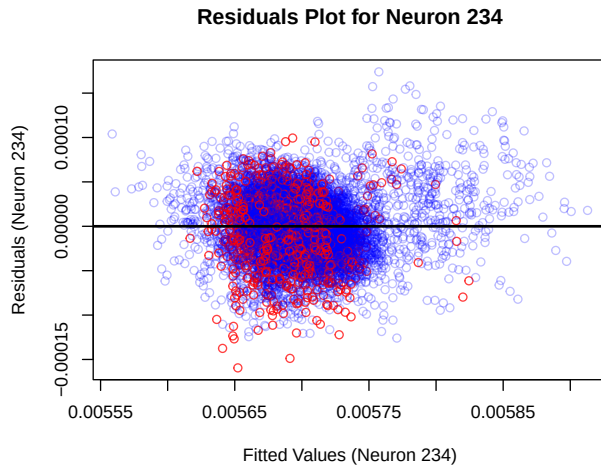


Figure 7: Residual Plots for Frequentist Models without Interaction Term

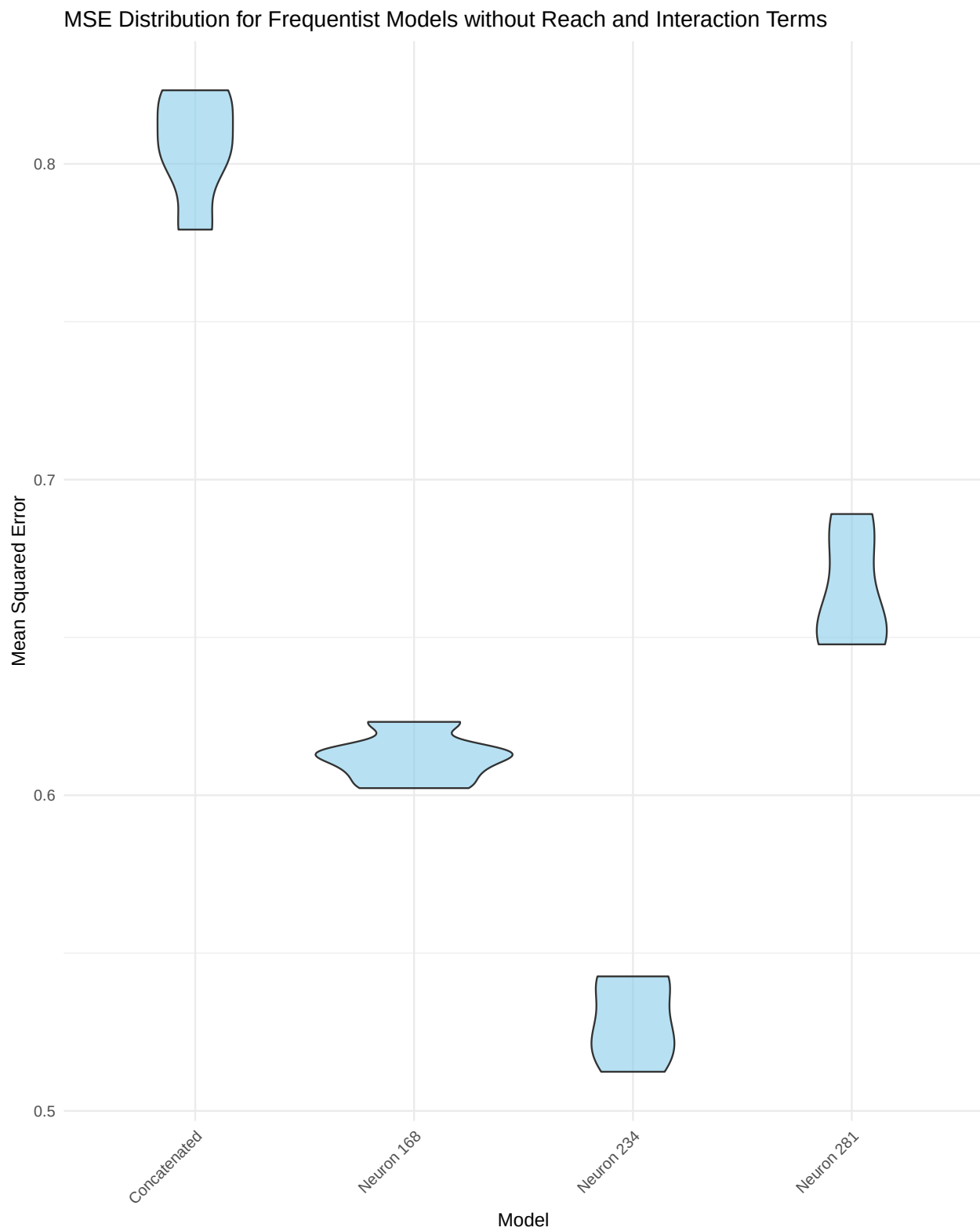


Figure 8: Violin Plot of Five-Fold CV MSEs for Frequentist Models without Interaction Term

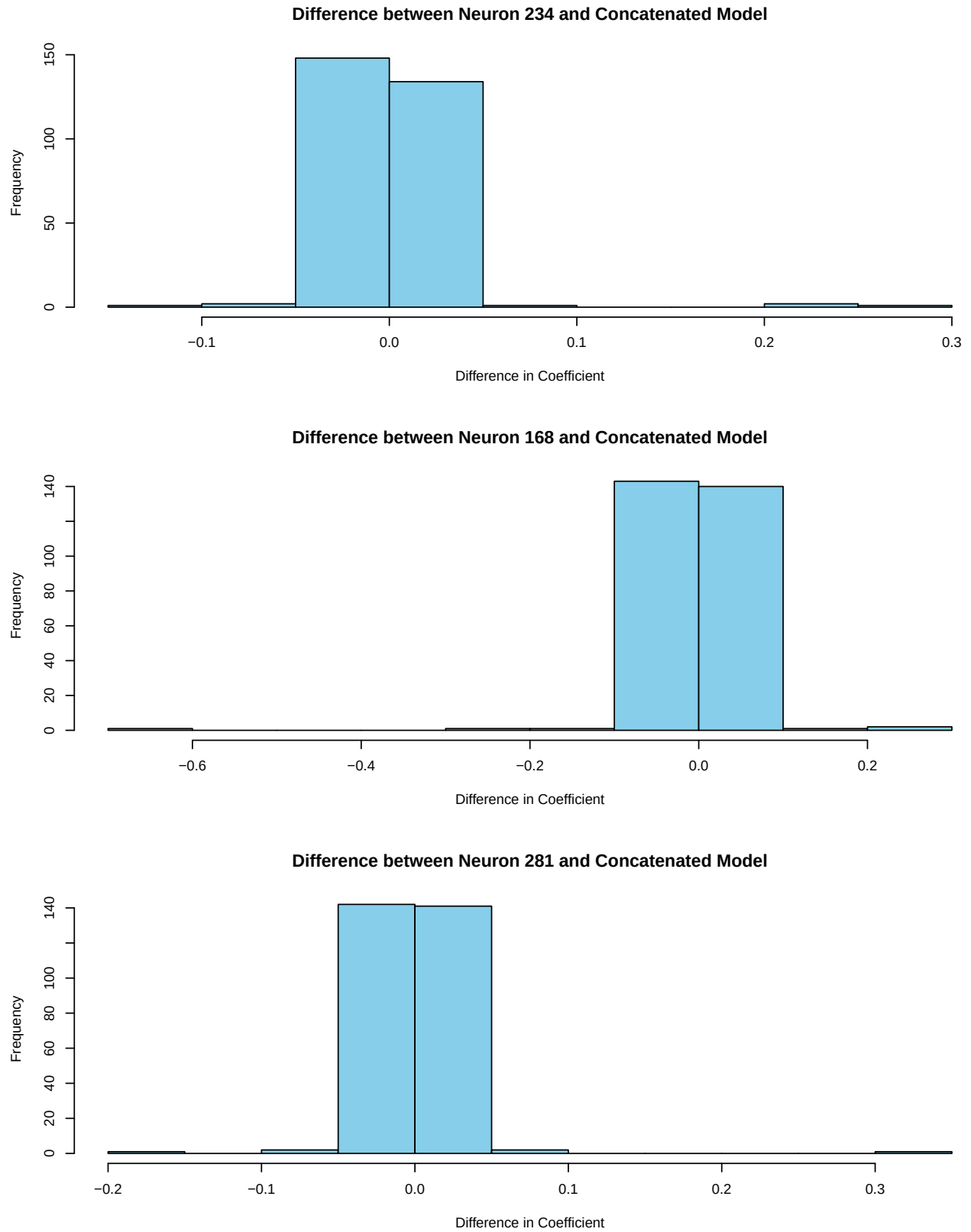


Figure 9: Histogram of Difference in Coefficients between Frequentist Models

3.4 Frequentist Model with Behavior and Interaction Terms

Our previous analysis suggested that we should fit separate models for different behaviors (i.e. reach and nonreach). We achieve this through adding a behavior term and an interaction term.

Model with Behavior and Interaction Terms:

Model 1:

$$Y_{it} = \alpha_i + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt} \beta_{ij} + \gamma_i Z_t + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt} Z_t \delta_{ij} + \varepsilon_{it}$$

Where:

- Y_{it} is the inferred spikes of neuron i at time t .
- X_{jt} is the inferred spikes of neuron j at time t .
- Z_t is the mouse behavior at time t . $Z_t = 1$ indicates that the mouse is reaching.
- α_i , γ_i , β_{ij} and γ_{ij} are parameters to be fitted.
- ε_{it} is the error term for the model, $\varepsilon_{it} \sim \mathcal{N}(0, \sigma_i^2)$.

Model 2:

$$Y = \alpha + X\beta + \gamma Z + \delta X \odot Z + \varepsilon$$

Where:

- Y is the time series of neurons 234, 168, and 281, concatenated vertically. It is an array of dimension ($3 \times \text{number of time bins}$) by 1
- X_t is the time series of all neurons other than neurons 234, 168, and 281, concatenated three times vertically. It is a matrix of dimension ($3 \times \text{number of time bins}$) by (number of other neurons).
- Z is a vector of length (number of time bins). Each entry indicates whether the mouse is reaching. $\odot$ indicates element-wise multiplication onto the rows of X .
- α , β , γ and δ are parameters to be fitted.
- ε is the error term for the model, where each entry follows $\mathcal{N}(0, \sigma^2)$ distribution.

Conclusion:

Figure 10 compares the MSE with and without the interaction terms. We observe that adding the interaction term slightly improved the models. This aligns with our hypothesis that considering behavior would improve the prediction. We also display the residual plots and the difference between the coefficients in Figure 11 and Figure 12. The observations are similar to the previous observations in section 3.3.

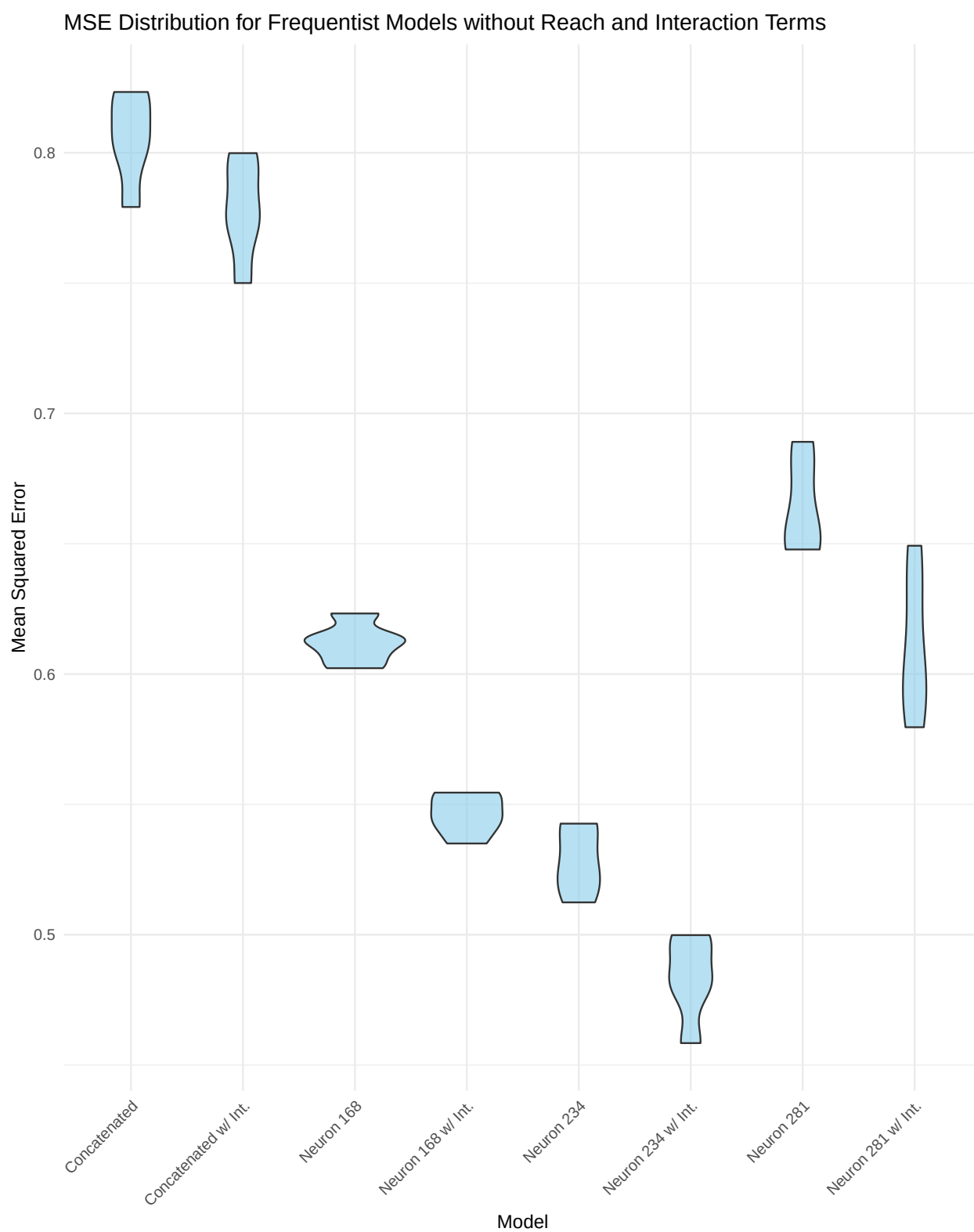


Figure 10: Violin Plot of Five-Fold CV MSEs for Frequentist Models

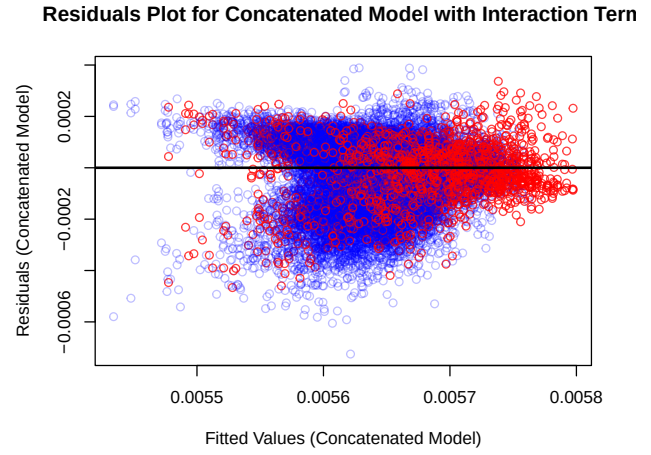
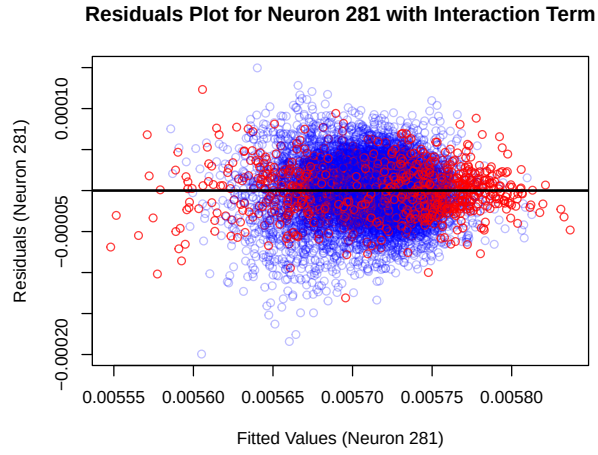
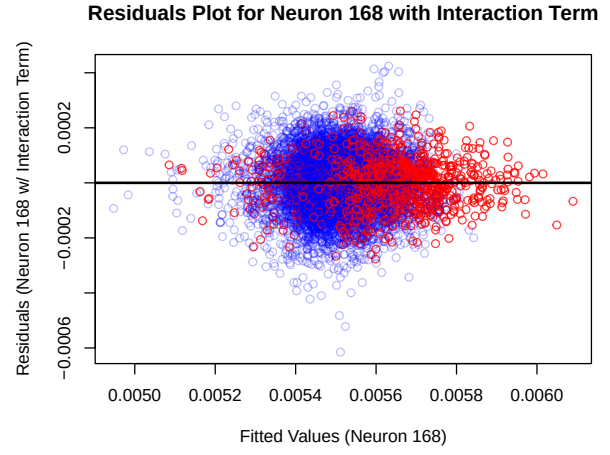
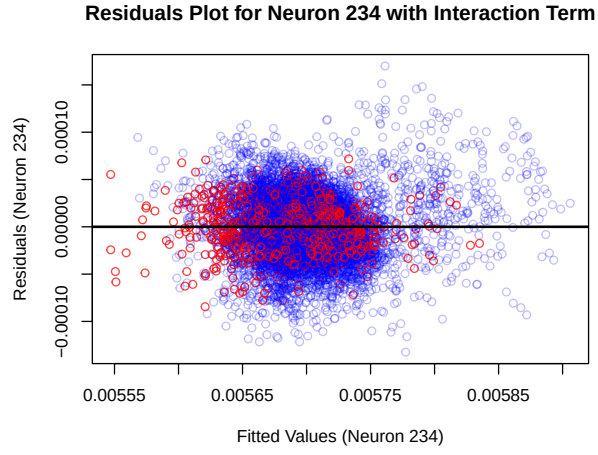


Figure 11: Residual Plots for Frequentist Models with Interaction Term

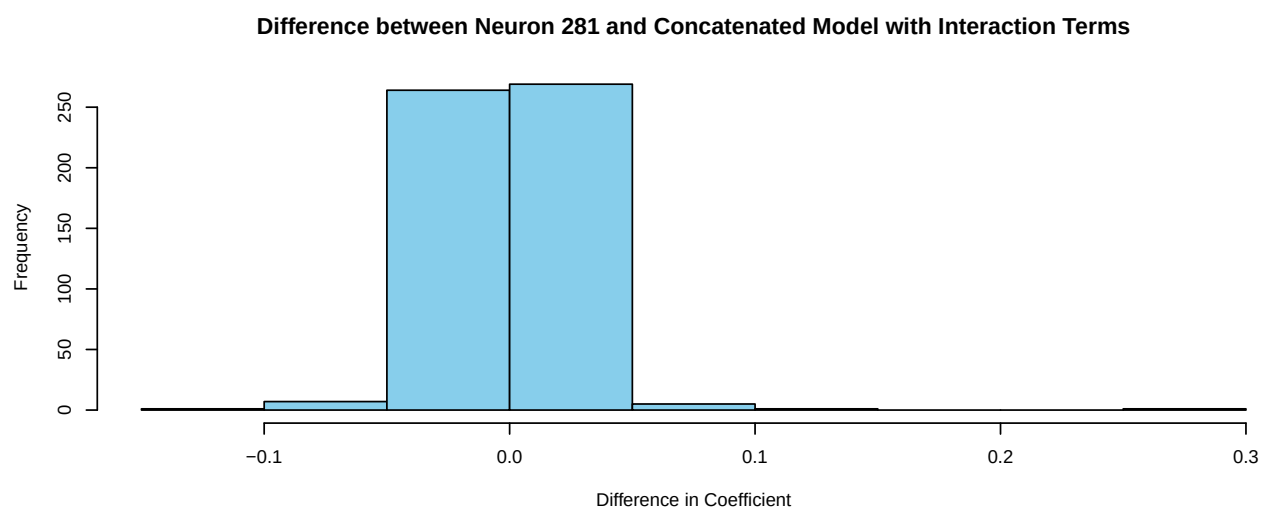
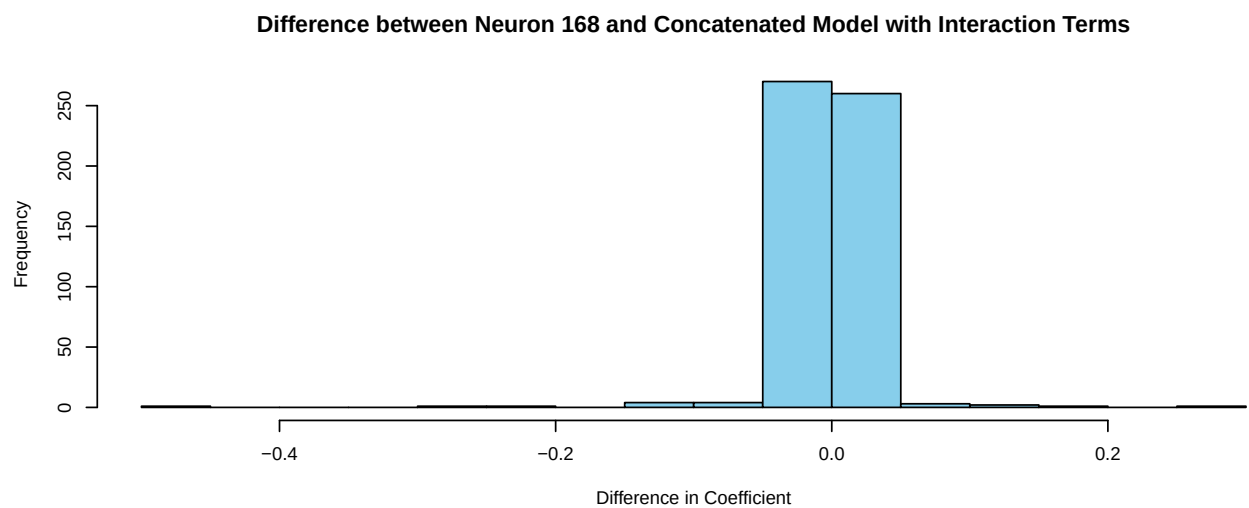
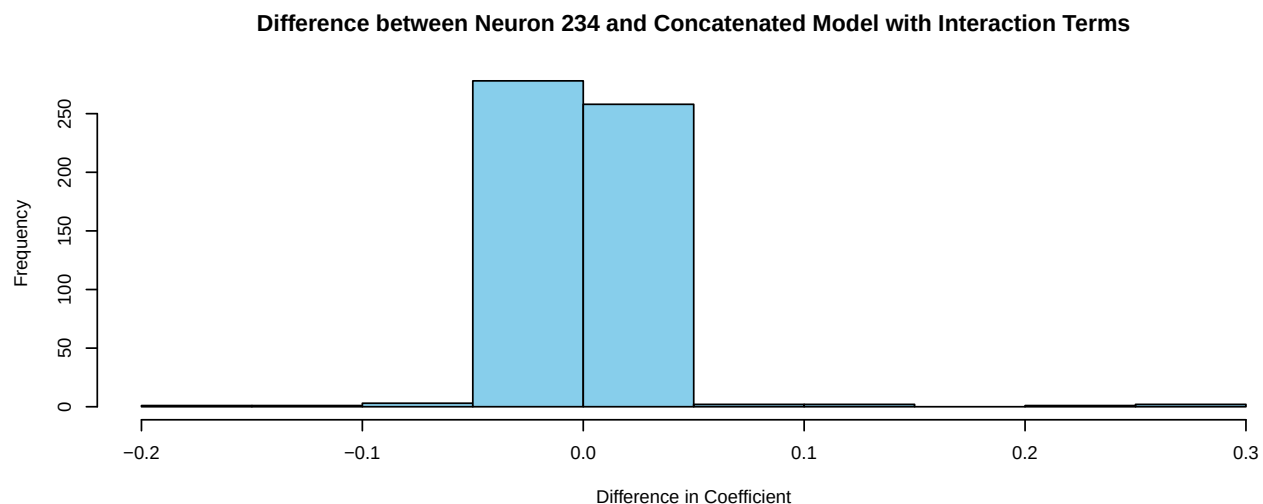


Figure 12: Histogram of Difference in Coefficients between Frequentist Models with Interaction Terms

3.5 Frequentist Linear Mixed Effects Model

In previous models, we fitted separate models for different neurons. Yet, neuron responses are often correlated, so we would like to capture both population-level effects and neuron-level effects. Therefore, we fit a linear mixed effects model in this section. The model is formulated as follows:

$$Y_{it} = \alpha + u_i + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt}(\beta + v_{ij}) + (\gamma + w_i)Z_t + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt}Z_t(\delta + r_{ij}) + \varepsilon_{it}$$

$$u_i \sim \mathcal{N}(0, \sigma_u^2), \quad v_{ij} \sim \mathcal{N}(0, \sigma_v^2), \quad w_i \sim \mathcal{N}(0, \sigma_w^2), \quad r_{ij} \sim \mathcal{N}(0, \sigma_r^2), \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$$

Where:

- α is the fixed intercept across all neurons.
- $u_i \sim \mathcal{N}(0, \sigma_u^2)$ is the random deviation for neuron i , indicating neuron-level heterogeneity.
- β is the fixed slope across all neurons.
- $v_{ij} \sim \mathcal{N}(0, \sigma_v^2)$ captures random slope for each neuron.
- γ is the fixed behaviorial effect across all neurons.
- $w_i \sim \mathcal{N}(0, \sigma_w^2)$ is the random effect of behavior for each neuron.
- δ is the fixed effect of interaction term.
- $r_{ij} \sim \mathcal{N}(0, \sigma_r^2)$ is the random effect of interaction term.
- $\varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$ models the noise.

However, due to the computational constraint, we could only model the random effects of the intercepts and the reach term. Hence, the model we implemented is as follows:

$$Y_{it} = \alpha + u_i + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt}\beta + (\gamma + w_i)Z_t + \sum_{j=1, j \notin \{234, 168, 281\}}^{315} X_{jt}Z_t\delta + \varepsilon_{it}$$

$$u_i \sim \mathcal{N}(0, \sigma_u^2), \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$$

The parameters we are fitting are α , σ_u , β , γ , δ , and σ .

Predictor	Neuron 234	Neuron 168	Neuron 281	Concatenated Model	Linear Mixed Effects Model
Intercept	(0.0021583739895, 0.004471207307)	(-0.01063487569, -0.003543537845)	(-0.001066063085, 0.0014223170948)	(-0.0036737543079, 0.001276228231)	(-0.065692799, 0.1455247155)
Neuron_3	(-0.0000780009480, -0.000031094434)	(-0.00007737601, 0.000066443228)	(-0.000057490714, -0.000070239391)	(-0.0000809523722, 0.000019438098)	(-0.039520983, -0.0022942587)
Neuron_4	(-0.0000771287406, -0.000032103193)	(0.00004983924, 0.000187891285)	(-0.000043561090, 0.000048819516)	(-0.0000332124807, 0.000063152299)	(-0.042981567, -0.0006984383)
Neuron_5	(-0.0000511415859, 0.000012723366)	(-0.00048650617, -0.000290690954)	(-0.000066471120, 0.000022412593)	(-0.0002149835349, -0.000078298199)	(-0.089987366, -0.0322766238)
Neuron_6	(0.0000094065782, 0.000036206673)	(-0.00004731870, 0.000034852600)	(-0.000048115324, -0.0000192810697)	(-0.0000343873172, 0.000022970902)	(-0.046223033, 0.0341552136)
Neuron_7	(-0.0000534134398, -0.000017292806)	(-0.00006293211, 0.000047816738)	(0.000009431073, 0.0000482933072)	(-0.000043360051, 0.000033970261)	(-0.023125219, 0.0159889453)
Neuron_8	(-0.0000201501730, 0.000015912228)	(0.00003237543, 0.000142945727)	(0.000005235175, 0.0000440347566)	(-0.0000018652937, 0.000075316341)	(-0.010583735, 0.0429009227)
Neuron_9	(0.0000186736986, 0.000098028082)	(0.00014532872, 0.000388635806)	(-0.000053085999, 0.0000322914790)	(0.0000200605323, 0.000189896729)	(-0.003580357, 0.0558150235)
Neuron_11	(-0.0000577119732, 0.000010887332)	(0.00021347274, 0.000423803874)	(-0.000032775416, 0.0000410306597)	(0.0000263755795, 0.000173193493)	(0.003592414, 0.0369334518)
Neuron_12	(-0.0000350784826, -0.000014753609)	(0.00002591826, 0.000088236004)	(0.000021802847, 0.0000436704030)	(-0.0000001173250, 0.000043382467)	(0.002110504, 0.0493843480)

Table 2: 95% Confidence Intervals for Predictor Coefficients for Models with Interaction Terms and Linear Mixed Effects Model

Results:

Table 2 displays the 95% confidence interval for the intercept and slope for each neuron.

After performing five-fold cross validation, Figure 13 showed that the linear mixed effect model has a higher MSE than all other frequentist models, except for the Concatenated Model without reach and interaction terms which it performed similarly to. This is likely because we could only take into account the random effect on the intercept and reach term, due to computational constraints. Additionally, the complexity of a neural network in combination with this data only being for a partial network might have caused the lme model to choose inaccurate random effects that actually decreased model performance.

Figure 14 displays the residual plot for the lme model. Compared to our previous two concatenated models, the residual plot for the lme model is more desirable and more similar to the residual plots for our separated models, showing the promise of using a hierarchical model when trying to create a single concatenated model for prediction.

In conclusion, for the lme model, we observe higher MSE for the testing data from five-fold cross validation, but less pattern in the residual plot. Hence, lme model could be a better fit, despite the higher MSE for the testing data from five-fold cross validation.

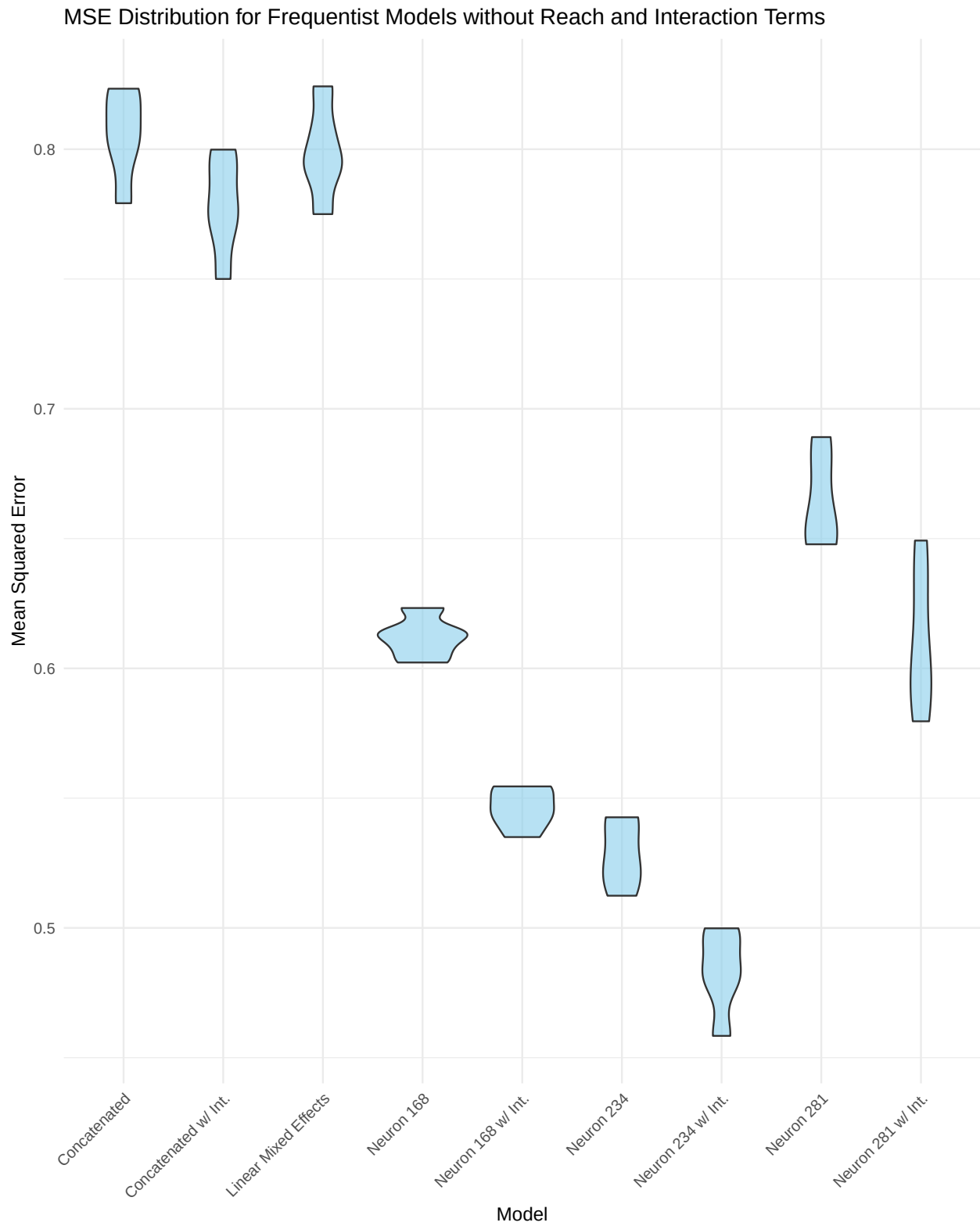


Figure 13: Violin Plot of Five-Fold CV MSEs for Frequentist Models

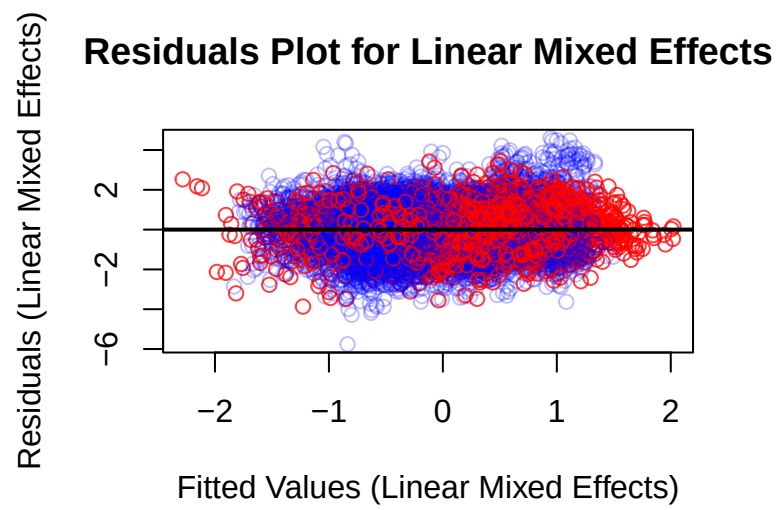


Figure 14: Residual Plot for Frequentist Linear Mixed Effects Model

4. Bayesian Analysis

In this section, we fit a Bayesian model to predict neuron activity. This allows us to account for both population-level information and neuron-level heterogeneity, as well as to incorporate prior information.

4.1 Proposed Bayesian Model(s)

We propose to model the neuron activity using a Bayesian linear regression model. This is because in our previous frequentist analysis, we observed both correlation across neuron population, but also heterogeneity for individual neurons. A Bayesian linear regression model is able to capture the pattern at both the population and individual level.

The first model is formulated as follows:

$$Y_{it} \sim \mathcal{N}(\mu_{it}, \sigma^2)$$

$$\mu_{it} = \alpha + u_i + \sum_{j=1}^p X_{jt}\beta_j + \gamma Z_t + \sum_{j=1}^p X_{jt}Z_t\delta_j$$

Random effects:

$$\pi(u_i) \sim \mathcal{N}(0, \sigma_u^2)$$

Fixed effect:

$$\pi(\beta_j) \propto 1, \pi(\delta_j) \propto 1, \pi(\gamma) \propto 1, \pi(\sigma_u) \sim t(3, 0, 2.5), \pi(\sigma) \sim t(3, 0, 2.5)$$

The second model is formulated as follows:

$$Y_{it} \sim \mathcal{N}(\mu_{it}, \sigma^2)$$

$$\mu_{it} = \alpha + u_i + \sum_{j=1}^p X_{jt}\beta_j + \gamma Z_t + \sum_{j=1}^p X_{jt}Z_t\delta_j$$

Random effects:

$$\pi(u_i) \sim \mathcal{N}(0, \sigma_u^2), \pi(w_i) \sim \mathcal{N}(0, \sigma_u^2)$$

Fixed effect:

$$\pi(\beta_j) \propto 1, \pi(\delta_j) \propto 1, \pi(\gamma) \propto 1, \pi(\sigma_u) \sim t(3, 0, 2.5), \pi(\sigma) \sim t(3, 0, 2.5)$$

We chose these priors because they are the default of the brms package.

4.2 Fitting the Bayesian model(s)

We fit the models by running MCMC chains, with 2 chains, 3000 iterations and 1000 iterations as warm up.

After fitting the models we performed Leave-One-Out cross validation to select the better model. We found that the better model was Model 2, the model that included random effects on the reach term. The elpd_diff for model 1 was -71.7 and the se_diff was 17.7, showing that model 2 is much better, so when moving forwards with sensitivity analysis, reach mixed effects will be included.

After selecting Bayesian Model 2 as the better Bayesian model, we performed sensitivity analysis by refitting the model with the following weakly informative priors:

$$\pi(\beta_j) \sim \mathcal{N}(0, 1), \pi(\delta_j) \sim \mathcal{N}(0, 1), \pi(\gamma) \sim \mathcal{N}(0, 1), \pi(\sigma_u) \sim t(3, 0, 1), \pi(\sigma) \sim t(3, 0, 1.5)$$

We found that the original model (Bayesian Model 2) was the better model. The elpd_diff of the model with the weakly informative prior was -177.2 and the sd_diff was 12.5.

Results:

Table 3 displays the 95% confidence interval for the coefficients β_j and the random effects u_i and w_i .

We check convergence by examining the trace plots in Figure 15, the acf plots in Figure 16, the Rhat Values shown in Figure 17, and the ESS values in Figure 18. Due to the high number of predictors, we chose to show the intercept, reach term, a few neuron activity terms, and a few interaction terms for the trace plots and acf plots in order to properly represent plots for each group of predictors within the limited space. The trace plots shows convergence in the neuron activity slopes and lack of convergence in the random effects and the intercept, reach, and interaction terms. The acf plots, Rhat plots, and ESS plots all support this. The MCMC simulation converged for the slopes of the neuron activity terms, indicated by a Rhat value close to 1, but not the random effects or the reach and interaction terms and the intercept.

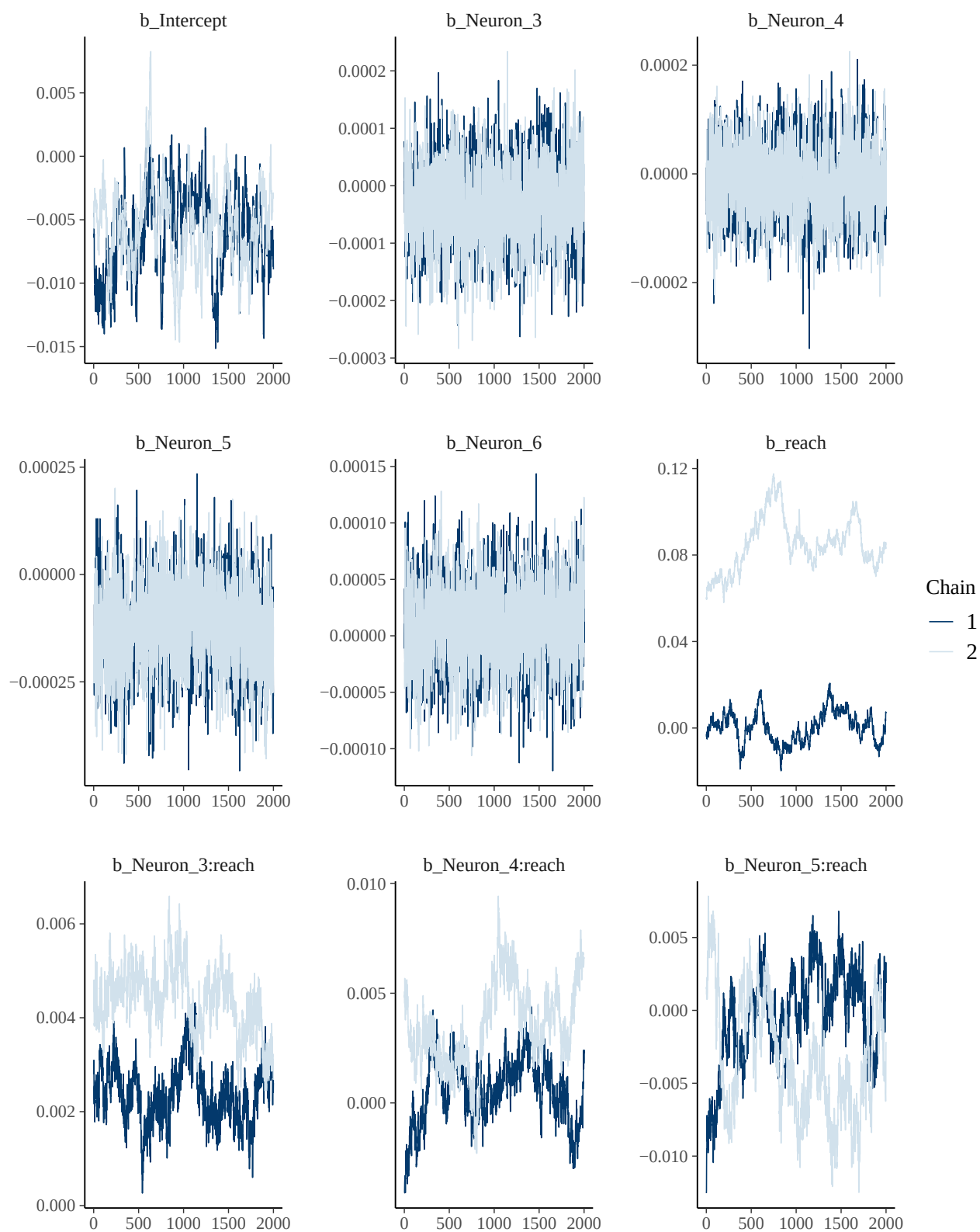


Figure 15: Trace Plots for Best Bayesian Model

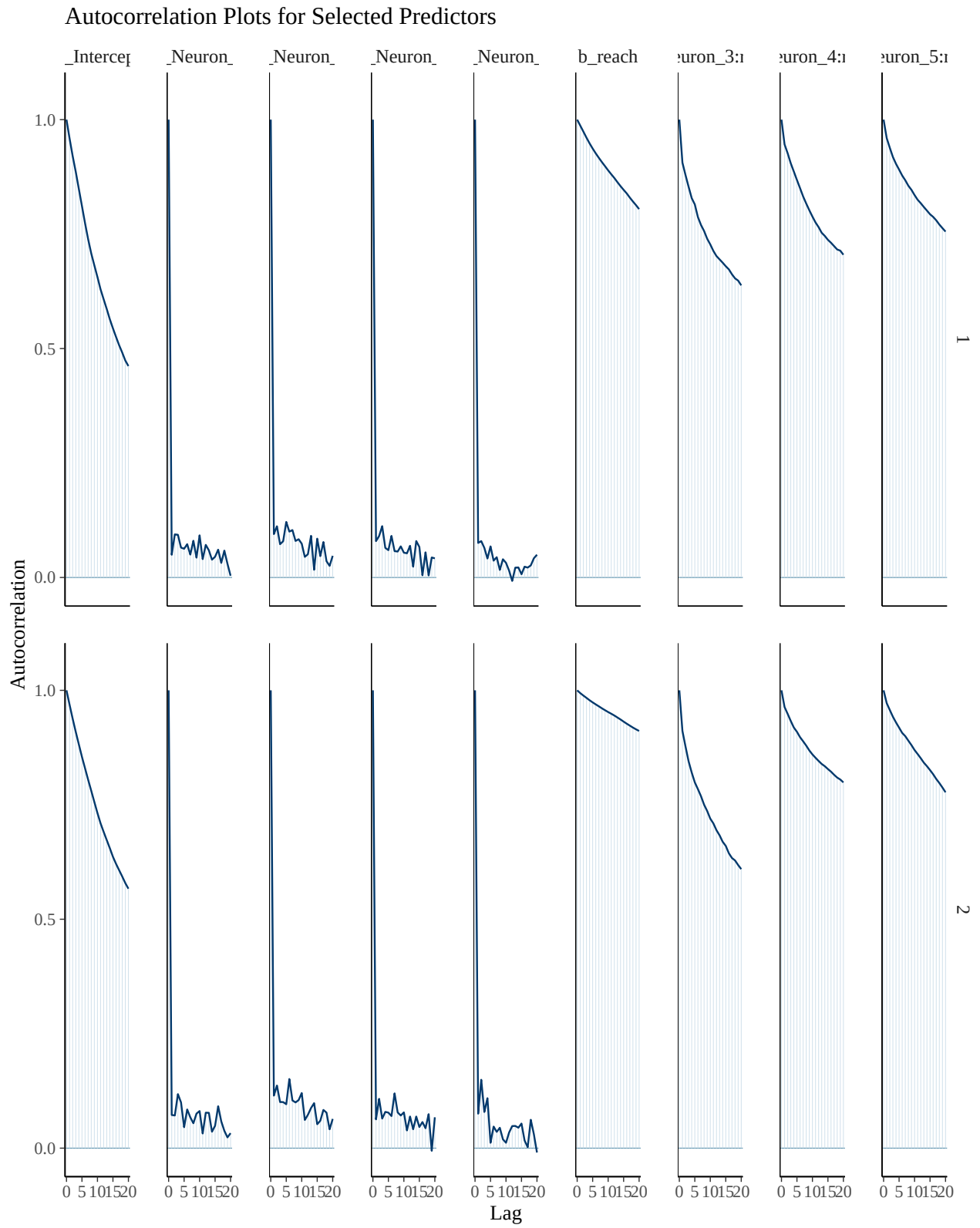


Figure 16: ACF Plots for Best Bayesian Model

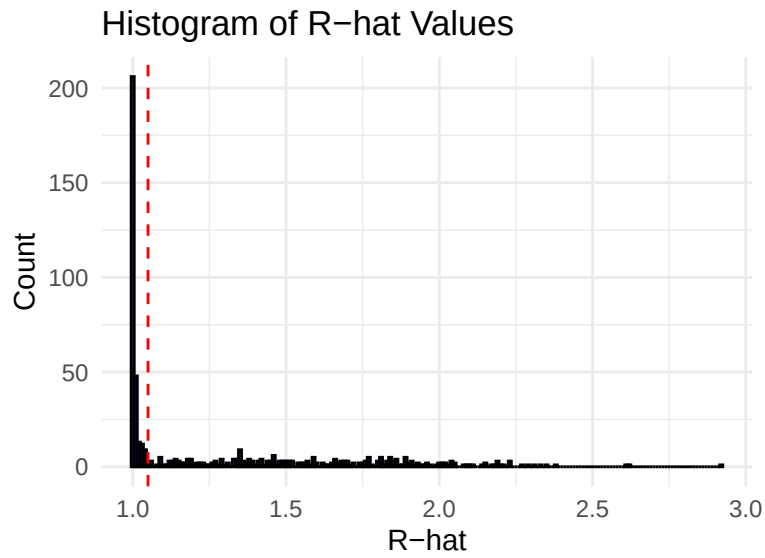


Figure 17: Histogram of R-hat Values for Best Bayesian Model

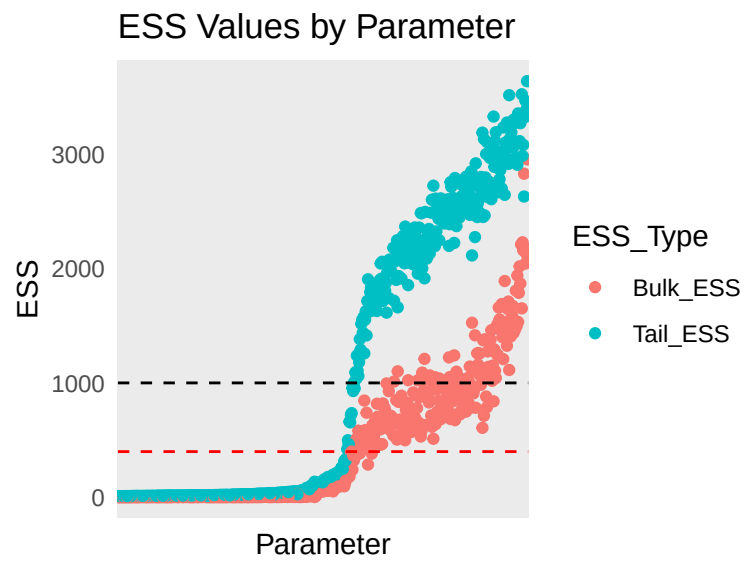


Figure 18: Scatterplot of ESS vs. Variable for Best Bayesian Model

Predictor	95% CI for Coefficients
Intercept	(-0.012315, 0.000253)
Neuron_3	(-0.000167, 0.000102)
Neuron_4	(-0.000138, 0.00011)
Neuron_5	(-0.000312, 0.000056)
Neuron_6	(-0.000061, 0.000078)
Neuron_7	(-0.000022, 0.000188)
Neuron_8	(-0.00005, 0.000151)
Neuron_9	(0.000082, 0.000578)
Neuron_11	(-0.000157, 0.000258)
Neuron_12	(-0.000026, 0.000083)
Random Effects	
Intercept (Neuron_168)	(-0.00015, 0.002167)
Intercept (Neuron_234)	(0.000044, 0.002361)
Intercept (Neuron_281)	(0.000056, 0.002375)
Reach (Neuron_168)	(-0.01908, 0.005007)
Reach (Neuron_234)	(-0.019113, 0.004952)
Reach (Neuron_281)	(-0.019066, 0.005024)

Table 3: 95% Confidence Intervals for Coefficients and Random Effects in the Bayesian Hierarchical Model

4.3 Prediction

To make posterior predictions, we sampled 25000 observations from the data that wasn't used to fit any of our models and split it into 5 different test sets. Using these 5 different test sets that none of our data was fit with will allow us to create comparable MSEs to compare the frequentist models to the Bayesian models. Additionally, we will use the predict function in the brms package to generate predictions for 1 of these test sets.

Figure 19 shows that the Bayesian model performed terribly compared to the Frequentist model on the testing data. We believe this is likely caused by the convergence issues of the random effects and reach and interaction terms. This means that for this analysis, the best overall model is the separated Frequentist model with reach and interaction terms.

The comparison between predicted data and actual data is displayed in Figure 20, showing that after removing the outliers, the models predictions are very weak. The points should ideally follow along the dotted line but they do not, showing the weakness of the model that is likely caused by the convergence issues of the random effects and reach and interaction terms.

We also did posterior predictive checking in Figure 21 for our model. It is clear that our Bayesian model is not very good at prediction as the posterior predictions do not properly align with the observed data. Further showing one of the weaknesses of our Bayesian hierarchical model.

Finally, after selecting our best overall model, the separated Frequentist model with reach and interaction terms, we created the predicted Neural Network for each neuron. Figure 22 shows these predicted Neural Networks, the neurons within each plot are the ones predicted by the model to have a non-zero connection with the neuron being predicted for. The true predicted Neural Network would show connection lines from each neuron to the neuron being predicted for, but for visibility reasons we only show the neurons with the strongest connections.

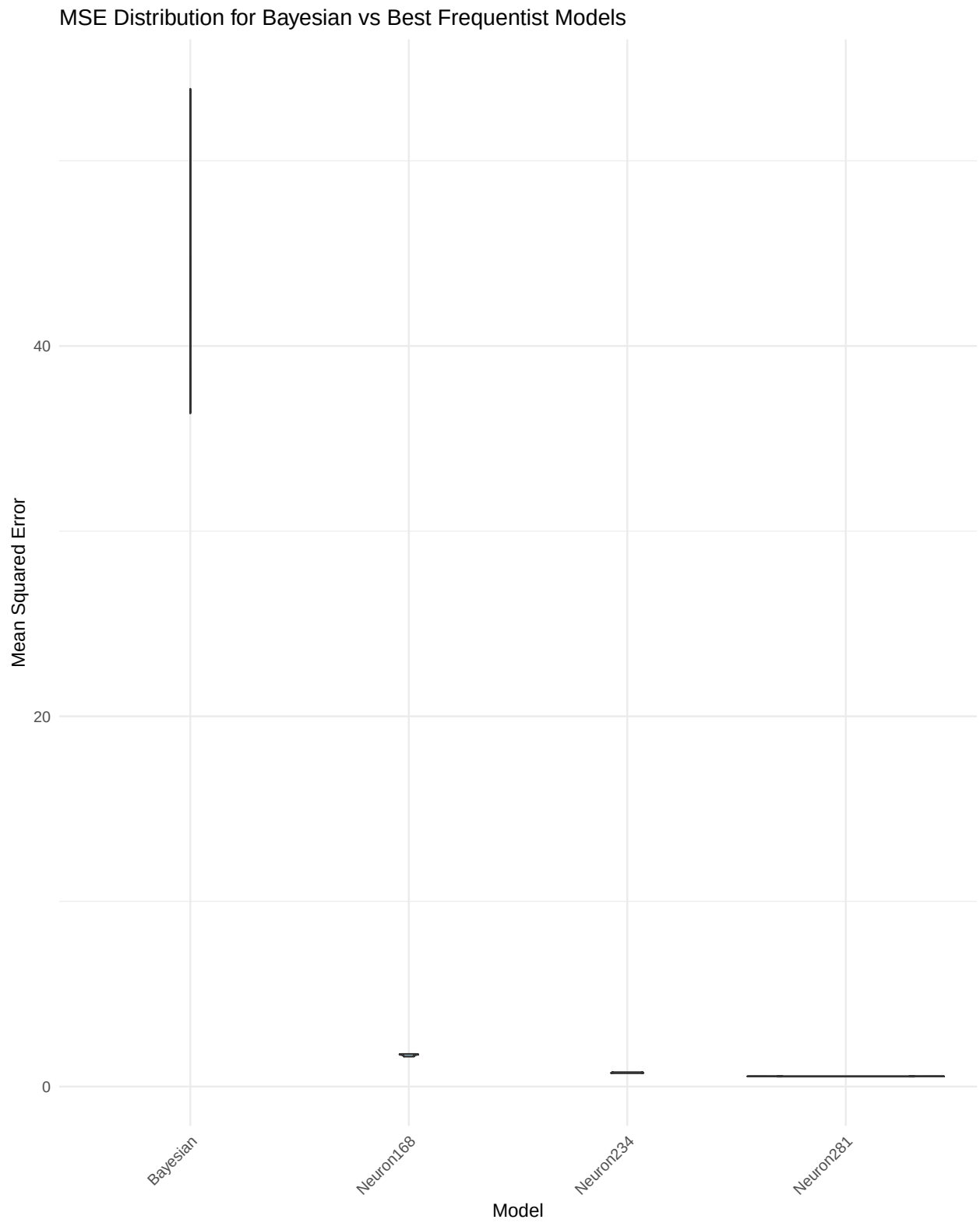


Figure 19: MSE Model Comparison between Best Bayesian Model and Best Frequentist Model

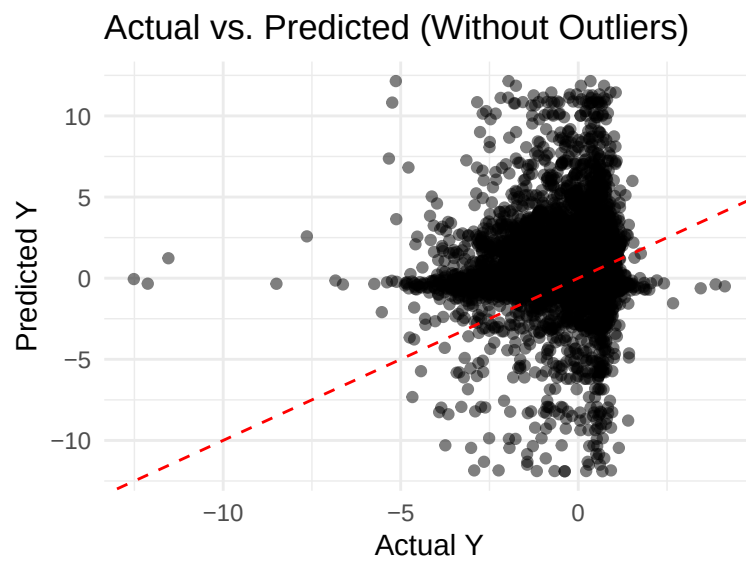


Figure 20: Actual Y Values vs. Predicted Y Value for Bayesian Heirarchical Model with Outliers Removed

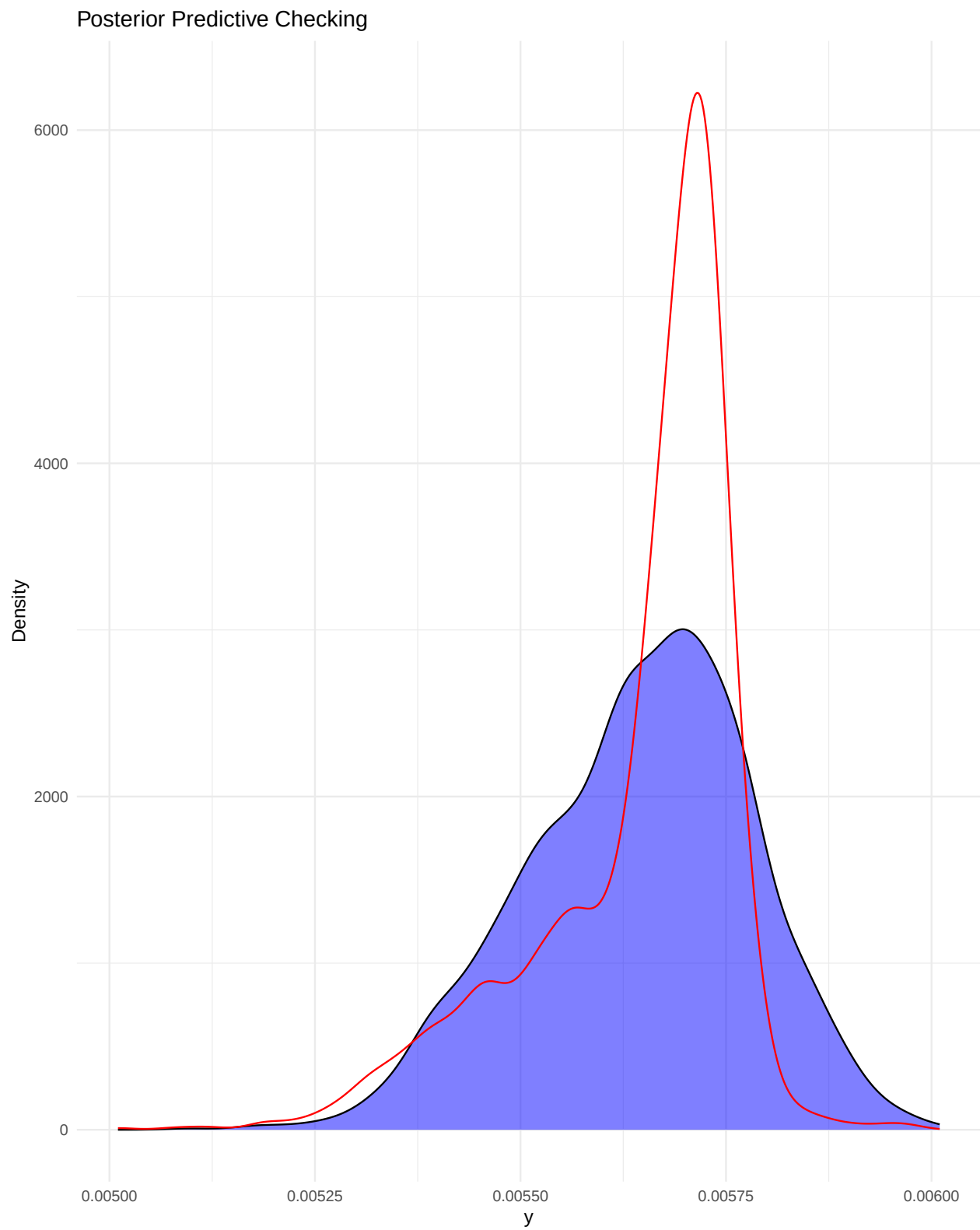


Figure 21: Posterior Predictive Checking for Best Bayesian Model

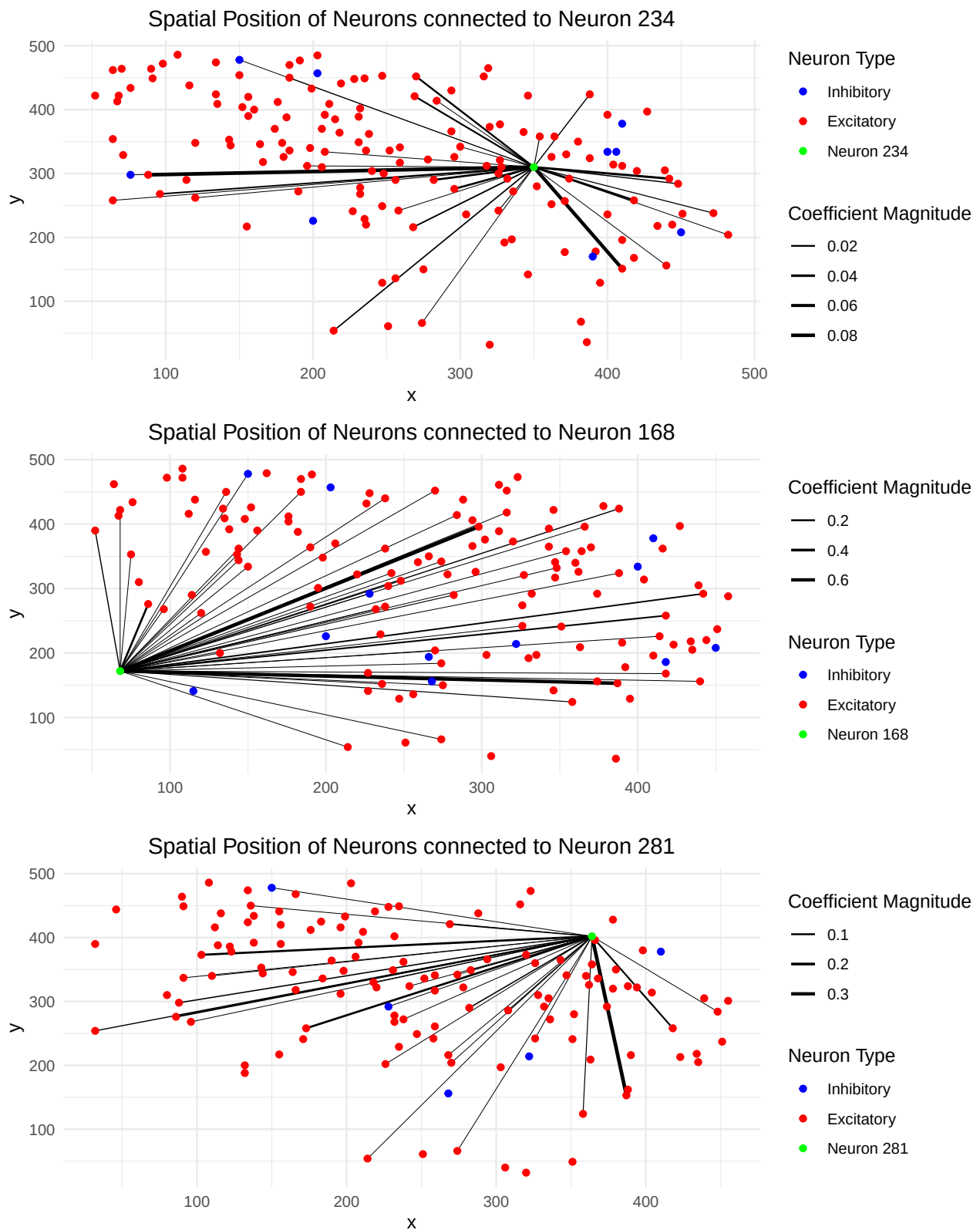


Figure 22: Predicted Neural Networks Using Best Model

5. Discussion

While the current model offers valuable and interesting insights, there are several challenges and limitations that should be addressed to improve the accuracy and generalizability of the findings. These limitations stem from both the data itself and the inherent complexity of neural networks, which can make accurate modeling difficult. Below, we detail the key issues and provide suggestions for future improvements:

5.1. Incomplete Network:

The dataset used for this model represents a small subset of neurons within the M1 cortex due to limitations in imaging techniques. As a result, we do not have access to the full network of neurons, and some of the neurons in the area of interest are not recorded. This incomplete network restricts the ability to generalize the results to the entire M1 cortex. To mitigate this, future studies could seek to expand the network coverage using more advanced imaging techniques that allow for a larger area to be recorded.

5.2. Upstream Signals:

The model does not account for shared upstream signals that influence the entire M1 cortex. The absence of data on these upstream signals is a significant limitation, as they likely play a key role in driving neural activity within the network. Incorporating upstream signal measurements, if available, would help create a more complete model.

5.3. Complexity of Neural Networks:

The complexity of neural networks, particularly in the brain, poses a major challenge for traditional modeling approaches. The M1 cortex consists of a vast and intricate network of interconnected neurons with diverse properties, which are not fully captured by the current models. Neural networks exhibit complex, non-linear relationships and dynamic behavior that are difficult to model accurately with simple linear or frequentist models. As such, the current models make simplifying assumptions about the network, such as treating neurons as homogenous or assuming that the effects of connections between neurons are linear. These assumptions lead to poor model performance and inaccurate conclusions.

5.4. Potential Confounding Factors:

Given the incomplete network and the lack of upstream signal information, there may be important confounding factors influencing neural activity that are not captured by the current model. These omitted variables could bias the results and limit the ability to draw valid conclusions.

5.5. Neuronal Heterogeneity:

Another limitation is that the current model does not fully account for the heterogeneity of neurons within the M1 cortex due to computation limitations. Ideally, random effects would account for this heterogeneity but computationally our models couldn't handle this. Neurons in the brain differ in their response patterns, firing rates, synaptic connections, and other properties. By treating all neurons as if they are homogenous, the model overlooks these important differences, which could impact the results.

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Appendix

R code for Exploratory Data Analysis graphs:

```
#Neuron 234 Time Series Plot
spks_t = as.data.frame(t(spks))
neurons = paste("Neuron", 1:315, sep = "_")
```

```
colnames(spks_t) = neurons

time_points = seq(0, by = 30 / 1000, length.out = 148468)
spks_t$time = time_points

ggplot(spks_t, aes(x = time, y = Neuron_234)) +
  geom_line(size = 1) +
  labs(title = "Time Series", x = "Time", y = "Neuron 234") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5))
```

#Neuron Type Distribution

```
colnames(redlabels) = c("neuron_type")
redlabels$neuron_type = factor(redlabels$neuron_type, levels= c(TRUE, FALSE), labels=c('Excitatory', 'Inhibitory'))
ggplot(redlabels, aes(x = neuron_type)) +
  geom_bar() +
  theme_minimal() +
  labs(title= "Number of Excitatory and Inhibitory Neurons", x = "Neuron Type", y = "Number of Neurons") +
  theme(plot.title = element_text(hjust = 0.5))
```

#Pearson Correlation Heatmap

```
spks_t = as.data.frame(t(spks))
cor_matrix = cor(spks_t, use = "pairwise.complete.obs")
diag(cor_matrix) = NA

library(pheatmap)
pheatmap(cor_matrix,
  color = colorRampPalette(c("blue", "white", "red"))(50),
  main = "Pearson Correlation Heatmap",
  treeheight_row = 0,
  treeheight_col = 0,
  show_rownames = FALSE,
  show_colnames = FALSE
)
```

#Spatial Position Plot

```
colnames(cell_pos) = c("x", "y")
color_pos = cbind(cell_pos, redlabels)
ggplot(color_pos, aes(x = x, y = y, color = as.factor(neuron_type))) +
  geom_point() +
  theme_minimal() +
  scale_color_manual(values = c("blue", "red")) +
  labs(title="Spatial Position of Neurons", color = "Neuron Type") +
  theme(plot.title = element_text(hjust = 0.5))
```

```

#Distribution of Reach Times
colnames(calcium_reach_times) = c("start", "end")
calcium_reach_times$reach_time = calcium_reach_times$end - calcium_reach_times$start
ggplot(calcium_reach_times, aes(x=reach_time)) +
  geom_histogram(binwidth=20) +
  theme_minimal() +
  labs(title= "Distribution of Reach Times", x = "Reach Time", y = "Count") +
  theme(plot.title = element_text(hjust = 0.5))

```

R code for Frequentist Models Section:

```

#Histogram of R^2 Values
load("r_squared_list.RData")
r_squared_vector <- unlist(r_squared_list)

hist(r_squared_vector,
     main = "Histogram of R-squared values",
     xlab = "R-squared",
     col = "skyblue",
     border = "black")

#Create 4 Frequentist Models without Interaction Terms
spks_t = as.data.frame(t(spks))
neurons = paste("Neuron", 1:315, sep = "_")
colnames(spks_t) = neurons

response_neurons = c("Neuron_234", "Neuron_168", "Neuron_281")
predictor_neurons = setdiff(neurons, response_neurons)
load("high_vif.RData")

X = as.matrix(spks_t[, predictor_neurons])

X_filter = X[, !(colnames(X) %in% high_vif)]

Y1 = spks_t[, "Neuron_234"]
model1 = lm(Y1 ~ X_filter)

Y2 = spks_t[, "Neuron_168"]
model2 = lm(Y2 ~ X_filter)

Y3 = spks_t[, "Neuron_281"]
model3 = lm(Y3 ~ X_filter)

Y = c(Y1, Y2, Y3)

```



```

X3 = rbind(X_filter, X_filter, X_filter)
model4 = lm(Y ~ X3)

#Residual Plots for 4 Frequentist Models without Interaction Terms
par(mfrow=c(2, 2))

spks_t = spks_t %>%
  mutate(reach = as.integer(row_number() %in% unlist(apply(calcium_reach_times, 1, func

colors = ifelse(spks_t$reach == 1, rgb(1, 0, 0, alpha = 0.75), rgb(0, 0, 1, alpha = 0.25)

residuals1 = residuals(model1)
residuals2 = residuals(model2)
residuals3 = residuals(model3)
residuals4 = residuals(model4)

fitted1 = fitted(model1)
fitted2 = fitted(model2)
fitted3 = fitted(model3)
fitted4 = fitted(model4)

plot(fitted1, residuals1, col = colors,
     xlab = "Fitted Values (Neuron 234)", ylab = "Residuals (Neuron 234)",
     main = "Residuals Plot for Neuron 234")
abline(h = 0, col = "black", lwd = 2)

plot(fitted2, residuals2, col = colors,
     xlab = "Fitted Values (Neuron 168)", ylab = "Residuals (Neuron 168)",
     main = "Residuals Plot for Neuron 168")
abline(h = 0, col = "black", lwd = 2)

plot(fitted3, residuals3, col = colors,
     xlab = "Fitted Values (Neuron 281)", ylab = "Residuals (Neuron 281)",
     main = "Residuals Plot for Neuron 281")
abline(h = 0, col = "black", lwd = 2)

plot(fitted4, residuals4, col = colors,
     xlab = "Fitted Values (Concatenated Model)", ylab = "Residuals (Concatenated Model)",
     main = "Residuals Plot for Concatenated Model")
abline(h = 0, col = "black", lwd = 2)

par(mfrow=c(1,1))

```

#Histograms showing difference between coefficients for 4 Models without Interaction Terms

```
beta = coef(model4)
beta1 = coef(model1)
beta2 = coef(model2)
beta3 = coef(model3)
beta1dif = beta - beta1
beta2dif = beta - beta2
beta3dif = beta - beta3
par(mfrow=c(3,1))
hist(beta1dif,
      main = "Difference between Neuron 234 and Concatenated Model",
      xlab = "Difference in Coefficient",
      col = "skyblue",
      border = "black")

hist(beta2dif,
      main = "Difference between Neuron 168 and Concatenated Model",
      xlab = "Difference in Coefficient",
      col = "skyblue",
      border = "black")

hist(beta3dif,
      main = "Difference between Neuron 281 and Concatenated Model",
      xlab = "Difference in Coefficient",
      col = "skyblue",
      border = "black")
par(mfrow=c(1,1))
```

#Create 4 Frequentist Models with Interaction Terms

```
X_int = model.matrix(~ . * reach, data = spks_t[, c(predictor_neurons, "reach")])
X_int = X_int[, -1]
Y1_int = spks_t$Neuron_234

X_int_filter = X_int[, !(colnames(X_int) %in% high_vif)]

model1_int = lm(Y1_int ~ X_int_filter)

Y2_int = spks_t$Neuron_168

model2_int = lm(Y2_int ~ X_int_filter)

Y3_int = spks_t$Neuron_281

model3_int = lm(Y3_int ~ X_int_filter)
```

```
Y_combined = c(spks_t$Neuron_234, spks_t$Neuron_168, spks_t$Neuron_281)
```

```
X_combined = rbind(X_int_filter, X_int_filter, X_int_filter)
```

```
model4_int = lm(Y_combined ~ X_combined)
```

```
#Residual Plots for 4 Frequentist Models without Interaction Terms
```

```
par(mfrow=c(2, 2))
```

```
colors = ifelse(spks_t$reach == 1, rgb(1, 0, 0, alpha = 0.75), rgb(0, 0, 1, alpha = 0.25))
```

```
residuals1_int = residuals(model1_int)
```

```
residuals2_int = residuals(model2_int)
```

```
residuals3_int = residuals(model3_int)
```

```
residuals4_int = residuals(model4_int)
```

```
fitted1_int = fitted(model1_int)
```

```
fitted2_int = fitted(model2_int)
```

```
fitted3_int = fitted(model3_int)
```

```
fitted4_int = fitted(model4_int)
```

```
plot(fitted1_int, residuals1_int, col = colors,  
     xlab = "Fitted Values (Neuron 234)", ylab = "Residuals (Neuron 234)",  
     main = "Residuals Plot for Neuron 234 with Interaction Term")  
abline(h = 0, col = "black", lwd = 2)
```

```
plot(fitted2_int, residuals2_int, col = colors,  
     xlab = "Fitted Values (Neuron 168)", ylab = "Residuals (Neuron 168 w/ Interaction T",  
     main = "Residuals Plot for Neuron 168 with Interaction Term")  
abline(h = 0, col = "black", lwd = 2)
```

```
plot(fitted3_int, residuals3_int, col = colors,  
     xlab = "Fitted Values (Neuron 281)", ylab = "Residuals (Neuron 281)",  
     main = "Residuals Plot for Neuron 281 with Interaction Term")  
abline(h = 0, col = "black", lwd = 2)
```

```
plot(fitted4_int, residuals4_int, col = colors,  
     xlab = "Fitted Values (Concatenated Model)", ylab = "Residuals (Concatenated Model)",  
     main = "Residuals Plot for Concatenated Model with Interaction Term")  
abline(h = 0, col = "black", lwd = 2)
```

```
par(mfrow=c(1,1))
```

```

#Histograms showing difference between coefficients for 4 Models without Interaction Terms
beta_int = coef(model4_int)
beta1_int = coef(model1_int)
beta2_int = coef(model2_int)
beta3_int = coef(model3_int)
beta1dif_int = beta_int - beta1_int
beta2dif_int = beta_int - beta2_int
beta3dif_int = beta_int - beta3_int
par(mfrow=c(3,1))
hist(beta1dif_int,
      main = "Difference between Neuron 234 and Concatenated Model with Interaction Terms",
      xlab = "Difference in Coefficient",
      col = "skyblue",
      border = "black")

hist(beta2dif_int,
      main = "Difference between Neuron 168 and Concatenated Model with Interaction Terms",
      xlab = "Difference in Coefficient",
      col = "skyblue",
      border = "black")

hist(beta3dif_int,
      main = "Difference between Neuron 281 and Concatenated Model with Interaction Terms",
      xlab = "Difference in Coefficient",
      col = "skyblue",
      border = "black")
par(mfrow=c(1,1))

```

```

#Create Linear Mixed Effects Model
spks_t = as.data.frame(t(spks))
neurons = paste("Neuron", 1:315, sep = "_")
colnames(spks_t) = neurons

response_neurons = c("Neuron_234", "Neuron_168", "Neuron_281")
predictor_neurons = setdiff(neurons, response_neurons)
load("high_vif.RData")

spks_t = spks_t %>%
  mutate(reach = as.integer(row_number()) %in% unlist(apply(calcium_reach_times, 1, func
spks_t = as.data.frame(scale(spks_t))

neuron_labels <- rep(c("Neuron_234", "Neuron_168", "Neuron_281"), each = nrow(spks_t))

X_int = model.matrix(~ . * reach, data = spks_t[, c(predictor_neurons, "reach")])

```

```

X_int = X_int[, -1]
Y1_int = spks_t$Neuron_234

X_int_filter = X_int[, !(colnames(X_int) %in% high_vif)]
Y_combined = c(spks_t$Neuron_234, spks_t$Neuron_168, spks_t$Neuron_281)

X_combined = rbind(X_int_filter, X_int_filter, X_int_filter)

X1_df = as.data.frame(X_combined)
X1_df$Y = Y_combined
X1_df$neuron = neuron_labels
X1_df$reach = c(spks_t$reach, spks_t$reach, spks_t$reach)

random_formula <- "(1 + reach | neuron)"

fixed_formula <- "Y ~ . - neuron +"

full_formula <- as.formula(paste(fixed_formula, random_formula))

modellme <- lmer(full_formula, data = X1_df)

```

```
#Residual Plots for Linear Mixed Effects Model
```

```

res = residuals(modellme)
fit = fitted(modellme)

colors = ifelse(X1_df$reach == 1, rgb(1, 0, 0, alpha = 0.75), rgb(0, 0, 1, alpha = 0.25))

plot(fit, res, col = colors,
     xlab = "Fitted Values (Linear Mixed Effects)", ylab = "Residuals (Linear Mixed Effects)",
     main = "Residuals Plot for Linear Mixed Effects")
abline(h = 0, col = "black", lwd = 2)

```

```
#Five-fold CV for MSE calculation
```

```

library(caret)
library(lme4)
set.seed(1)

spks_t = as.data.frame(t(spks))
neurons = paste("Neuron", 1:315, sep = "_")
colnames(spks_t) = neurons

response_neurons = c("Neuron_234", "Neuron_168", "Neuron_281")
predictor_neurons = setdiff(neurons, response_neurons)
load("high_vif.RData")

```

```

high_vif <- setdiff(high_vif, "reach")

spks_t = spks_t %>%
  mutate(reach = as.integer(row_number() %in% unlist(apply(calcium_reach_times, 1, func
spks_t = as.data.frame(spks_t)

X_int = model.matrix(~ . * reach, data = spks_t[, c(predictor_neurons, "reach")])
X_int = X_int[, -1]

k = 5
folds = createFolds(spks_t$Neuron_234, k = k, list = TRUE)

compute_mse = function(model, test_data, test_Y) {
  preds = predict(model, newdata = test_data, allow.new.levels = TRUE)
  mean((test_Y - preds)^2)
}

mse_results = data.frame(Model = character(), Fold = integer(), MSE = numeric())

for (i in 1:k) {
  test_idx = folds[[i]]
  train_idx = setdiff(1:nrow(spks_t), test_idx)

  train_data = as.data.frame(X_int[train_idx, ])
  test_data = as.data.frame(X_int[test_idx, ])

  Y1 = spks_t$Neuron_234[train_idx]
  Y2 = spks_t$Neuron_168[train_idx]
  Y3 = spks_t$Neuron_281[train_idx]
  Y4 = c(Y1, Y2, Y3)

  X_noint = train_data[, predictor_neurons]
  X_wint = model.matrix(~ . * reach, data = train_data[, c(predictor_neurons, "reach")])
  X_wint <- X_wint[, !(colnames(X_wint) == "(Intercept)")]

  X_comb = rbind(X_wint, X_wint, X_wint)
  X_combnoint = rbind(X_noint, X_noint, X_noint)

  X_noint_df <- as.data.frame(X_noint)
  X_wint_df <- as.data.frame(X_wint)
  X_combnoint_df <- as.data.frame(X_combnoint)
  X_comb_df <- as.data.frame(X_comb)

  colnames(X_noint_df) <- predictor_neurons

```

```

colnames(X_wint_df) <- colnames(X_wint)
colnames(X_combnoint_df) <- predictor_neurons
colnames(X_comb_df) <- colnames(X_wint)

X_noint_df <- X_noint_df[, !(colnames(X_noint_df) %in% high_vif)]
X_combnoint_df <- X_combnoint_df[, !(colnames(X_combnoint_df) %in% high_vif)]
X_wint_df <- X_wint_df[, !(colnames(X_wint_df) %in% high_vif)]
X_comb_df <- X_comb_df[, !(colnames(X_wint_df) %in% high_vif)]

train_data1 <- cbind(Y1 = Y1, X_noint_df)
train_data2 <- cbind(Y2 = Y2, X_noint_df)
train_data3 <- cbind(Y3 = Y3, X_noint_df)
train_data4 <- cbind(Y4 = Y4, X_combnoint_df)

train_data1_int <- cbind(Y1 = Y1, X_wint_df)
train_data2_int <- cbind(Y2 = Y2, X_wint_df)
train_data3_int <- cbind(Y3 = Y3, X_wint_df)
train_data4_int <- cbind(Y4 = Y4, X_comb_df)

msemodel1 <- lm(Y1 ~ ., data = train_data1)
msemodel2 <- lm(Y2 ~ ., data = train_data2)
msemodel3 <- lm(Y3 ~ ., data = train_data3)
msemodel4 <- lm(Y4 ~ ., data = train_data4)

msemodel1_int <- lm(Y1 ~ ., data = train_data1_int)
msemodel2_int <- lm(Y2 ~ ., data = train_data2_int)
msemodel3_int <- lm(Y3 ~ ., data = train_data3_int)
msemodel4_int <- lm(Y4 ~ ., data = train_data4_int)

neuron_labels <- rep(c("Neuron_234", "Neuron_168", "Neuron_281"), each = nrow(X_comb_df))
X_comb_df$neuron = neuron_labels
X_comb_df$neuron = as.factor(X_comb_df$neuron)

msemodel1me = lmer(Y4 ~ . - neuron + (1 + reach | neuron), data = X_comb_df)

test_Y1 = spks_t$Neuron_234[test_idx]
test_Y2 = spks_t$Neuron_168[test_idx]
test_Y3 = spks_t$Neuron_281[test_idx]
test_Y4 = c(test_Y1, test_Y2, test_Y3)

mse1kf <- compute_mse(msemodel1, test_data, test_Y1)
mse2kf <- compute_mse(msemodel2, test_data, test_Y2)
mse3kf <- compute_mse(msemodel3, test_data, test_Y3)
mse4kf <- compute_mse(msemodel4, test_data, test_Y4)

```

```

mse5kf <- compute_mse(msemodel1_int, test_data, test_Y1)
mse6kf <- compute_mse(msemodel2_int, test_data, test_Y2)
mse7kf <- compute_mse(msemodel3_int, test_data, test_Y3)
mse8kf <- compute_mse(msemodel4_int, test_data, test_Y4)

neuron_labels <- rep(c("Neuron_234", "Neuron_168", "Neuron_281"), length.out = nrow(test_data))
test_data$neuron = neuron_labels
test_data$neuron = as.factor(test_data$neuron)
test_data <- test_data[, colnames(X_comb_df), drop = FALSE]

mse9kf = compute_mse(msemodel1me, test_data, test_Y4)

mse_results = rbind(mse_results, data.frame(
  Model = rep(c("LM1", "LM2", "LM3", "LM4", "LM5", "LM6", "LM7", "LM8", "LM9"), each = 5),
  Fold = i,
  MSE = c(mse1kf, mse2kf, mse3kf, mse4kf, mse5kf, mse6kf, mse7kf, mse8kf, mse9kf)
))
}

summary_mse <- mse_results %>%
  group_by(Model) %>%
  summarize(Mean_MSE = mean(MSE), SD_MSE = sd(MSE))

print(summary_mse)

```

#MSE Violin Plots for Five-fold Cross Validation

```

set.seed(123)

mse_results <- mse_results %>%
  mutate(Model = recode(Model,
    "LM1" = "Neuron 234",
    "LM2" = "Neuron 168",
    "LM3" = "Neuron 281",
    "LM4" = "Concatenated",
    "LM5" = "Neuron 234 w/ Int.",
    "LM6" = "Neuron 168 w/ Int.",
    "LM7" = "Neuron 281 w/ Int.",
    "LM8" = "Concatenated w/ Int.",
    "LM9" = "Linear Mixed Effects"
  ))
ggplot(mse_results, aes(x = Model, y = MSE)) +
  geom_violin(fill = "skyblue", alpha = 0.6) +
  theme_minimal() +

```



```
labs(title = "MSE Distribution for Frequentist Models",
     x = "Model",
     y = "Mean Squared Error") +
theme(axis.text.x = element_text(angle = 45, hjust = 1))
```

R code for Bayesian Section:

```
#Implementation of Bayesian models with 2000 data points
load("high_vif.RData")
spks_t = as.data.frame(t(spks))
neurons = paste("Neuron", 1:315, sep = "_")
colnames(spks_t) = neurons

response_neurons = c("Neuron_234", "Neuron_168", "Neuron_281")
predictor_neurons = setdiff(neurons, response_neurons)

spks_t = spks_t %>%
  mutate(reach = as.integer(row_number() %in% unlist(apply(calcium_reach_times, 1, func

X_int = model.matrix(~ . * reach, data = spks_t[, c(predictor_neurons, "reach")])
X_int = X_int[, -1]
X_int_filter = X_int[, !(colnames(X_int) %in% high_vif)]
X_combined = rbind(X_int_filter, X_int_filter, X_int_filter)

neuron_labels <- rep(c("Neuron_234", "Neuron_168", "Neuron_281"), each = nrow(spks_t))

X1_df = as.data.frame(X_combined)
Y_combined = c(spks_t$Neuron_234, spks_t$Neuron_168, spks_t$Neuron_281)
reach_combined = c(spks_t$reach, spks_t$reach, spks_t$reach)
X1_df$reach = reach_combined
X1_df$Y = Y_combined
X1_df$neuron = neuron_labels

sample_size <- 667

X1_sampled <- X1_df %>%
  group_by(neuron) %>%
  sample_n(sample_size, replace = FALSE) %>%
  ungroup()

# mixed effect on intercept, default prior
fixed_effects <- setdiff(names(X1_df), c("Y", "neuron"))
formula_str <- paste("Y ~", paste(fixed_effects, collapse = " + "), "+ (1 | neuron)")
```

```
brmodel1 <- brm(formula = formula_str,
  data = X1_sampled,
  warmup = 1000, iter = 3000,
  cores = 4, chains = 2, seed = 123)

fixed_effects <- setdiff(names(X1_df), c("Y", "neuron"))
formula_str <- paste("Y ~", paste(fixed_effects, collapse = " + "), "+ (1 + reach | neuron)")

# mixed effect on reach and intercept, default prior
brmodel2 <- brm(formula = formula_str,
  data = X1_sampled,
  warmup = 1000, iter = 3000,
  cores = 4, chains = 2, seed = 123)
```

#Leave-One-Out CV for Bayesian Models

```
loo1 = loo(brmodel1)
loo2 = loo(brmodel2)

loo_compare(loo1, loo2)
```

#Sensitivity Analysis

```
priors <- c(
  set_prior("normal(0, 1)", class = "b"),
  set_prior("student_t(3, 0, 1)", class = "sd"),
  set_prior("student_t(3, 0, 1.5)", class = "sigma")
)

fixed_effects <- setdiff(names(X1_df), c("Y", "neuron"))
formula_str <- paste("Y ~", paste(fixed_effects, collapse = " + "), "+ (1 + reach | neuron)")

brmodel_weak <- brm(formula = formula_str,
  data = X1_sampled,
  prior = priors,
  warmup = 1000, iter = 3000,
  cores = 4, chains = 2, seed = 123)
```

#Leave-One-Out CV for Sensitivity Analysis

```
loo_weak = loo(brmodel_weak)

loo_compare(loo2, loo_weak)
```

#Trace Plots

```
posterior_samples <- as.array(brmodel2)
```

```

chosen_predictors <- c("b_Intercept", "b_Neuron_3", "b_Neuron_4", "b_Neuron_5",
                      "b_Neuron_6", "b_reach", "b_Neuron_3:reach", "b_Neuron_4:reach",

trace_plots <- mcmc_trace(posterior_samples, pars = chosen_predictors) +
  ggtitle("Traceplots for Selected Predictors")

trace_plots <- mcmc_trace(posterior_samples, pars = chosen_predictors, facet_args = list

trace_plots

```

#ACF Plots

```

acf_plots <- mcmc_acf(posterior_samples, pars = chosen_predictors) +
  ggtitle("Autocorrelation Plots for Selected Predictors")

acf_plots

```

#Rhat Histogram

```

fitsummary = summary(brmodel2)
fixed = fitsummary$fixed
random = fitsummary$random$neuron
rhats = c(fixed$Rhat, random$Rhat)
params = c(rownames(fixed), rownames(random))

rhat_df = data.frame(Parameter = params, Rhat = rhats)

ggplot(rhat_df, aes(x = Rhat)) +
  geom_histogram(binwidth = 0.01, fill = "blue", alpha = 0.7, color = "black") +
  geom_vline(xintercept=1.05, linetype="dashed", color="red") +
  labs(title="Histogram of R-hat Values", x="R-hat", y="Count") +
  theme_minimal()

```

#ESS Scatterplot

```

bulk_ess = c(fixed$Bulk_ESS, random$Bulk_ESS)
tail_ess = c(fixed$Tail_ESS, random$Tail_ESS)

ess_df = data.frame(Parameter = params, Bulk_ESS = bulk_ess, Tail_ESS = tail_ess)

ess_long <- ess_df %>%
  pivot_longer(cols = c("Bulk_ESS", "Tail_ESS"), names_to = "ESS_Type", values_to = "ESS

ess_long %>%
  ggplot(aes(x = reorder(Parameter, ESS_Value), y = ESS_Value, color = ESS_Type)) +

```

```
geom_point() +
labs(title = "ESS Values by Parameter", x = "Parameter", y = "ESS") +
geom_hline(yintercept=1000, linetype="dashed", color="black") +
geom_hline(yintercept=400, linetype="dashed", color="red") +
theme_minimal() +
theme(axis.text.x = element_blank())
```

```
library(dplyr)
#MSE calculation on Testing Data for Bayesian Model and Best Frequentist Model

load("sample_spks.RData")
spks_sample_t = as.data.frame(t(sample_spks))
neurons = paste("Neuron", 1:315, sep = "_")
colnames(spks_sample_t) = neurons

response_neurons = c("Neuron_234", "Neuron_168", "Neuron_281")
predictor_neurons = setdiff(neurons, response_neurons)
load("high_vif.RData")

spks_sample_t = spks_sample_t %>%
  mutate(reach = as.integer(row_number()) %in% unlist(apply(calcium_reach_times, 1, func

X_int_sample = model.matrix(~ . * reach, data = spks_sample_t[, c(predictor_neurons, "re
X_int_sample = X_int_sample[, -1]
X_int_sample_filter = X_int_sample[, !(colnames(X_int_sample) %in% high_vif)]
X_sample_combined = rbind(X_int_sample_filter, X_int_sample_filter, X_int_sample_filter)

neuron_labels <- rep(c("Neuron_234", "Neuron_168", "Neuron_281"), each = nrow(spks_sampl

X1_sample_df = as.data.frame(X_sample_combined)
Y_sample_combined = c(spks_sample_t$Neuron_234, spks_sample_t$Neuron_168, spks_sample_t$
X1_sample_df$Y = Y_sample_combined
X1_sample_df$reach = c(spks_sample_t$reach, spks_sample_t$reach, spks_sample_t$reach)
X1_sample_df = as.data.frame(scale(X1_sample_df))
X1_sample_df$neuron = neuron_labels

X1_sample_df <- X1_sample_df %>%
  group_by(neuron) %>%
  mutate(row_id = row_number())

X1_sample_df <- X1_sample_df %>%
  mutate(fold = (row_id - 1) %% 5 + 1) %>%
  ungroup()
```

```

test_splits <- split(X1_sample_df, X1_sample_df$fold)

test_splits <- lapply(test_splits, function(df) df %>% select(-fold, -row_id))

compute_mse = function(model, test_data, test_Y) {
  preds = predict(model, newdata = test_data, allow.new.levels = TRUE)
  mean((test_Y - preds)^2)
}

brms_mse_results <- lapply(test_splits, function(test_df) {
  test_Y <- test_df$Y
  test_data <- test_df %>% select(-Y)

  compute_mse(brmodel2, test_data, test_Y)
})

mse_model1 <- sapply(test_splits, function(test_df) {
  test_subset <- test_df[1:5000, ]
  test_Y <- test_subset$Y
  test_data <- test_subset %>% select(-Y, -neuron)

  compute_mse(model1_int, test_data, test_Y)
})

mse_model2 <- sapply(test_splits, function(test_df) {
  test_subset <- test_df[5001:10000, ]
  test_Y <- test_subset$Y
  test_data <- test_subset %>% select(-Y, -neuron)

  compute_mse(model2_int, test_data, test_Y)
})

mse_model3 <- sapply(test_splits, function(test_df) {
  test_subset <- test_df[10001:15000, ]
  test_Y <- test_subset$Y
  test_data <- test_subset %>% select(-Y, -neuron)

  compute_mse(model3_int, test_data, test_Y)
})

brms_mse_results <- unlist(brms_mse_results)

full_mse_results <- data.frame(Fold = 1:5, Neuron234 = mse_model1, Neuron168 = mse_model

```

```
#Posterior Predictive Checking
```

```
posterior_predictive_samples <- posterior_predict(brmodel2)
```

```
observed_data <- X1_sampled$Y
```

```
ggplot() +
```

```
  geom_density(aes(x = posterior_predictive_samples[1, ]), fill = "blue", alpha = 0.5) +
```

```
  geom_density(aes(x = observed_data), color = "red") +
```

```
  labs(title = "Posterior Predictive Checking", x = "y", y = "Density") +
```

```
  theme_minimal()
```

```
#Find Non-Zero Coefs for Neural Network Plot
```

```
library(broom)
```

```
library(dplyr)
```

```
library(broom.mixed)
```

```
freq_coefs1 <- tidy(model1_int, conf.int = TRUE) %>%
```

```
  select(term, estimate, conf.low, conf.high) %>%
```

```
  mutate(model = "Frequentist Neuron 234")
```

```
freq_coefs2 <- tidy(model2_int, conf.int = TRUE) %>%
```

```
  select(term, estimate, conf.low, conf.high) %>%
```

```
  mutate(model = "Frequentist Neuron 168")
```

```
freq_coefs3 <- tidy(model3_int, conf.int = TRUE) %>%
```

```
  select(term, estimate, conf.low, conf.high) %>%
```

```
  mutate(model = "Frequentist Neuron 281")
```

```
freq_coefs <- bind_rows(freq_coefs1, freq_coefs2, freq_coefs3)
```

```
coef_no_zero_ci <- freq_coefs %>%
```

```
  filter((conf.low > 0 | conf.high < 0) & !grepl(":reach", term))
```

```
neuron_234_connections = coef_no_zero_ci %>% filter(model == "Frequentist Neuron 234")
```

```
neuron_168_connections = coef_no_zero_ci %>% filter(model == "Frequentist Neuron 168")
```

```
neuron_281_connections = coef_no_zero_ci %>% filter(model == "Frequentist Neuron 281")
```

```
#Create Neural Network Plot for Neuron 234
```

```
neuron_234_filtered <- neuron_234_connections %>%
```

```
  slice(-1)
```

```
neuron_numbers234 <- gsub("X_int_filterNeuron_", "", neuron_234_filtered$term)
```

```

neuron_numbers234 <- as.integer(neuron_numbers234)
neuron_numbers234 = c(neuron_numbers234, 234)

color_pos_234 <- color_pos[neuron_numbers234, ]

neuron_234_coords <- color_pos_234[color_pos_234$neuron_type == "Neuron 234", c("x", "y")]

color_pos_234$distance_to_neuron_234 <- sqrt((color_pos_234$x - neuron_234_coords$x)^2 + (color_pos_234$y - neuron_234_coords$y)^2)

color_pos_234$line_thickness <- abs(c(neuron_234_filtered$estimate, 0))

top_10_neurons234 <- color_pos_234 %>%
  filter(line_thickness >= 0.0005)

network234 = ggplot(color_pos_234, aes(x = x, y = y, color = as.factor(neuron_type))) +
  geom_segment(data = top_10_neurons234[color_pos_234$neuron_type != "Neuron 234", ],
    aes(xend = neuron_234_coords$x, yend = neuron_234_coords$y,
      size = line_thickness),
    color = "black") +
  geom_point() +
  theme_minimal() +
  scale_color_manual(values = c("blue", "red", "green", "green", "green")) +
  labs(title = "Spatial Position of Neurons connected to Neuron 234", color = "Neuron Type") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_size_continuous(name = "Coefficient Magnitude", range = c(0.01, 1))

#Create Neural Network Plot for Neuron 168
neuron_168_filtered <- neuron_168_connections %>%
  slice(-1)

neuron_numbers168 <- gsub("X_int_filterNeuron_", "", neuron_168_filtered$term)

neuron_numbers168 <- as.integer(neuron_numbers168)
neuron_numbers168 = c(neuron_numbers168, 168)

color_pos_168 <- color_pos[neuron_numbers168, ]

neuron_168_coords <- color_pos_168[color_pos_168$neuron_type == "Neuron 168", c("x", "y")]

color_pos_168$distance_to_neuron_168 <- sqrt((color_pos_168$x - neuron_168_coords$x)^2 + (color_pos_168$y - neuron_168_coords$y)^2)

color_pos_168$line_thickness <- abs(c(neuron_168_filtered$estimate, 0))

top_10_neurons168 <- color_pos_168 %>%

```

```

filter(line_thickness >= 0.0005)

network168 = ggplot(color_pos_168, aes(x = x, y = y, color = as.factor(neuron_type))) +
  geom_segment(data = top_10_neurons168[color_pos_234$neuron_type != "Neuron 234", ],
    aes(xend = neuron_168_coords$x, yend = neuron_168_coords$y,
      size = line_thickness),
    color = "black") +
  geom_point() +
  theme_minimal() +
  scale_color_manual(values = c("blue", "red", "green", "green", "green")) +
  labs(title = "Spatial Position of Neurons connected to Neuron 168", color = "Neuron Type") +
  theme(plot.title = element_text(hjust = 0.5)) +
  scale_size_continuous(name = "Coefficient Magnitude", range = c(0.01, 1))

#Create Neural Network Plot for Neuron 281
neuron_281_filtered <- neuron_281_connections %>%
  slice(-1)

neuron_numbers281 <- gsub("X_int_filterNeuron_", "", neuron_281_filtered$term)

neuron_numbers281 <- as.integer(neuron_numbers281)
neuron_numbers281 = c(neuron_numbers281, 281)

color_pos_281 <- color_pos[neuron_numbers281, ]

neuron_281_coords <- color_pos_281[color_pos_281$neuron_type == "Neuron 281", c("x", "y")]

color_pos_281$distance_to_neuron_281 <- sqrt((color_pos_281$x - neuron_281_coords$x)^2 +
  (color_pos_281$y - neuron_281_coords$y)^2)

color_pos_281$line_thickness <- abs(c(neuron_281_filtered$estimate, 0))

top_10_neurons281 <- color_pos_281 %>%
  filter(line_thickness >= 0.0005)

network281 = ggplot(color_pos_281, aes(x = x, y = y, color = as.factor(neuron_type))) +
  geom_segment(data = top_10_neurons281[color_pos_281$neuron_type != "Neuron 281", ],
    aes(xend = neuron_281_coords$x, yend = neuron_281_coords$y,
      size = line_thickness),
    color = "black") +
  geom_point() +
  theme_minimal() +
  scale_color_manual(values = c("blue", "red", "green", "green", "green")) +
  labs(title = "Spatial Position of Neurons connected to Neuron 281", color = "Neuron Type") +
  theme(plot.title = element_text(hjust = 0.5)) +

```



```
scale_size_continuous(name = "Coefficient Magnitude", range = c(0.01, 1))
```

```
#Display all Neural Network Plots
```

```
library(gridExtra)
```

```
grid.arrange(network234, network168, network281, ncol = 2)
```

R code for Graphs used in Presentation:

```
#Presentation Graph
```

```
library(broom)
```

```
library(dplyr)
```

```
library(broom.mixed)
```

```
models <- list(modellme, model4_int)
```

```
model_names <- c("Linear Mixed Effects Model", "Concatenated Model")
```

```
coef_df <- bind_rows(lapply(seq_along(models), function(i) {
  tidy(models[[i]], conf.int = TRUE) %>%
    mutate(model = model_names[i],
           term = gsub("^(X3|X_combined|^)", "", term))
}))
```

```
ggplot(coef_df, aes(x = estimate, y = term)) +
  geom_point(aes(color = model)) +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high, color = model), width = 0.2) +
  facet_wrap(~model, scales = "fixed") +
  ggtitle("Coefficient Comparison by Model") +
  theme_minimal() +
  theme(axis.text.y = element_blank())
```

```
#Presentation Graph
```

```
library(ggplot2)
```

```
library(broom)
```

```
library(dplyr)
```

```
models <- list(model1_int, model2_int, model3_int)
```

```
model_names <- c("Neuron 234", "Neuron 168", "Neuron 281")
```

```
coef_df <- bind_rows(lapply(seq_along(models), function(i) {
  tidy(models[[i]], conf.int = TRUE) %>%
    mutate(model = model_names[i],
           term = gsub("^(X_combined|X_int_filter)", "", term))
}))
```

```

model_colors <- c("Neuron 234" = "#00BFC4",
                  "Neuron 168" = "#7CAE00",
                  "Neuron 281" = "#C77CFF")

ggplot(coef_df, aes(x = estimate, y = term, color = model)) +
  geom_point(position = position_dodge(width = 0.5)) +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high),
                position = position_dodge(width = 0.5), width = 0.2) +
  ggtitle("Coefficient Estimates with 95% CIs with Interaction Terms") +
  theme_minimal() +
  scale_color_manual(values = model_colors) +
  theme(axis.text.y = element_blank(),
        legend.position = "none")

```

#Presentation Graph

```

library(broom)
library(ggplot2)
library(dplyr)

models <- list(model1, model2, model3, model4)
model_names <- c("Neuron 234", "Neuron 168", "Neuron 281", "Concatenated Model")

coef_df <- bind_rows(lapply(seq_along(models), function(i) {
  tidy(models[[i]], conf.int = TRUE) %>%
    mutate(model = model_names[i],
           term = gsub("(X3|X_filter)", "", term))
}))

ggplot(coef_df, aes(x = estimate, y = term, color = model)) +
  geom_point(position = position_dodge(width = 0.5)) +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high),
                position = position_dodge(width = 0.5), width = 0.2) +
  ggtitle("Coefficient Estimates with 95% CIs") +
  theme_minimal() +
  theme(axis.text.y = element_blank())

```

#Presentation Graph

```

library(broom)
library(dplyr)
library(broom.mixed)

freq_coefs1 <- tidy(model1_int, conf.int = TRUE) %>%
  select(term, estimate, conf.low, conf.high) %>%

```

```

mutate(model = "Frequentist Neuron 234")

freq_coefs2 <- tidy(model2_int, conf.int = TRUE) %>%
  select(term, estimate, conf.low, conf.high) %>%
  mutate(model = "Frequentist Neuron 168")

freq_coefs3 <- tidy(model3_int, conf.int = TRUE) %>%
  select(term, estimate, conf.low, conf.high) %>%
  mutate(model = "Frequentist Neuron 281")

freq_coefs <- bind_rows(freq_coefs1, freq_coefs2, freq_coefs3)

library(broom.mixed)

bayes_coefs <- tidy(brmodel2, effects = "fixed", conf.int = TRUE) %>%
  select(term, estimate, conf.low, conf.high) %>%
  mutate(model = "Bayesian")

coef_df <- bind_rows(freq_coefs, bayes_coefs)
coef_df$term <- gsub("^X_int_filter", "", coef_df$term)

library(ggplot2)

ggplot(coef_df, aes(x = estimate, y = term, color = model)) +
  geom_point(position = position_dodge(width = 0.4), size = 3) +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high),
               position = position_dodge(width = 0.4), width = 0.2) +
  ggtitle("Comparison of 95% CIs for Coefficients: Bayesian vs. Frequentist") +
  theme_minimal() +
  theme(axis.text.y = element_blank())

ggplot(coef_df, aes(x = estimate, y = term)) +
  geom_point(aes(color = model)) +
  geom_errorbar(aes(xmin = conf.low, xmax = conf.high, color = model), width = 0.2) +
  facet_wrap(~model, scales = "fixed") +
  ggtitle("Coefficient Comparison by Model") +
  theme_minimal() +
  theme(axis.text.y = element_blank())

```

Tables of Coefficients, Rhat, and ESS:

##	Neuron	model1
## 1	(Intercept)	(0.001982, 0.004297)
## 2	Neuron_3	(-0.000048, -0.000012)

## 3	Neuron_4	(-0.000039, 0.000003)
## 4	Neuron_5	(-0.000064, 0.000001)
## 5	Neuron_6	(-0.000002, 0.000025)
## 6	Neuron_7	(-0.000057, -0.000027)
## 7	Neuron_8	(0.000001, 0.000035)
## 8	Neuron_9	(-0.000007, 0.000067)
## 9	Neuron_11	(-0.000092, -0.000037)
## 10	Neuron_12	(-0.000034, -0.000014)
## 11	Neuron_13	(-0.000088, -0.000027)
## 12	Neuron_14	(0.000063, 0.000124)
## 13	Neuron_15	(-0.002347, 0.00176)
## 14	Neuron_16	(-0.000109, -0.000061)
## 15	Neuron_17	(-0.000033, 0.000004)
## 16	Neuron_18	(-0.000058, 0.000025)
## 17	Neuron_19	(-0.000005, 0.000027)
## 18	Neuron_20	(0.000017, 0.000056)
## 19	Neuron_21	(0.000055, 0.00011)
## 20	Neuron_22	(-0.000041, 0.000029)
## 21	Neuron_23	(-0.000017, 0.000291)
## 22	Neuron_24	(-0.000091, -0.000021)
## 23	Neuron_25	(0.000006, 0.000029)
## 24	Neuron_26	(-0.000054, 0.000009)
## 25	Neuron_27	(0.000063, 0.000129)
## 26	Neuron_28	(-0.00002, 0.000061)
## 27	Neuron_29	(-0.000148, 0.000109)
## 28	Neuron_30	(-0.00003, 0.000007)
## 29	Neuron_31	(0.01448, 0.032104)
## 30	Neuron_32	(-0.000044, -0.000018)
## 31	Neuron_33	(-0.000069, -0.000037)
## 32	Neuron_34	(-0.000039, -0.000008)
## 33	Neuron_35	(-0.000122, -0.000041)
## 34	Neuron_36	(-0.000003, 0.000021)
## 35	Neuron_37	(-0.000056, -0.000012)
## 36	Neuron_38	(-0.000138, -0.00005)
## 37	Neuron_39	(-0.000019, 0.000073)
## 38	Neuron_40	(-0.000014, 0.000029)
## 39	Neuron_41	(-0.00039, 0.000191)
## 40	Neuron_42	(-0.000011, 0.000063)
## 41	Neuron_43	(-0.000002, 0.000024)
## 42	Neuron_44	(0.000491, 0.011264)
## 43	Neuron_45	(-0.000022, 0.000036)
## 44	Neuron_46	(-0.000016, 0.000021)
## 45	Neuron_47	(-0.000229, 0.000395)
## 46	Neuron_48	(-0.001206, -0.000386)
## 47	Neuron_50	(-0.000059, -0.000001)

## 48	Neuron_51	(-0.000032, -0.000013)
## 49	Neuron_52	(-0.000008, 0.000019)
## 50	Neuron_53	(-0.000048, -0.000014)
## 51	Neuron_54	(0.000069, 0.000141)
## 52	Neuron_55	(0.000015, 0.000068)
## 53	Neuron_56	(0.000054, 0.000191)
## 54	Neuron_57	(0.000034, 0.000094)
## 55	Neuron_58	(0.000006, 0.000007)
## 56	Neuron_59	(-0.000019, 0.000007)
## 57	Neuron_60	(0.000006, 0.000028)
## 58	Neuron_61	(-0.000045, -0.000001)
## 59	Neuron_62	(0.000055, 0.000091)
## 60	Neuron_63	(-0.000042, -0.000004)
## 61	Neuron_64	(-0.000009, 0.000014)
## 62	Neuron_65	(-0.000003, 0.000048)
## 63	Neuron_66	(-0.000011, 0.000015)
## 64	Neuron_67	(-0.000015, 0.000001)
## 65	Neuron_68	(-0.000017, 0.000021)
## 66	Neuron_69	(-0.000006, -0.000003)
## 67	Neuron_70	(0.002241, 0.07029)
## 68	Neuron_71	(-0.000029, -0.000004)
## 69	Neuron_72	(-0.000032, -0.000007)
## 70	Neuron_73	(0.000006, 0.000039)
## 71	Neuron_74	(-0.000047, 0.000044)
## 72	Neuron_75	(-0.039023, 0.124377)
## 73	Neuron_76	(-0.00065, 0.000371)
## 74	Neuron_77	(-0.00011, -0.000043)
## 75	Neuron_78	(0.000062, 0.000148)
## 76	Neuron_79	(-0.000023, 0.000022)
## 77	Neuron_80	(-0.000355, -0.000234)
## 78	Neuron_81	(-0.001092, 0.000424)
## 79	Neuron_82	(-0.00001, 0.000024)
## 80	Neuron_83	(-0.000055, 0.000018)
## 81	Neuron_84	(-0.000031, 0.000005)
## 82	Neuron_85	(-0.000047, -0.000022)
## 83	Neuron_86	(-0.000027, 0)
## 84	Neuron_87	(-0.000191, -0.000094)
## 85	Neuron_88	(0.000005, 0.000053)
## 86	Neuron_89	(-0.161514, 0.050489)
## 87	Neuron_90	(-0.000044, -0.000019)
## 88	Neuron_91	(-0.000012, 0.000038)
## 89	Neuron_93	(-0.000022, 0.000001)
## 90	Neuron_94	(0.000033, 0.000073)
## 91	Neuron_95	(-0.000065, 0.000013)
## 92	Neuron_96	(0.000065, 0.000109)

## 93	Neuron_97	(-0.001893, 0.004615)
## 94	Neuron_98	(0.000027, 0.000054)
## 95	Neuron_99	(-0.000347, -0.000214)
## 96	Neuron_100	(-0.000065, -0.000018)
## 97	Neuron_101	(-0.002077, 0.000223)
## 98	Neuron_102	(-0.000047, -0.000014)
## 99	Neuron_103	(0.000029, 0.000065)
## 100	Neuron_104	(-0.000079, 0.000035)
## 101	Neuron_105	(0.000006, 0.000039)
## 102	Neuron_106	(-0.000045, -0.000012)
## 103	Neuron_107	(-0.000031, -0.000011)
## 104	Neuron_108	(-0.000184, -0.000106)
## 105	Neuron_109	(-0.000523, -0.000205)
## 106	Neuron_110	(-0.000155, -0.000072)
## 107	Neuron_111	(-0.000006, 0.000006)
## 108	Neuron_112	(-0.000098, -0.000038)
## 109	Neuron_113	(-0.000036, 0.000059)
## 110	Neuron_114	(-0.000014, 0.000049)
## 111	Neuron_115	(-0.000317, 0.000122)
## 112	Neuron_116	(-0.000011, 0.000001)
## 113	Neuron_117	(0.000014, 0.000034)
## 114	Neuron_118	(-0.000002, 0.000006)
## 115	Neuron_119	(-0.000036, -0.000005)
## 116	Neuron_120	(0.000241, 0.000891)
## 117	Neuron_121	(-0.000003, 0.000027)
## 118	Neuron_122	(-0.000179, -0.000042)
## 119	Neuron_123	(-0.000065, 0.000029)
## 120	Neuron_124	(0.000034, 0.000103)
## 121	Neuron_125	(0.000866, 0.00176)
## 122	Neuron_127	(-0.000017, 0.000017)
## 123	Neuron_128	(0.000031, 0.000077)
## 124	Neuron_129	(-0.000017, 0.000024)
## 125	Neuron_130	(-0.000155, -0.000017)
## 126	Neuron_131	(0.000164, 0.000221)
## 127	Neuron_132	(-0.000059, -0.000026)
## 128	Neuron_133	(-0.000051, -0.000014)
## 129	Neuron_134	(0.000009, 0.000054)
## 130	Neuron_135	(-0.000015, 0.000058)
## 131	Neuron_136	(-0.002048, -0.000702)
## 132	Neuron_137	(-0.000056, -0.000012)
## 133	Neuron_138	(-0.000069, -0.000009)
## 134	Neuron_139	(0.000003, 0.000057)
## 135	Neuron_140	(-0.000002, 0.000003)
## 136	Neuron_141	(0.002353, 0.007121)
## 137	Neuron_142	(0.000022, 0.000079)

## 138	Neuron_143	(-0.000096, -0.000065)
## 139	Neuron_144	(-0.000091, -0.000002)
## 140	Neuron_145	(-0.000068, 0.000653)
## 141	Neuron_146	(-0.000057, -0.000026)
## 142	Neuron_147	(-0.000002, 0.000007)
## 143	Neuron_148	(-0.000028, 0.000014)
## 144	Neuron_149	(0.000291, 0.000369)
## 145	Neuron_150	(-0.000147, -0.000089)
## 146	Neuron_151	(0.000045, 0.000227)
## 147	Neuron_152	(0.000099, 0.00024)
## 148	Neuron_153	(0.000365, 0.000585)
## 149	Neuron_154	(-0.000005, 0.000036)
## 150	Neuron_155	(-0.000032, 0.000005)
## 151	Neuron_156	(0.000002, 0.000028)
## 152	Neuron_157	(-0.000002, 0.000009)
## 153	Neuron_158	(-0.000005, 0.000112)
## 154	Neuron_159	(-0.000046, 0.000008)
## 155	Neuron_160	(0.000001, 0.000043)
## 156	Neuron_161	(0.000042, 0.000071)
## 157	Neuron_162	(0.000004, 0.000038)
## 158	Neuron_163	(0, 0.000053)
## 159	Neuron_164	(-0.010838, 0.021367)
## 160	Neuron_165	(-0.000166, -0.000097)
## 161	Neuron_166	(-0.000485, -0.000071)
## 162	Neuron_167	(-0.000065, 0.000024)
## 163	Neuron_169	(-0.000133, -0.000066)
## 164	Neuron_170	(-0.000008, -0.000024)
## 165	Neuron_171	(0.000245, 0.000329)
## 166	Neuron_172	(-0.000125, -0.000022)
## 167	Neuron_173	(-0.000047, 0.00019)
## 168	Neuron_174	(-0.000016, 0.000022)
## 169	Neuron_175	(-0.000018, 0.000052)
## 170	Neuron_176	(0.000002, 0.000053)
## 171	Neuron_177	(-0.000013, 0.000012)
## 172	Neuron_180	(0.000066, 0.000109)
## 173	Neuron_181	(-0.000188, -0.000027)
## 174	Neuron_182	(-0.00012, 0.000088)
## 175	Neuron_183	(0.000068, 0.00015)
## 176	Neuron_184	(-0.000035, -0.000005)
## 177	Neuron_185	(-0.000334, 0.00368)
## 178	Neuron_186	(-0.02138, 0.019849)
## 179	Neuron_187	(-0.000026, -0.000006)
## 180	Neuron_189	(0.000141, 0.013678)
## 181	Neuron_190	(0.000027, 0.000056)
## 182	Neuron_191	(0.007453, 0.026506)

## 183	Neuron_192	(0.000018, 0.000055)
## 184	Neuron_193	(0.005936, 0.015686)
## 185	Neuron_194	(-0.000389, 0.000743)
## 186	Neuron_196	(0.021146, 0.027577)
## 187	Neuron_197	(-0.001737, 0.003436)
## 188	Neuron_198	(-0.000038, 0.000024)
## 189	Neuron_199	(0.000029, 0.000094)
## 190	Neuron_200	(0.000006, 0.000079)
## 191	Neuron_202	(-0.000122, 0.000011)
## 192	Neuron_204	(0.000005, 0.000102)
## 193	Neuron_205	(-0.000096, 0.000141)
## 194	Neuron_207	(0.00344, 0.006165)
## 195	Neuron_208	(-0.000047, 0.000046)
## 196	Neuron_209	(0.041165, 0.107381)
## 197	Neuron_210	(0.036929, 0.102189)
## 198	Neuron_211	(-0.000078, 0.000003)
## 199	Neuron_212	(-0.000046, 0.000009)
## 200	Neuron_213	(0.000089, 0.00018)
## 201	Neuron_214	(-0.000011, 0.000015)
## 202	Neuron_215	(-0.000046, 0.000008)
## 203	Neuron_216	(0.000001, 0.000064)
## 204	Neuron_218	(-0.000039, -0.000004)
## 205	Neuron_219	(-0.001361, 0.013344)
## 206	Neuron_220	(-0.000022, 0.000005)
## 207	Neuron_221	(-0.000096, 0.000226)
## 208	Neuron_222	(0.000055, 0.000092)
## 209	Neuron_223	(-0.003346, 0.051483)
## 210	Neuron_224	(-0.000777, 0.000172)
## 211	Neuron_225	(-0.000274, -0.000171)
## 212	Neuron_227	(-0.000036, 0.000009)
## 213	Neuron_228	(-0.136114, -0.035323)
## 214	Neuron_229	(0.000065, 0.000756)
## 215	Neuron_230	(-0.000031, 0.000032)
## 216	Neuron_231	(0.004934, 0.022574)
## 217	Neuron_232	(-0.003928, 0.010778)
## 218	Neuron_233	(0.000044, 0.000085)
## 219	Neuron_235	(-0.000003, 0.000033)
## 220	Neuron_236	(-0.00587, 0.168116)
## 221	Neuron_237	(0.000015, 0.000054)
## 222	Neuron_238	(-0.000014, 0.000014)
## 223	Neuron_239	(0.000059, 0.000097)
## 224	Neuron_240	(-0.00012, -0.000063)
## 225	Neuron_241	(-0.000007, 0.000058)
## 226	Neuron_242	(-0.000021, 0.000016)
## 227	Neuron_244	(-0.000017, 0.000092)

## 228	Neuron_246	(-0.000057, -0.000013)
## 229	Neuron_247	(-0.000501, 0.000356)
## 230	Neuron_248	(-0.000051, 0.000056)
## 231	Neuron_249	(-0.000284, -0.000105)
## 232	Neuron_250	(0.000004, 0.000048)
## 233	Neuron_251	(-0.000025, 0.000024)
## 234	Neuron_253	(-0.008224, 0.034894)
## 235	Neuron_254	(0.002769, 0.005689)
## 236	Neuron_255	(-0.00006, -0.000012)
## 237	Neuron_256	(-0.000214, 0.000363)
## 238	Neuron_257	(-0.000025, 0.000022)
## 239	Neuron_258	(0.000111, 0.000176)
## 240	Neuron_259	(0.000063, 0.000206)
## 241	Neuron_260	(0.000204, 0.002848)
## 242	Neuron_261	(-0.000053, 0.000005)
## 243	Neuron_262	(0.000124, 0.000707)
## 244	Neuron_264	(-0.000012, 0.000035)
## 245	Neuron_265	(-0.00021, 0.000053)
## 246	Neuron_266	(-0.000082, 0.000002)
## 247	Neuron_267	(-0.000018, 0.000019)
## 248	Neuron_268	(0.00413, 0.009672)
## 249	Neuron_269	(-0.024127, 0.000676)
## 250	Neuron_270	(-0.074478, 0.127495)
## 251	Neuron_271	(-0.003915, 0.025042)
## 252	Neuron_273	(-0.000097, 0.000006)
## 253	Neuron_274	(-0.008298, -0.0022)
## 254	Neuron_275	(-0.00023, 0.000575)
## 255	Neuron_276	(-0.000029, 0.000012)
## 256	Neuron_277	(-0.000046, 0.000044)
## 257	Neuron_278	(0.001324, 0.00396)
## 258	Neuron_279	(0.000001, 0.000039)
## 259	Neuron_280	(-0.000037, 0.000011)
## 260	Neuron_282	(0.011347, 0.036334)
## 261	Neuron_283	(-0.000166, -0.000068)
## 262	Neuron_284	(-0.000269, -0.000007)
## 263	Neuron_285	(-0.005094, -0.000641)
## 264	Neuron_286	(-0.000062, -0.000015)
## 265	Neuron_287	(-0.000057, 0.000031)
## 266	Neuron_288	(-0.000077, 0.000014)
## 267	Neuron_289	(-0.000168, 0.01046)
## 268	Neuron_290	(0.008522, 0.019091)
## 269	Neuron_291	(-0.000055, -0.000008)
## 270	Neuron_292	(0.000206, 0.000496)
## 271	Neuron_293	(0.000012, 0.000085)
## 272	Neuron_294	(-0.000207, 0.000092)

## 273	Neuron_295	(-0.000066, 0.000023)
## 274	Neuron_296	(0.025118, 0.042746)
## 275	Neuron_297	(-0.000067, 0.000283)
## 276	Neuron_298	(-0.011351, 0.015335)
## 277	Neuron_299	(0.000053, 0.000952)
## 278	Neuron_300	(0.000091, 0.000173)
## 279	Neuron_301	(-0.000096, 0.000369)
## 280	Neuron_302	(0.000279, 0.00119)
## 281	Neuron_303	(-0.000002, 0.000073)
## 282	Neuron_304	(-0.001351, 0.002466)
## 283	Neuron_306	(-0.000901, 0.012171)
## 284	Neuron_308	(0.000439, 0.001064)
## 285	Neuron_309	(-0.000003, 0.000085)
## 286	Neuron_311	(-0.000354, 0.000431)
## 287	Neuron_312	(-0.000314, -0.000124)
## 288	Neuron_314	(-0.00055, -0.000282)
## 289	Neuron_315	(-0.000016, 0.000017)
## 290	Neuron_3:reach	<NA>
## 291	Neuron_4:reach	<NA>
## 292	Neuron_5:reach	<NA>
## 293	Neuron_6:reach	<NA>
## 294	Neuron_7:reach	<NA>
## 295	Neuron_8:reach	<NA>
## 296	Neuron_9:reach	<NA>
## 297	Neuron_12:reach	<NA>
## 298	Neuron_13:reach	<NA>
## 299	Neuron_14:reach	<NA>
## 300	Neuron_16:reach	<NA>
## 301	Neuron_17:reach	<NA>
## 302	Neuron_18:reach	<NA>
## 303	Neuron_19:reach	<NA>
## 304	Neuron_20:reach	<NA>
## 305	Neuron_21:reach	<NA>
## 306	Neuron_22:reach	<NA>
## 307	Neuron_23:reach	<NA>
## 308	Neuron_24:reach	<NA>
## 309	Neuron_25:reach	<NA>
## 310	Neuron_26:reach	<NA>
## 311	Neuron_27:reach	<NA>
## 312	Neuron_28:reach	<NA>
## 313	Neuron_29:reach	<NA>
## 314	Neuron_30:reach	<NA>
## 315	Neuron_32:reach	<NA>
## 316	Neuron_33:reach	<NA>
## 317	Neuron_34:reach	<NA>

## 318	Neuron_35:reach	<NA>
## 319	Neuron_36:reach	<NA>
## 320	Neuron_37:reach	<NA>
## 321	Neuron_38:reach	<NA>
## 322	Neuron_39:reach	<NA>
## 323	Neuron_40:reach	<NA>
## 324	Neuron_41:reach	<NA>
## 325	Neuron_42:reach	<NA>
## 326	Neuron_43:reach	<NA>
## 327	Neuron_44:reach	<NA>
## 328	Neuron_45:reach	<NA>
## 329	Neuron_46:reach	<NA>
## 330	Neuron_47:reach	<NA>
## 331	Neuron_48:reach	<NA>
## 332	Neuron_50:reach	<NA>
## 333	Neuron_51:reach	<NA>
## 334	Neuron_52:reach	<NA>
## 335	Neuron_53:reach	<NA>
## 336	Neuron_54:reach	<NA>
## 337	Neuron_55:reach	<NA>
## 338	Neuron_56:reach	<NA>
## 339	Neuron_57:reach	<NA>
## 340	Neuron_58:reach	<NA>
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## 343	Neuron_61:reach	<NA>
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## 348	Neuron_66:reach	<NA>
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## 360	Neuron_79:reach	<NA>
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## 363	Neuron_82:reach	<NA>
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## 365	Neuron_84:reach	<NA>
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## 368	Neuron_87:reach	<NA>
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## 513	Neuron_266:reach	<NA>
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## 515	Neuron_273:reach	<NA>
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## 517	Neuron_276:reach	<NA>
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## 539	Neuron_301:reach	<NA>
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## 542	Neuron_306:reach	<NA>

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## 543      Neuron_308:reach      <NA>
## 544      Neuron_309:reach      <NA>
## 545      Neuron_311:reach      <NA>
## 546      Neuron_312:reach      <NA>
## 547      Neuron_314:reach      <NA>
## 548      Neuron_315:reach      <NA>
## 549      reach      <NA>
## 550      Intercept      <NA>
## 551      sd_neuron__Intercept      <NA>
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## 553      cor_neuron__Intercept__reach      <NA>
## 554      sigma      <NA>
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## 557 r_neuron[Neuron_234,Intercept]      <NA>
## 558 r_neuron[Neuron_281,Intercept]      <NA>
## 559      r_neuron[Neuron_168,reach]      <NA>
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## 561      r_neuron[Neuron_281,reach]      <NA>
## 562      lprior      <NA>
## 563      lp__      <NA>
##      model2      model3      model4
## 1  (-0.007771, -0.000531) (-0.000947, 0.001542) (-0.002589, 0.002113)
## 2  (0.000116, 0.000227) (-0.000054, -0.000016) (-0.000001, 0.000072)
## 3  (0.00005, 0.000181) (-0.000049, -0.000004) (-0.000019, 0.000066)
## 4  (-0.000504, -0.000301) (-0.000072, -0.000003) (-0.000223, -0.000091)
## 5  (-0.000008, 0.000076) (-0.000046, -0.000017) (-0.000023, 0.000032)
## 6  (-0.000076, 0.00002) (-0.000001, 0.000032) (-0.00005, 0.000013)
## 7  (0.000046, 0.000153) (-0.000003, 0.000034) (0.000009, 0.000079)
## 8  (0.000132, 0.000361) (-0.000077, 0.000002) (0.000005, 0.000154)
## 9  (0.00019, 0.000361) (-0.000024, 0.000034) (0.000017, 0.000128)
## 10 (0.000039, 0.000103) (0.000019, 0.00004) (0.000005, 0.000046)
## 11 (0.000083, 0.000273) (0.000043, 0.000108) (0.000004, 0.000127)
## 12 (-0.000015, 0.000177) (-0.000004, 0.000062) (0.000006, 0.00013)
## 13 (0.003796, 0.016638) (0.002664, 0.007079) (0.000761, 0.009102)
## 14 (-0.000062, 0.000089) (-0.00005, 0.000002) (-0.000081, 0.000017)
## 15 (0.00005, 0.000166) (-0.000052, -0.000012) (-0.000017, 0.000059)
## 16 (-0.000146, 0.000113) (-0.000144, -0.000055) (-0.000128, 0.00004)
## 17 (-0.00003, 0.000071) (-0.000032, 0.000003) (-0.000027, 0.000038)
## 18 (-0.000091, 0.000031) (-0.000041, 0.000001) (-0.000044, 0.000035)
## 19 (-0.00007, 0.000101) (-0.000081, -0.000023) (-0.00004, 0.000071)
## 20 (-0.000558, -0.000339) (-0.000122, -0.000046) (-0.000251, -0.000108)
## 21 (-0.001832, -0.00087) (-0.00023, 0.000101) (-0.000739, -0.000113)
## 22 (-0.000213, 0.000005) (-0.000023, 0.000052) (-0.000119, 0.000022)
## 23 (-0.000048, 0.000024) (0.000023, 0.000048) (-0.00001, 0.000037)

```


## 24	(-0.000199, -0.000005)	(-0.000049, 0.000018)	(-0.00011, 0.000017)
## 25	(-0.000229, -0.000025)	(-0.000139, -0.000068)	(-0.000111, 0.000022)
## 26	(-0.000314, -0.00006)	(-0.0001, -0.000013)	(-0.000157, 0.000008)
## 27	(0.000022, 0.000828)	(-0.000055, 0.000222)	(-0.000099, 0.000425)
## 28	(-0.000111, 0.000005)	(-0.000033, 0.000007)	(-0.000064, 0.000011)
## 29	(-0.047697, 0.00741)	(-0.015605, 0.00334)	(-0.01889, 0.016901)
## 30	(0.000064, 0.000143)	(-0.000012, 0.000015)	(-0.000001, 0.00005)
## 31	(-0.000056, 0.000046)	(0.000006, 0.000041)	(-0.000044, 0.000021)
## 32	(-0.000191, -0.000095)	(0.000008, 0.000041)	(-0.000079, -0.000016)
## 33	(0.000183, 0.000437)	(-0.000107, -0.000019)	(-0.000027, 0.000138)
## 34	(-0.000055, 0.000021)	(-0.000024, 0.000003)	(-0.000031, 0.000018)
## 35	(-0.000129, 0.00001)	(-0.000017, 0.000031)	(-0.000074, 0.000016)
## 36	(-0.000418, -0.000146)	(-0.000034, 0.00006)	(-0.000209, -0.000033)
## 37	(0.000075, 0.000361)	(-0.00008, 0.000018)	(-0.000022, 0.000164)
## 38	(-0.000128, 0.000009)	(-0.000027, 0.00002)	(-0.000063, 0.000026)
## 39	(0.003916, 0.005733)	(0.000578, 0.001202)	(0.001282, 0.002462)
## 40	(-0.000222, 0.000008)	(-0.000008, 0.000071)	(-0.000091, 0.000058)
## 41	(-0.00004, 0.000042)	(0.00001, 0.000038)	(-0.000014, 0.000039)
## 42	(0.018624, 0.052308)	(0.000045, 0.011624)	(0.004787, 0.026665)
## 43	(-0.000118, 0.000064)	(-0.000051, 0.000012)	(-0.000072, 0.000046)
## 44	(-0.000112, 0.000003)	(-0.000028, 0.000011)	(-0.000057, 0.000017)
## 45	(-0.00126, 0.000692)	(-0.000154, 0.000517)	(-0.00064, 0.000628)
## 46	(-0.005535, -0.002974)	(-0.000899, -0.000018)	(-0.002668, -0.001005)
## 47	(-0.000055, 0.000127)	(-0.00005, 0.000012)	(-0.000064, 0.000055)
## 48	(0.00003, 0.000089)	(-0.000004, 0.000016)	(-0.000005, 0.000034)
## 49	(-0.0001, -0.000016)	(-0.000022, 0.000007)	(-0.000047, 0.000007)
## 50	(-0.000048, 0.000058)	(-0.000034, 0.000003)	(-0.000048, 0.00002)
## 51	(-0.000109, 0.000115)	(0.000009, 0.000086)	(-0.000021, 0.000124)
## 52	(-0.00007, 0.000098)	(0.000065, 0.000123)	(-0.000005, 0.000104)
## 53	(-0.000035, 0.000394)	(-0.00011, 0.000038)	(-0.000051, 0.000228)
## 54	(0.000012, 0.000199)	(-0.000007, 0.000057)	(0.000004, 0.000126)
## 55	(-0.000238, -0.000037)	(-0.000023, 0.000046)	(-0.000095, 0.000036)
## 56	(-0.000146, -0.000065)	(-0.000023, 0.000005)	(-0.000067, -0.000014)
## 57	(-0.000066, 0.000003)	(-0.000014, 0.00001)	(-0.000028, 0.000017)
## 58	(0.00012, 0.000228)	(0, 0.000037)	(0.00002, 0.00009)
## 59	(-0.000211, -0.000101)	(-0.000049, -0.000011)	(-0.000073, -0.000002)
## 60	(-0.000101, 0.000018)	(-0.000002, 0.000039)	(-0.000054, 0.000023)
## 61	(-0.000081, -0.000009)	(-0.000015, 0.00001)	(-0.000038, 0.000009)
## 62	(-0.000011, 0.000148)	(-0.000027, 0.000028)	(-0.000021, 0.000082)
## 63	(-0.00002, 0.00006)	(0.000018, 0.000046)	(-0.000008, 0.000044)
## 64	(-0.000077, 0.000002)	(-0.000027, 0.000001)	(-0.000043, 0.000008)
## 65	(-0.000134, -0.000016)	(0.00002, 0.000061)	(-0.000049, 0.000027)
## 66	(-0.000183, -0.000005)	(-0.00002, 0.000041)	(-0.000096, 0.00002)
## 67	(-0.049025, 0.163743)	(0.038883, 0.112029)	(-0.012736, 0.125456)
## 68	(0.000017, 0.000097)	(-0.000022, 0.000006)	(-0.000015, 0.000037)

## 69	(-0.000054, 0.000023)	(-0.000004, 0.000023)	(-0.000034, 0.000017)
## 70	(-0.000113, -0.000011)	(-0.000013, 0.000022)	(-0.000045, 0.000021)
## 71	(-0.000253, 0.00003)	(-0.000079, 0.000018)	(-0.00014, 0.000044)
## 72	(0.696233, 1.207136)	(-0.066426, 0.109212)	(0.17267, 0.5045)
## 73	(0.001167, 0.004358)	(0.001745, 0.002842)	(0.000602, 0.002675)
## 74	(-0.000195, 0.000017)	(-0.000098, -0.000025)	(-0.000145, -0.000007)
## 75	(-0.000067, 0.000202)	(-0.000088, 0.000005)	(-0.000044, 0.000131)
## 76	(-0.000167, -0.000029)	(-0.000051, -0.000003)	(-0.000087, 0.000003)
## 77	(0.000194, 0.000574)	(0.00011, 0.000241)	(-0.000035, 0.000211)
## 78	(0.000427, 0.005166)	(-0.00113, 0.000499)	(-0.000823, 0.002255)
## 79	(-0.000012, 0.000095)	(0.000023, 0.000059)	(-0.000005, 0.000064)
## 80	(-0.000226, 0.000002)	(-0.000067, 0.000012)	(-0.000126, 0.000022)
## 81	(0.000017, 0.000128)	(0.000017, 0.000055)	(-0.000004, 0.000068)
## 82	(-0.000041, 0.000039)	(-0.000006, 0.000021)	(-0.000035, 0.000017)
## 83	(-0.000093, -0.00001)	(-0.000031, -0.000002)	(-0.000054, 0)
## 84	(-0.00033, -0.000029)	(-0.000119, -0.000016)	(-0.000228, -0.000032)
## 85	(-0.000224, -0.000076)	(-0.000057, -0.000006)	(-0.000099, -0.000003)
## 86	(-0.610846, 0.052026)	(-0.016738, 0.211145)	(-0.294506, 0.136027)
## 87	(-0.000004, 0.000076)	(-0.000047, -0.000019)	(-0.000035, 0.000017)
## 88	(-0.000187, -0.000032)	(-0.000054, 0)	(-0.000092, 0.000009)
## 89	(0.000023, 0.000122)	(0.000058, 0.000092)	(0.000015, 0.000079)
## 90	(-0.00003, 0.000095)	(-0.000044, -0.000001)	(-0.000019, 0.000061)
## 91	(-0.000348, -0.000106)	(-0.000109, -0.000026)	(-0.000186, -0.000028)
## 92	(-0.00014, -0.000003)	(0.000004, 0.000051)	(-0.000031, 0.000059)
## 93	(0.042366, 0.062714)	(-0.008042, -0.001047)	(0.009844, 0.02306)
## 94	(-0.000115, -0.000033)	(-0.000028, 0)	(-0.000042, 0.000011)
## 95	(-0.000409, 0.000007)	(0.000042, 0.000185)	(-0.000258, 0.000012)
## 96	(-0.00005, 0.000095)	(0.000014, 0.000064)	(-0.000041, 0.000054)
## 97	(0.001358, 0.008549)	(-0.001184, 0.001289)	(-0.000975, 0.003695)
## 98	(-0.000015, 0.000085)	(0.000024, 0.000059)	(-0.000018, 0.000048)
## 99	(-0.000105, 0.000006)	(-0.000057, -0.000019)	(-0.00005, 0.000023)
## 100	(-0.000165, 0.000192)	(-0.000007, 0.000115)	(-0.000101, 0.000131)
## 101	(-0.000072, 0.000033)	(-0.000058, -0.000022)	(-0.000047, 0.000022)
## 102	(0.000103, 0.000205)	(-0.000093, -0.000058)	(-0.000017, 0.00005)
## 103	(-0.000025, 0.000037)	(-0.000001, 0.00002)	(-0.000022, 0.000018)
## 104	(0.000038, 0.00028)	(-0.000013, 0.00007)	(-0.000064, 0.000093)
## 105	(-0.001924, -0.00093)	(0.000056, 0.000397)	(-0.000844, -0.000199)
## 106	(-0.00045, -0.000188)	(-0.000018, 0.000072)	(-0.00022, -0.00005)
## 107	(-0.000155, 0.000049)	(-0.000042, 0.000028)	(-0.000077, 0.000055)
## 108	(-0.000034, 0.000153)	(-0.00005, 0.000014)	(-0.00007, 0.000052)
## 109	(-0.000104, 0.000191)	(-0.00002, 0.000082)	(-0.000067, 0.000124)
## 110	(-0.000105, 0.000093)	(-0.000022, 0.000046)	(-0.000056, 0.000072)
## 111	(-0.000849, 0.000525)	(-0.001013, -0.000541)	(-0.000792, 0.000101)
## 112	(-0.000131, -0.000066)	(-0.00004, -0.000017)	(-0.000064, -0.000021)
## 113	(0.000013, 0.000076)	(-0.000025, -0.000004)	(-0.000002, 0.000038)

```

## 114 (-0.000181, 0.000011) (-0.000028, 0.000038) (-0.000079, 0.000045)
## 115 (-0.000018, 0.000079) (0.000031, 0.000064) (-0.000012, 0.000005)
## 116 (-0.002218, -0.000185) (-0.001736, -0.001037) (-0.001334, -0.000014)
## 117 (-0.000082, 0.000011) (-0.000033, 0) (-0.000044, 0.000017)
## 118 (-0.000509, -0.000082) (-0.000019, 0.000128) (-0.000256, 0.000022)
## 119 (-0.000067, 0.000229) (-0.000116, -0.000014) (-0.000097, 0.000096)
## 120 (0.000007, 0.000223) (-0.000054, 0.00002) (-0.000015, 0.000126)
## 121 (0.001048, 0.003841) (-0.000472, 0.000488) (0.000348, 0.002162)
## 122 (0.000075, 0.000183) (-0.000026, 0.000011) (0.000006, 0.000075)
## 123 (-0.00001, 0.000133) (-0.000031, 0.000018) (-0.00001, 0.000083)
## 124 (0.000153, 0.000278) (-0.000058, -0.000015) (0.00002, 0.000102)
## 125 (0.000341, 0.000771) (-0.000042, 0.000106) (0.000028, 0.000307)
## 126 (-0.000014, 0.000163) (-0.000058, 0.000003) (0.000022, 0.000137)
## 127 (-0.000072, 0.000032) (0.000012, 0.000048) (-0.000044, 0.000023)
## 128 (0.000087, 0.000202) (-0.00007, -0.000031) (-0.000017, 0.000058)
## 129 (-0.000235, -0.000095) (-0.000038, 0.00001) (-0.000095, -0.000004)
## 130 (-0.000135, 0.000094) (-0.000078, 0) (-0.000087, 0.000061)
## 131 (-0.003426, 0.000782) (-0.001645, -0.000198) (-0.002572, 0.000161)
## 132 (-0.000165, -0.000028) (-0.00001, 0.000037) (-0.000083, 0.000005)
## 133 (0.00001, 0.000198) (0.000014, 0.000078) (-0.000024, 0.000098)
## 134 (-0.000146, 0.000024) (0.000015, 0.000073) (-0.000051, 0.000059)
## 135 (-0.000287, -0.000132) (-0.000171, -0.000118) (-0.000166, -0.000066)
## 136 (-0.013547, 0.001361) (-0.000501, 0.004624) (-0.004606, 0.005076)
## 137 (0.000211, 0.000389) (-0.000006, 0.000055) (0.000067, 0.000183)
## 138 (-0.000086, 0.000012) (0.000006, 0.000039) (-0.000063, 0)
## 139 (-0.000109, 0.000114) (0.000022, 0.000098) (-0.00007, 0.000075)
## 140 (0.000631, 0.002886) (0.000174, 0.000949) (0.000139, 0.001603)
## 141 (-0.000004, 0.000092) (-0.000021, 0.000012) (-0.000032, 0.000003)
## 142 (-0.000142, -0.000056) (-0.000014, 0.000015) (-0.000063, -0.000007)
## 143 (0.000006, 0.000192) (-0.000038, 0.000007) (-0.000008, 0.000077)
## 144 (0.000054, 0.000296) (0.000044, 0.000128) (0.000118, 0.000276)
## 145 (-0.000133, 0.000048) (-0.000115, -0.000052) (-0.00014, -0.000023)
## 146 (0.000301, 0.000869) (0.000109, 0.000305) (0.000125, 0.000494)
## 147 (-0.000634, -0.000194) (-0.000218, -0.000067) (-0.000272, 0.000014)
## 148 (-0.00046, 0.000227) (-0.000233, 0.000003) (-0.000142, 0.000304)
## 149 (-0.000067, 0.000202) (-0.000019, 0.000074) (-0.000058, 0.000117)
## 150 (-0.000113, 0.000001) (-0.000009, 0.000003) (-0.000057, 0.000017)
## 151 (0.000001, 0.000085) (-0.000003, -0.000001) (-0.000013, 0.000041)
## 152 (-0.000129, -0.000036) (-0.000006, 0.000025) (-0.000056, 0.000004)
## 153 (-0.00019, 0.000318) (-0.000072, 0.000103) (-0.000128, 0.000202)
## 154 (-0.000087, 0.000008) (-0.000073, -0.000015) (-0.000077, 0.000032)
## 155 (0.000157, 0.000289) (-0.000032, 0.000013) (0.000035, 0.000121)
## 156 (-0.000142, -0.000005) (0.000003, 0.000062) (-0.000028, 0.000032)
## 157 (-0.000231, -0.000125) (0.000008, 0.000044) (-0.000078, -0.000009)
## 158 (-0.000082, 0.000083) (-0.000005, 0.000007) (-0.000052, 0.000055)

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## 159	(-0.037573, 0.063123)	(0.002928, 0.037545)	(-0.019942, 0.045459)
## 160	(0.000067, 0.000284)	(-0.000032, 0.000042)	(-0.000054, 0.000087)
## 161	(-0.000843, 0.000452)	(-0.000939, -0.000494)	(-0.000817, 0.000024)
## 162	(0.000036, 0.000317)	(-0.000101, -0.000004)	(-0.000057, 0.000126)
## 163	(-0.000056, 0.000153)	(0.000047, 0.000119)	(-0.000057, 0.000078)
## 164	(0.000032, 0.000206)	(-0.000042, 0.000017)	(-0.000038, 0.000075)
## 165	(-0.000177, 0.000086)	(-0.000032, 0.000059)	(0, 0.00017)
## 166	(-0.000146, 0.000176)	(-0.000142, -0.000031)	(-0.000153, 0.000056)
## 167	(0.000674, 0.001417)	(-0.000316, -0.000061)	(0.000068, 0.000551)
## 168	(0.000019, 0.000139)	(0.000016, 0.000057)	(0.000001, 0.000078)
## 169	(0.000193, 0.000411)	(-0.000023, 0.000052)	(0.000004, 0.000182)
## 170	(0.000069, 0.000175)	(-0.000052, -0.000016)	(0.000007, 0.000076)
## 171	(-0.000018, 0.000006)	(-0.000013, 0.000014)	(-0.000018, 0.000032)
## 172	(0.000123, 0.000258)	(-0.000014, 0.000033)	(0.000052, 0.00014)
## 173	(-0.000489, 0.000014)	(-0.000087, 0.000085)	(-0.000278, 0.000048)
## 174	(-0.000631, 0.000018)	(0.000179, 0.000403)	(-0.000221, 0.000201)
## 175	(-0.000027, 0.000228)	(-0.000064, 0.000024)	(-0.000002, 0.000146)
## 176	(-0.000054, 0.000004)	(-0.000033, -0.000001)	(-0.000045, 0.000016)
## 177	(-0.006012, 0.00654)	(0.00172, 0.006035)	(-0.002138, 0.006014)
## 178	(-0.046803, 0.082107)	(-0.001281, 0.043036)	(-0.029275, 0.054451)
## 179	(-0.000032, 0.000031)	(-0.000003, 0.000019)	(-0.000023, 0.000017)
## 180	(-0.040475, 0.001852)	(-0.019375, -0.004823)	(-0.021913, 0.005579)
## 181	(-0.000046, 0.000047)	(-0.000024, 0.000007)	(-0.000019, 0.000041)
## 182	(-0.066353, -0.00678)	(0.030797, 0.051277)	(-0.012196, 0.026496)
## 183	(-0.000167, -0.000052)	(0.000019, 0.000058)	(-0.000049, 0.000026)
## 184	(-0.01215, 0.018336)	(0.010041, 0.020522)	(-0.000172, 0.019629)
## 185	(-0.002534, 0.001003)	(-0.001252, -0.000036)	(-0.00156, 0.000738)
## 186	(0.010164, 0.030274)	(-0.000953, 0.00596)	(0.009164, 0.022225)
## 187	(-0.000496, 0.015678)	(-0.003699, 0.001861)	(-0.002745, 0.00776)
## 188	(-0.000277, -0.000084)	(-0.000039, 0.000027)	(-0.000127, -0.000002)
## 189	(-0.000221, -0.000018)	(-0.000053, 0.000017)	(-0.000091, 0.000041)
## 190	(-0.00003, 0.000196)	(-0.000066, 0.000012)	(-0.000041, 0.000106)
## 191	(-0.000599, -0.000183)	(0.000092, 0.000235)	(-0.00023, 0.00004)
## 192	(-0.00009, 0.000073)	(-0.000016, 0.00004)	(-0.000026, 0.000079)
## 193	(-0.000791, -0.00005)	(-0.000213, 0.000041)	(-0.000402, 0.000079)
## 194	(0.007963, 0.016484)	(-0.002903, 0.000026)	(0.002428, 0.007963)
## 195	(0.000285, 0.000574)	(-0.00007, 0.00003)	(0.000043, 0.00023)
## 196	(-0.146216, 0.060821)	(-0.071348, -0.000173)	(-0.06863, 0.06584)
## 197	(-0.22521, -0.02116)	(0.010281, 0.08043)	(-0.069022, 0.063508)
## 198	(-0.000005, 0.000248)	(-0.000026, 0.000061)	(-0.000048, 0.000116)
## 199	(-0.00009, 0.000082)	(-0.000062, -0.000003)	(-0.000074, 0.000038)
## 200	(-0.00046, -0.000174)	(-0.000078, 0.00002)	(-0.000163, 0.000022)
## 201	(0.000039, 0.00012)	(-0.000032, -0.000004)	(-0.000005, 0.000047)
## 202	(0.000022, 0.000191)	(0.000014, 0.000072)	(-0.000012, 0.000098)
## 203	(-0.000277, -0.000111)	(0.000026, 0.000084)	(-0.000088, 0.00002)

## 204	(0.000083, 0.000193)	(-0.000055, -0.000017)	(-0.000009, 0.000063)
## 205	(-0.028694, 0.017286)	(-0.001768, 0.014039)	(-0.012791, 0.017073)
## 206	(-0.000072, 0.000013)	(-0.000011, 0.000018)	(-0.000039, 0.000016)
## 207	(-0.000059, 0.000948)	(-0.000077, 0.000269)	(-0.000125, 0.000529)
## 208	(0.000055, 0.000169)	(0.000005, 0.000044)	(0.000033, 0.000107)
## 209	(0.095096, 0.266531)	(0.051998, 0.110934)	(0.039776, 0.151122)
## 210	(-0.005662, -0.002694)	(-0.000041, 0.000979)	(-0.002301, -0.000374)
## 211	(-0.000215, 0.000107)	(-0.000048, 0.000062)	(-0.000194, 0.000015)
## 212	(-0.000152, -0.000001)	(-0.000082, -0.000033)	(-0.000097, -0.000004)
## 213	(0.283042, 0.598188)	(0.025088, 0.133429)	(0.042375, 0.247061)
## 214	(-0.000472, 0.001687)	(-0.000889, -0.000147)	(-0.000534, 0.000868)
## 215	(-0.000165, 0.000033)	(-0.000046, 0.000022)	(-0.00009, 0.000038)
## 216	(-0.019033, 0.036119)	(-0.002632, 0.016328)	(-0.008196, 0.027626)
## 217	(-0.003107, 0.042876)	(-0.015836, -0.000028)	(-0.009807, 0.020059)
## 218	(-0.00008, 0.000048)	(-0.000007, 0.000037)	(-0.000002, 0.000063)
## 219	(0.000013, 0.000126)	(0.000006, 0.000045)	(0, 0.000073)
## 220	(0.217338, 0.761343)	(0.292057, 0.479075)	(0.142012, 0.495342)
## 221	(-0.000086, 0.000035)	(0.00004, 0.000082)	(-0.000016, 0.000063)
## 222	(-0.000078, 0.000009)	(-0.000021, 0.000009)	(-0.000042, 0.000015)
## 223	(-0.000211, -0.000092)	(0.000005, 0.000045)	(-0.000055, 0.000022)
## 224	(-0.000216, -0.000038)	(-0.00007, -0.000009)	(-0.000144, -0.000028)
## 225	(-0.000051, 0.000153)	(-0.00006, 0.00001)	(-0.000049, 0.000084)
## 226	(-0.000001, 0.000113)	(-0.00002, 0.000019)	(-0.000019, 0.000055)
## 227	(-0.000123, 0.000218)	(-0.000031, 0.000087)	(-0.000073, 0.000149)
## 228	(0.000034, 0.00017)	(-0.000025, 0.000022)	(-0.000022, 0.000066)
## 229	(-0.002957, -0.000278)	(-0.000251, 0.00067)	(-0.001363, 0.000376)
## 230	(-0.000072, 0.000262)	(0.000105, 0.00022)	(-0.000021, 0.000195)
## 231	(-0.000288, 0.000272)	(-0.000375, -0.000182)	(-0.000342, 0.000021)
## 232	(0.000003, 0.000143)	(-0.000001, 0.000047)	(-0.000005, 0.000086)
## 233	(0.00006, 0.000213)	(-0.000022, 0.000031)	(-0.000003, 0.000096)
## 234	(-0.037207, 0.097611)	(-0.00188, 0.044468)	(-0.022171, 0.065392)
## 235	(-0.010641, -0.001513)	(-0.000517, 0.002621)	(-0.00323, 0.002699)
## 236	(-0.000385, -0.000235)	(0.000053, 0.000104)	(-0.000138, -0.00004)
## 237	(-0.000334, 0.00147)	(-0.00049, 0.00013)	(-0.000432, 0.00074)
## 238	(0.000161, 0.000307)	(-0.000038, 0.000012)	(0.000026, 0.000121)
## 239	(0.000074, 0.000279)	(0.000001, 0.000071)	(0.000052, 0.000185)
## 240	(0.000417, 0.000866)	(-0.000324, -0.00017)	(0.000031, 0.000322)
## 241	(-0.009272, -0.001007)	(-0.001877, 0.000964)	(-0.004041, 0.001327)
## 242	(-0.000008, 0.000174)	(-0.000081, -0.000019)	(-0.000056, 0.000062)
## 243	(0.000356, 0.002178)	(-0.000591, 0.000035)	(-0.000124, 0.00106)
## 244	(-0.000309, -0.000164)	(-0.000065, -0.000016)	(-0.000135, -0.000042)
## 245	(0.000348, 0.001171)	(-0.000176, 0.000106)	(-0.000052, 0.000482)
## 246	(0.000586, 0.000852)	(-0.000132, -0.000041)	(0.000111, 0.000284)
## 247	(-0.000025, 0.000091)	(0.000013, 0.000052)	(-0.000016, 0.00006)
## 248	(-0.027217, -0.009889)	(-0.00052, 0.005437)	(-0.008692, 0.002563)

## 249	(-0.030607, 0.046945)	(0.00408, 0.030741)	(-0.020567, 0.029803)
## 250	(-0.601657, 0.029854)	(-0.096381, 0.12072)	(-0.28749, 0.122675)
## 251	(-0.074439, 0.016101)	(-0.025489, 0.005637)	(-0.038913, 0.019893)
## 252	(0.000188, 0.000511)	(-0.000057, 0.000054)	(-0.000004, 0.000206)
## 253	(-0.005569, 0.013498)	(-0.00438, 0.002175)	(-0.006988, 0.005396)
## 254	(-0.006596, -0.004078)	(-0.000285, 0.00058)	(-0.00249, -0.000854)
## 255	(-0.000103, 0.000024)	(-0.000083, -0.000039)	(-0.000078, 0.000005)
## 256	(-0.00013, 0.000151)	(-0.000034, 0.000063)	(-0.000083, 0.000099)
## 257	(-0.011452, -0.003212)	(-0.001488, 0.001345)	(-0.004263, 0.001089)
## 258	(-0.000076, 0.000043)	(-0.000038, 0.000004)	(-0.000043, 0.000034)
## 259	(-0.000047, 0.000104)	(0.000037, 0.000089)	(-0.000023, 0.000075)
## 260	(-0.078741, -0.000613)	(0.03345, 0.060309)	(-0.015024, 0.03572)
## 261	(-0.000398, -0.000091)	(-0.00007, 0.000036)	(-0.000226, -0.000027)
## 262	(-0.00058, 0.000239)	(-0.000183, 0.000099)	(-0.000383, 0.000149)
## 263	(-0.001924, 0.012002)	(0.000583, 0.005371)	(-0.002806, 0.006239)
## 264	(-0.000164, -0.000015)	(-0.000052, -0.000001)	(-0.0001, -0.000003)
## 265	(-0.000444, -0.000168)	(-0.000107, -0.000012)	(-0.000216, -0.000037)
## 266	(-0.000136, 0.000148)	(0.000061, 0.000159)	(-0.000064, 0.000121)
## 267	(-0.005718, 0.027515)	(-0.011549, -0.000124)	(-0.00739, 0.014195)
## 268	(0.011301, 0.044348)	(0.008961, 0.020321)	(0.008026, 0.029489)
## 269	(-0.000194, -0.000047)	(-0.000018, 0.000032)	(-0.000096, -0.000001)
## 270	(-0.000172, 0.000737)	(-0.000142, 0.000171)	(-0.000079, 0.000511)
## 271	(-0.000166, 0.000062)	(-0.000006, 0.000072)	(-0.000064, 0.000084)
## 272	(-0.000086, 0.000847)	(-0.000244, 0.000077)	(-0.000223, 0.000383)
## 273	(0.000238, 0.000518)	(-0.000125, -0.000028)	(0.000002, 0.000184)
## 274	(0.082067, 0.137184)	(-0.022911, -0.003963)	(0.025474, 0.061273)
## 275	(-0.000179, 0.000916)	(-0.000037, 0.000339)	(-0.000147, 0.000565)
## 276	(-0.052124, 0.031314)	(-0.006162, 0.022522)	(-0.027174, 0.027019)
## 277	(-0.004276, -0.001464)	(0.000329, 0.001296)	(-0.001432, 0.000395)
## 278	(-0.000336, -0.00008)	(-0.000022, 0.000066)	(-0.000101, 0.000065)
## 279	(-0.000747, 0.000708)	(-0.000067, 0.000433)	(-0.000372, 0.000573)
## 280	(-0.002389, 0.000461)	(-0.000295, 0.000685)	(-0.000937, 0.000914)
## 281	(-0.000282, -0.000049)	(-0.000094, -0.000014)	(-0.000137, 0.000014)
## 282	(-0.009943, 0.00199)	(-0.004703, -0.000601)	(-0.005899, 0.001851)
## 283	(0.037447, 0.078319)	(-0.007733, 0.006319)	(0.007664, 0.03421)
## 284	(-0.002763, -0.000809)	(-0.00066, 0.000012)	(-0.001087, 0.000182)
## 285	(-0.00002, 0.000253)	(-0.000068, 0.000026)	(-0.000043, 0.000134)
## 286	(-0.003943, -0.00149)	(-0.00066, 0.000183)	(-0.001768, -0.000175)
## 287	(-0.000978, -0.000385)	(-0.000064, 0.000139)	(-0.00048, -0.000095)
## 288	(-0.001218, -0.00038)	(-0.000232, 0.000057)	(-0.000706, -0.000162)
## 289	(-0.000046, 0.000057)	(-0.000077, -0.000042)	(-0.000051, 0.000016)
## 290	<NA>	<NA>	<NA>
## 291	<NA>	<NA>	<NA>
## 292	<NA>	<NA>	<NA>
## 293	<NA>	<NA>	<NA>

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

##	model1_int	model2_int	model3_int
## 1	(0.002158, 0.004471)	(-0.010635, -0.003544)	(-0.001066, 0.001422)
## 2	(-0.000078, -0.000031)	(-0.000077, 0.000066)	(-0.000057, -0.000007)
## 3	(-0.000077, -0.000032)	(0.00005, 0.000188)	(-0.000044, 0.000005)
## 4	(-0.000051, 0.000013)	(-0.000487, -0.000291)	(-0.000066, 0.000002)
## 5	(0.000009, 0.000036)	(-0.000047, 0.000035)	(-0.000048, -0.000019)
## 6	(-0.000053, -0.000017)	(-0.000063, 0.000048)	(0.000009, 0.000048)
## 7	(-0.00002, 0.000016)	(0.000032, 0.000143)	(0.000005, 0.000044)
## 8	(0.000019, 0.000098)	(0.000145, 0.000389)	(-0.000053, 0.000032)
## 9	(-0.000058, 0.000011)	(0.000213, 0.000424)	(-0.000033, 0.000041)
## 10	(-0.000035, -0.000015)	(0.000026, 0.000088)	(0.000022, 0.000044)
## 11	(-0.000091, -0.000032)	(0.000093, 0.000274)	(0.000042, 0.000105)
## 12	(0.00005, 0.00011)	(-0.000023, 0.000162)	(-0.000004, 0.000061)
## 13	(-0.000388, 0.003789)	(-0.002831, 0.009977)	(0.000916, 0.00541)
## 14	(-0.000106, -0.000057)	(-0.000014, 0.000136)	(-0.000042, 0.000011)
## 15	(-0.00004, 0.000001)	(-0.000001, 0.000122)	(-0.000053, -0.00001)
## 16	(-0.000075, 0.000009)	(-0.000153, 0.000103)	(-0.000135, -0.000045)
## 17	(-0.000001, 0.000031)	(-0.00003, 0.000067)	(-0.000034, 0)
## 18	(0.000021, 0.00006)	(-0.000078, 0.000041)	(-0.000041, 0.000001)
## 19	(0.000021, 0.000081)	(-0.000021, 0.000164)	(-0.000054, 0.000011)
## 20	(-0.000056, 0.000015)	(-0.00044, -0.000223)	(-0.000106, -0.00003)
## 21	(-0.000112, 0.000189)	(-0.001506, -0.000583)	(-0.000238, 0.000086)
## 22	(-0.000099, -0.00003)	(-0.000178, 0.000034)	(-0.000007, 0.000068)
## 23	(0.000006, 0.000028)	(-0.000033, 0.000036)	(0.000017, 0.000042)
## 24	(-0.000028, 0.000051)	(-0.000044, 0.000198)	(0, 0.000085)
## 25	(0.00006, 0.000124)	(-0.000306, -0.00011)	(-0.000127, -0.000058)
## 26	(0.000006, 0.000084)	(-0.000338, -0.000098)	(-0.000086, -0.000002)
## 27	(-0.000178, 0.00008)	(-0.000217, 0.000572)	(-0.000072, 0.000205)
## 28	(-0.000026, 0.000011)	(-0.000074, 0.000039)	(-0.000047, -0.000007)
## 29	(0.016485, 0.033899)	(-0.052182, 0.00121)	(-0.01419, 0.004546)
## 30	(-0.000031, -0.000003)	(0.000054, 0.00014)	(-0.000008, 0.000023)
## 31	(-0.000064, -0.000031)	(-0.000067, 0.000036)	(0, 0.000036)
## 32	(-0.000028, 0.000005)	(-0.000129, -0.000029)	(-0.000001, 0.000034)
## 33	(-0.000155, -0.000074)	(0.000228, 0.000474)	(-0.00011, -0.000023)
## 34	(-0.00001, 0.000014)	(-0.00002, 0.000056)	(-0.000028, -0.000001)
## 35	(-0.000062, -0.000019)	(-0.000126, 0.000007)	(-0.000009, 0.000038)
## 36	(-0.000142, -0.000057)	(-0.000481, -0.000222)	(-0.000046, 0.000045)
## 37	(-0.00003, 0.000058)	(0.000035, 0.000303)	(-0.000087, 0.000008)
## 38	(-0.000004, 0.00004)	(-0.000133, 0.000005)	(-0.000015, 0.000034)
## 39	(-0.00061, 0.000002)	(0.003237, 0.005115)	(0.00037, 0.001029)
## 40	(-0.000004, 0.000068)	(-0.000229, -0.000006)	(-0.000011, 0.000067)
## 41	(-0.000008, 0.000018)	(-0.000017, 0.000063)	(0.000015, 0.000043)
## 42	(-0.008132, 0.002723)	(0.005, 0.038283)	(0.00065, 0.012329)
## 43	(-0.000028, 0.00003)	(-0.000141, 0.000038)	(-0.000039, 0.000024)
## 44	(-0.000032, 0.000004)	(-0.000109, 0.000002)	(-0.000026, 0.000013)

## 45	(-0.000043, 0.000601)	(-0.002505, -0.000529)	(-0.00056, 0.000133)
## 46	(-0.001128, -0.000268)	(-0.005717, -0.00308)	(-0.000363, 0.000562)
## 47	(-0.000044, 0.000022)	(-0.000031, 0.000171)	(-0.000057, 0.000014)
## 48	(-0.000023, 0.000001)	(0.000091, 0.000164)	(-0.000002, 0.000023)
## 49	(-0.000005, 0.000022)	(-0.000096, -0.000013)	(-0.000024, 0.000005)
## 50	(-0.000049, -0.000015)	(-0.000104, 0.000001)	(-0.000047, -0.000001)
## 51	(0.000058, 0.00013)	(-0.000102, 0.000119)	(0.00003, 0.000108)
## 52	(0.000043, 0.000096)	(-0.000161, 0.000003)	(0.000049, 0.000107)
## 53	(0.000078, 0.000211)	(-0.00008, 0.000327)	(-0.000096, 0.000047)
## 54	(0.000039, 0.000098)	(0.000067, 0.000249)	(-0.000017, 0.000046)
## 55	(-0.000028, 0.000006)	(-0.000317, -0.000044)	(-0.000105, -0.000001)
## 56	(-0.000028, -0.000003)	(-0.000127, -0.000049)	(-0.000021, 0.000006)
## 57	(0.000003, 0.000025)	(-0.000064, 0.000002)	(-0.000001, 0.000013)
## 58	(-0.000031, 0.000005)	(0.000121, 0.00023)	(0.000001, 0.000039)
## 59	(0.000061, 0.000096)	(-0.000182, -0.000077)	(-0.000037, 0)
## 60	(-0.000038, 0)	(-0.000058, 0.000057)	(0.000004, 0.000044)
## 61	(0.000018, 0.000047)	(-0.000152, -0.000063)	(-0.000022, 0.000001)
## 62	(-0.000024, 0.000026)	(-0.000057, 0.000098)	(-0.000033, 0.000021)
## 63	(-0.000006, 0.000019)	(-0.000025, 0.000005)	(0.000018, 0.000045)
## 64	(-0.000002, 0.000025)	(-0.000066, 0.000018)	(-0.000032, -0.000003)
## 65	(-0.000057, -0.000013)	(-0.00026, -0.000127)	(0.000026, 0.000073)
## 66	(-0.000058, -0.000002)	(-0.000161, 0.000001)	(-0.000018, 0.000042)
## 67	(-0.013673, 0.061902)	(-0.109367, 0.122354)	(0.081723, 0.163035)
## 68	(-0.000039, -0.000012)	(-0.000001, 0.000073)	(-0.000025, 0.000004)
## 69	(-0.000037, -0.000013)	(-0.000037, 0.000037)	(0.0000001, 0.000027)
## 70	(0.000017, 0.000048)	(-0.00013, -0.000034)	(-0.000016, 0.000018)
## 71	(-0.000039, 0.000053)	(-0.000335, -0.000056)	(-0.00011, -0.000012)
## 72	(-0.061385, 0.101359)	(0.492438, 0.991424)	(-0.121883, 0.053213)
## 73	(-0.000179, 0.001327)	(0.00352, 0.008137)	(0.002355, 0.003975)
## 74	(-0.000087, -0.000002)	(-0.000216, -0.000001)	(-0.000009, -0.000018)
## 75	(0.000004, 0.000124)	(-0.000071, 0.000187)	(-0.000079, 0.000012)
## 76	(-0.000036, 0.000014)	(-0.000347, -0.000192)	(-0.000069, -0.000014)
## 77	(-0.000407, -0.000279)	(0.000353, 0.000745)	(0.000114, 0.000252)
## 78	(-0.001138, 0.000447)	(-0.000498, 0.004364)	(-0.001884, -0.000178)
## 79	(-0.000005, 0.000028)	(0.000005, 0.000108)	(0.000022, 0.000058)
## 80	(-0.000037, 0.000034)	(-0.0001, 0.000119)	(-0.000008, -0.000003)
## 81	(-0.000033, 0.000006)	(-0.00004, 0.000077)	(-0.000018, 0.000023)
## 82	(-0.000055, -0.000003)	(-0.000055, 0.000021)	(-0.000005, 0.000021)
## 83	(-0.000032, -0.000006)	(-0.000071, 0.000009)	(-0.000003, -0.000002)
## 84	(-0.000167, -0.000072)	(-0.000173, 0.000118)	(-0.000087, 0.000016)
## 85	(-0.000004, 0.000042)	(-0.000174, -0.000031)	(-0.000048, 0.000002)
## 86	(-0.167317, 0.045889)	(-0.230792, 0.422916)	(0.009291, 0.238679)
## 87	(-0.000051, -0.000025)	(-0.000002, 0.000076)	(-0.000048, -0.000021)
## 88	(-0.000007, 0.000042)	(-0.000151, -0.000001)	(-0.000061, -0.000009)
## 89	(-0.000024, 0.000007)	(0.000041, 0.000135)	(0.000066, 0.000099)

## 90	(0.000026, 0.000066)	(-0.00003, 0.000094)	(-0.00005, -0.000007)
## 91	(-0.000082, -0.000006)	(-0.00021, 0.000023)	(-0.000066, 0.000016)
## 92	(0.000067, 0.000112)	(-0.000112, 0.000025)	(0.000001, 0.000049)
## 93	(-0.000183, 0.007782)	(0.054324, 0.078745)	(-0.007986, 0.000583)
## 94	(0.000015, 0.000044)	(-0.000126, -0.000037)	(-0.000051, -0.000002)
## 95	(-0.000257, -0.000123)	(-0.000351, 0.000059)	(0.000027, 0.000171)
## 96	(-0.000067, -0.000021)	(-0.000085, 0.000057)	(0.000028, 0.000078)
## 97	(-0.003366, -0.000505)	(-0.003665, 0.005107)	(-0.000633, 0.002445)
## 98	(-0.000052, -0.000016)	(0.000012, 0.000121)	(0.000001, 0.000049)
## 99	(0.000028, 0.000064)	(-0.000117, -0.000008)	(-0.000062, -0.000024)
## 100	(-0.000131, 0.000008)	(-0.000325, 0.000322)	(0.000027, 0.000254)
## 101	(0.000022, 0.000058)	(-0.000135, -0.000026)	(-0.000065, -0.000027)
## 102	(-0.000047, -0.000009)	(0.000069, 0.000186)	(-0.000095, -0.000054)
## 103	(-0.000021, -0.000001)	(-0.000043, 0.000018)	(-0.000009, 0.000013)
## 104	(-0.000214, -0.000135)	(0.000024, 0.000266)	(-0.000013, 0.000072)
## 105	(-0.000561, -0.000251)	(-0.00194, -0.00099)	(0.000053, 0.000386)
## 106	(-0.000124, -0.000038)	(-0.000375, -0.000111)	(-0.000034, 0.000058)
## 107	(-0.000007, 0.000056)	(-0.000144, 0.000049)	(-0.000026, 0.000041)
## 108	(-0.000093, -0.000033)	(-0.000029, 0.000154)	(-0.000058, 0.000007)
## 109	(-0.000051, 0.000042)	(-0.00015, 0.000132)	(-0.000022, 0.000077)
## 110	(-0.000011, 0.000053)	(0.000079, 0.000275)	(-0.000009, 0.000006)
## 111	(-0.000405, 0.000069)	(-0.000143, 0.001308)	(-0.000873, -0.000364)
## 112	(-0.000007, 0.000014)	(-0.000094, -0.000029)	(-0.000031, -0.000008)
## 113	(0.000008, 0.000029)	(0.000006, 0.000007)	(-0.000027, -0.000005)
## 114	(-0.000005, 0.000055)	(-0.000213, -0.000029)	(-0.000043, 0.000021)
## 115	(-0.000034, -0.000002)	(-0.000029, 0.000066)	(0.000042, 0.000076)
## 116	(0.000211, 0.000091)	(-0.002995, -0.000853)	(-0.001498, -0.000747)
## 117	(-0.000004, 0.000026)	(-0.000102, -0.000011)	(-0.000031, 0.000001)
## 118	(-0.000143, -0.000009)	(-0.000599, -0.000187)	(-0.000014, 0.000131)
## 119	(-0.000051, 0.000045)	(-0.000149, 0.000143)	(-0.000092, 0.000001)
## 120	(0.000001, 0.000115)	(-0.000466, -0.000147)	(-0.000009, 0.000023)
## 121	(0.000516, 0.001448)	(0.000022, 0.002881)	(-0.000089, 0.000113)
## 122	(-0.000034, 0.000003)	(0.000045, 0.000157)	(-0.000025, 0.000014)
## 123	(0.000045, 0.000009)	(-0.000032, 0.000107)	(-0.000003, 0.000019)
## 124	(0.000004, 0.000047)	(0.000024, 0.000155)	(-0.000068, -0.000022)
## 125	(-0.000168, 0.000004)	(-0.00025, 0.000276)	(0.000004, 0.000189)
## 126	(0.000144, 0.000199)	(0.000002, 0.00017)	(-0.000047, 0.000012)
## 127	(-0.00007, -0.000033)	(-0.0001, 0.000014)	(0.000019, 0.000059)
## 128	(-0.000059, -0.000021)	(0.000062, 0.000176)	(-0.000074, -0.000034)
## 129	(0.000031, 0.000078)	(-0.000164, -0.000022)	(-0.000011, 0.000039)
## 130	(-0.000024, 0.000048)	(-0.000197, 0.000025)	(-0.000084, -0.000006)
## 131	(-0.001984, -0.000552)	(-0.004131, 0.000258)	(-0.001366, 0.000173)
## 132	(-0.000063, -0.000018)	(-0.000182, -0.000044)	(-0.000024, 0.000025)
## 133	(-0.000083, -0.000017)	(0.000025, 0.000225)	(0.000002, 0.000091)
## 134	(-0.000058, 0.000038)	(-0.000008, 0.000288)	(-0.000056, 0.000048)

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## 135 (-0.000039, 0.000055) (-0.000209, 0.000082) (-0.00025, -0.000148)
## 136 (0.003352, 0.008103) (-0.017682, -0.003113) (-0.00222, 0.002893)
## 137 (0.000024, 0.000082) (0.000214, 0.000392) (-0.000022, 0.00004)
## 138 (-0.000091, -0.00006) (-0.000108, -0.000012) (0.000009, 0.000042)
## 139 (-0.000102, -0.000031) (-0.000027, 0.000189) (0.000009, 0.000085)
## 140 (-0.000115, 0.000651) (0.00022, 0.002567) (-0.000107, 0.000716)
## 141 (-0.000053, -0.000021) (-0.000024, 0.000075) (-0.000017, 0.000018)
## 142 (-0.000015, 0.000012) (-0.000138, -0.000054) (-0.000023, 0.000007)
## 143 (-0.000022, 0.000019) (0.000056, 0.000181) (-0.000033, 0.000011)
## 144 (0.000251, 0.000327) (0.000066, 0.000296) (0.000017, 0.000098)
## 145 (-0.000153, -0.000097) (-0.000109, 0.000063) (-0.000103, -0.000043)
## 146 (0.000005, 0.000187) (0.000206, 0.000763) (0.000032, 0.000228)
## 147 (0.000047, 0.000187) (-0.000425, 0.000006) (-0.000128, 0.000023)
## 148 (0.000556, 0.000892) (-0.001404, -0.000374) (-0.000174, 0.000187)
## 149 (-0.000034, 0.000049) (-0.000077, 0.000175) (-0.000017, 0.000071)
## 150 (-0.000032, 0.000004) (-0.000132, -0.000021) (-0.000023, 0.000016)
## 151 (0, 0.000027) (0.000016, 0.000102) (-0.00003, 0)
## 152 (-0.000013, 0.000019) (-0.000126, -0.000027) (-0.000005, 0.000029)
## 153 (-0.000106, 0.000052) (-0.000161, 0.000325) (-0.000016, 0.000155)
## 154 (-0.000054, -0.000001) (-0.000066, 0.000095) (-0.000067, -0.00001)
## 155 (-0.000063, -0.000004) (-0.000068, 0.000113) (-0.000051, 0.000013)
## 156 (0.000051, 0.000081) (-0.000103, -0.000011) (0.000021, 0.000053)
## 157 (-0.000005, 0.000028) (-0.0002, -0.000098) (0.000019, 0.000054)
## 158 (-0.000004, 0.000047) (-0.000067, 0.000093) (-0.000068, -0.000012)
## 159 (-0.001235, 0.032222) (-0.010694, 0.091887) (0.006652, 0.042648)
## 160 (-0.000169, -0.000101) (0.000068, 0.000276) (-0.000038, 0.000035)
## 161 (-0.000373, 0.000028) (-0.000595, 0.000633) (-0.000867, -0.000437)
## 162 (-0.000057, 0.00003) (0.000046, 0.000312) (-0.000122, -0.000029)
## 163 (-0.000104, -0.000038) (0.000015, 0.000216) (0.000065, 0.000136)
## 164 (-0.000135, -0.000065) (-0.000003, 0.000212) (-0.000009, 0.000066)
## 165 (0.000223, 0.000305) (-0.000108, 0.000143) (-0.000036, 0.000052)
## 166 (-0.00015, -0.000048) (-0.000093, 0.000222) (-0.00015, -0.000039)
## 167 (-0.000064, 0.000166) (0.000635, 0.001338) (-0.000318, -0.000072)
## 168 (-0.00001, 0.000029) (-0.000015, 0.000106) (0.000023, 0.000066)
## 169 (-0.00001, 0.000058) (0.000156, 0.000366) (-0.000052, 0.000022)
## 170 (0.00002, 0.000058) (-0.000026, 0.000094) (-0.000067, -0.000025)
## 171 (-0.000029, -0.000002) (-0.000032, 0.000049) (-0.000019, 0.000009)
## 172 (0.0001, 0.000148) (0.000054, 0.000199) (-0.00003, 0.000021)
## 173 (-0.000144, 0.000016) (-0.000476, 0.000015) (-0.000065, 0.000107)
## 174 (-0.000134, 0.000077) (-0.000634, 0.000016) (0.000244, 0.000472)
## 175 (0.00008, 0.000161) (-0.000104, 0.000144) (-0.00007, 0.000017)
## 176 (-0.000029, 0.000001) (-0.000064, 0.000028) (-0.000028, 0.000005)
## 177 (-0.001659, 0.00238) (-0.001986, 0.010396) (0.001144, 0.005489)
## 178 (-0.00931, 0.033542) (-0.021248, 0.110138) (-0.012877, 0.033227)
## 179 (-0.00003, -0.000006) (-0.000003, 0.00007) (-0.000011, 0.000015)

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## 180	(-0.007115, 0.006692)	(-0.033953, 0.00838)	(-0.009238, 0.005617)
## 181	(0.000036, 0.000065)	(-0.000073, 0.000016)	(-0.000024, 0.000007)
## 182	(0.009029, 0.027809)	(-0.055498, 0.002081)	(0.0274, 0.047604)
## 183	(0.000089, 0.00013)	(-0.000164, -0.000037)	(0.000023, 0.000067)
## 184	(0.00966, 0.019888)	(-0.032836, -0.001476)	(0.003477, 0.014481)
## 185	(0.000107, 0.001228)	(-0.001889, 0.001546)	(-0.000869, 0.000337)
## 186	(0.018554, 0.024882)	(0.013263, 0.032663)	(-0.000589, 0.006219)
## 187	(-0.000575, 0.005117)	(0.014276, 0.031727)	(-0.002202, 0.003922)
## 188	(-0.000069, -0.000007)	(-0.000175, 0.000017)	(-0.00003, 0.000038)
## 189	(0.000033, 0.000098)	(-0.000223, -0.000023)	(-0.000067, 0.000003)
## 190	(0.000026, 0.000119)	(0.000057, 0.000342)	(-0.000107, -0.000007)
## 191	(-0.000261, -0.000004)	(-0.000627, 0.000159)	(-0.000192, 0.000084)
## 192	(0.000029, 0.000083)	(-0.000075, 0.000089)	(-0.00004, 0.000018)
## 193	(-0.000017, 0.000226)	(-0.000258, 0.000485)	(-0.000079, 0.000182)
## 194	(0.003907, 0.006793)	(0.008194, 0.017041)	(-0.00354, -0.000435)
## 195	(-0.000069, 0.000021)	(0.000264, 0.00054)	(-0.000048, 0.000049)
## 196	(0.044205, 0.11121)	(-0.052557, 0.152887)	(-0.068172, 0.003919)
## 197	(0.05187, 0.119042)	(-0.128567, 0.077389)	(0.005752, 0.078023)
## 198	(-0.000064, 0.000015)	(0.000032, 0.000273)	(-0.000016, 0.000069)
## 199	(-0.000058, -0.000003)	(-0.00013, 0.00004)	(-0.000051, 0.000008)
## 200	(0.000093, 0.000181)	(-0.000444, -0.000175)	(-0.000076, 0.000018)
## 201	(-0.00002, 0.000007)	(0.00004, 0.000125)	(-0.00004, -0.00001)
## 202	(-0.000036, 0.000022)	(0.000137, 0.000316)	(-0.000031, 0.000032)
## 203	(0.000007, 0.000059)	(-0.000234, -0.000076)	(0.000037, 0.000093)
## 204	(-0.000034, 0.000003)	(0.000053, 0.000168)	(-0.000043, -0.000002)
## 205	(0.00076, 0.015906)	(-0.015929, 0.03051)	(0.000873, 0.017169)
## 206	(-0.000017, 0.000014)	(-0.000014, 0.000083)	(0.000005, 0.000039)
## 207	(-0.000111, 0.000211)	(-0.000423, 0.000565)	(-0.000196, 0.000151)
## 208	(0.000036, 0.000075)	(0.00007, 0.000188)	(-0.000015, 0.000027)
## 209	(0.004185, 0.059323)	(0.049084, 0.218142)	(0.04782, 0.107143)
## 210	(-0.001129, -0.000091)	(-0.002285, 0.000899)	(-0.000009, 0.001109)
## 211	(-0.000274, -0.000168)	(-0.000209, 0.000115)	(-0.000023, 0.00009)
## 212	(-0.000038, 0.000006)	(-0.000104, 0.000032)	(-0.000077, -0.000029)
## 213	(-0.110042, 0.009329)	(0.039695, 0.405696)	(0.022712, 0.151143)
## 214	(0.000014, 0.000698)	(0.000617, 0.002715)	(-0.00078, -0.000044)
## 215	(-0.00007, -0.000004)	(-0.000156, 0.000046)	(-0.000063, 0.000008)
## 216	(0.005161, 0.022633)	(-0.013314, 0.040256)	(0.000021, 0.01882)
## 217	(-0.011643, 0.003173)	(0.005693, 0.051122)	(-0.014659, 0.001282)
## 218	(0.00005, 0.00009)	(-0.000059, 0.000065)	(-0.000012, 0.000032)
## 219	(-0.000025, 0.00001)	(-0.000003, 0.000105)	(0.000013, 0.00005)
## 220	(-0.02364, 0.153888)	(0.377194, 0.921509)	(0.279535, 0.470538)
## 221	(0.000019, 0.000057)	(-0.000075, 0.000043)	(0.000043, 0.000084)
## 222	(-0.000009, 0.000019)	(-0.000068, 0.000016)	(-0.000012, 0.000017)
## 223	(0.000056, 0.000094)	(-0.000178, -0.000063)	(0.000019, 0.000059)
## 224	(-0.000127, -0.000066)	(-0.000225, -0.000038)	(-0.000073, -0.000008)

## 225	(-0.000005, 0.000059)	(-0.000021, 0.000176)	(-0.000059, 0.000001)
## 226	(-0.000008, 0.000003)	(0.000048, 0.000164)	(-0.000029, 0.000012)
## 227	(-0.000065, 0.000044)	(-0.000038, 0.000295)	(-0.000045, 0.000072)
## 228	(-0.000056, -0.000012)	(-0.000016, 0.000119)	(-0.000034, 0.000013)
## 229	(-0.000718, 0.000183)	(-0.003998, -0.001236)	(0.000025, 0.000995)
## 230	(-0.000072, 0.000096)	(0.000435, 0.00095)	(0.000064, 0.000245)
## 231	(-0.000275, -0.000103)	(-0.000238, 0.000292)	(-0.000341, -0.000155)
## 232	(0.000007, 0.000051)	(0.000008, 0.000142)	(-0.000005, 0.000042)
## 233	(-0.000041, 0.000008)	(0.000023, 0.000173)	(0.000003, 0.000056)
## 234	(-0.038706, 0.006742)	(-0.032183, 0.107162)	(0.008276, 0.057173)
## 235	(0.001488, 0.004401)	(-0.011399, -0.002467)	(-0.000638, 0.002496)
## 236	(-0.000042, 0.000005)	(-0.000398, -0.000253)	(0.000045, 0.000096)
## 237	(-0.000227, 0.000374)	(-0.000007, 0.001837)	(-0.000521, 0.000126)
## 238	(-0.000034, 0.000013)	(0.000114, 0.000258)	(-0.000046, 0.000005)
## 239	(0.000103, 0.000167)	(0.000029, 0.000227)	(-0.000024, 0.000045)
## 240	(0.000009, 0.000178)	(-0.000296, 0.000223)	(-0.000375, -0.000193)
## 241	(-0.001432, 0.005619)	(-0.043178, -0.02156)	(-0.00845, -0.000864)
## 242	(-0.000049, 0.000007)	(0.000062, 0.000235)	(-0.000072, -0.000012)
## 243	(0.000254, 0.000881)	(-0.000635, 0.001286)	(-0.000677, -0.000003)
## 244	(-0.000005, 0.000004)	(-0.000257, -0.000118)	(-0.000072, -0.000024)
## 245	(-0.000199, 0.000061)	(0.000658, 0.001457)	(-0.000085, 0.000195)
## 246	(-0.000105, 0.000007)	(0.000752, 0.001095)	(-0.000102, 0.000019)
## 247	(0, 0.000005)	(-0.00009, 0.000063)	(-0.000034, 0.000002)
## 248	(0.001278, 0.006896)	(-0.019174, -0.001949)	(0.001196, 0.00724)
## 249	(-0.017723, 0.007063)	(0.006509, 0.082503)	(0.009627, 0.036294)
## 250	(-0.101593, 0.104425)	(-0.449442, 0.182225)	(-0.087773, 0.133881)
## 251	(-0.003296, 0.025919)	(-0.051147, 0.038429)	(-0.027641, 0.003791)
## 252	(-0.000109, -0.000009)	(0.000174, 0.000482)	(-0.000051, 0.000057)
## 253	(-0.009486, -0.003405)	(-0.007635, 0.011011)	(-0.005888, 0.000655)
## 254	(-0.000253, 0.000565)	(-0.005239, -0.002729)	(-0.000282, 0.000598)
## 255	(-0.00003, 0.000001)	(-0.000121, 0.000001)	(-0.000084, -0.000041)
## 256	(-0.000038, 0.000051)	(-0.00012, 0.000156)	(-0.000038, 0.000059)
## 257	(-0.000145, 0.002627)	(-0.011697, -0.003197)	(-0.001514, 0.001469)
## 258	(-0.000018, 0.000002)	(-0.000086, 0.000003)	(-0.000033, 0.000008)
## 259	(-0.000029, 0.000026)	(-0.000079, 0.000092)	(0.000021, 0.000081)
## 260	(0.013483, 0.039194)	(-0.079544, -0.000713)	(0.027597, 0.055259)
## 261	(-0.000148, -0.000052)	(-0.000365, -0.00007)	(-0.000053, 0.000005)
## 262	(-0.000276, -0.000022)	(-0.000378, 0.0004)	(-0.000157, 0.000116)
## 263	(-0.004108, 0.000239)	(-0.003786, 0.009543)	(-0.000071, 0.004606)
## 264	(-0.000069, -0.000023)	(-0.000189, -0.000046)	(-0.000039, 0.000012)
## 265	(-0.000029, 0.000059)	(-0.000379, -0.000108)	(-0.000093, 0.000002)
## 266	(-0.000061, 0.000029)	(-0.000082, 0.000192)	(0.000052, 0.000148)
## 267	(-0.000109, 0.010362)	(-0.009387, 0.022716)	(-0.010558, 0.000707)
## 268	(0.006005, 0.016735)	(0.002085, 0.034987)	(0.008796, 0.020341)
## 269	(-0.000027, 0.000019)	(-0.000165, -0.000024)	(-0.000013, 0.000036)

## 270	(0.000253, 0.00054)	(0.000079, 0.000959)	(-0.000091, 0.000218)
## 271	(0.000023, 0.000102)	(-0.000024, 0.000218)	(-0.000025, 0.00006)
## 272	(-0.000321, -0.000029)	(-0.000141, 0.000755)	(-0.000259, 0.000056)
## 273	(-0.000086, 0.000002)	(0.000231, 0.0005)	(-0.000105, -0.000011)
## 274	(0.022792, 0.041285)	(0.088662, 0.145363)	(-0.015085, 0.004811)
## 275	(0.000026, 0.000376)	(-0.000579, 0.000497)	(-0.000017, 0.00036)
## 276	(-0.010944, 0.021029)	(-0.023938, 0.074095)	(-0.021757, 0.012644)
## 277	(-0.000232, 0.00071)	(-0.004232, -0.001345)	(0.000801, 0.001814)
## 278	(0.000109, 0.000189)	(-0.00034, -0.000096)	(-0.000016, 0.00007)
## 279	(-0.000061, 0.000424)	(-0.001192, 0.000294)	(-0.000086, 0.000435)
## 280	(0.000142, 0.001116)	(-0.001541, 0.001444)	(-0.000318, 0.00073)
## 281	(-0.000033, 0.00004)	(-0.000228, -0.000003)	(-0.000102, -0.000023)
## 282	(-0.001907, 0.001941)	(-0.006124, 0.005674)	(-0.003867, 0.000273)
## 283	(0.001275, 0.014775)	(0.042731, 0.084126)	(-0.012461, 0.002064)
## 284	(0.000297, 0.000957)	(-0.001431, 0.000592)	(-0.000792, -0.000083)
## 285	(0.000018, 0.000104)	(0.000075, 0.000339)	(-0.000041, 0.000052)
## 286	(-0.000527, 0.000465)	(-0.005606, -0.002567)	(-0.000572, 0.000494)
## 287	(-0.000211, -0.000022)	(-0.000892, -0.000315)	(-0.00005, 0.000153)
## 288	(-0.000614, -0.000344)	(-0.001321, -0.000495)	(-0.000189, 0.000101)
## 289	(-0.000015, 0.000017)	(-0.00003, 0.000068)	(-0.000072, -0.000038)
## 290	(-0.00003, 0.000094)	(0.000252, 0.000631)	(-0.000004, 0.000129)
## 291	(0.000101, 0.000347)	(-0.001076, -0.000323)	(-0.000273, -0.000009)
## 292	(-0.000495, 0.000187)	(-0.000917, 0.001176)	(-0.000789, -0.000055)
## 293	(-0.000008, 0.000403)	(-0.000437, 0.000822)	(-0.000457, -0.000016)
## 294	(-0.000064, 0.000091)	(-0.000311, 0.000163)	(-0.000071, 0.000095)
## 295	(-0.00011, 0.00015)	(-0.000834, -0.000035)	(-0.000159, 0.000121)
## 296	(-0.000245, 0.000371)	(-0.001486, 0.000404)	(-0.000413, 0.00025)
## 297	(-0.00007, 0.00015)	(-0.000224, 0.00045)	(-0.000101, 0.000135)
## 298	(0.000918, 0.00478)	(-0.012078, -0.000235)	(-0.001219, 0.002937)
## 299	(-0.000299, 0.000648)	(-0.001687, 0.001219)	(-0.000163, 0.000857)
## 300	(-0.000165, 0.000121)	(-0.000345, 0.000531)	(-0.000206, 0.000101)
## 301	(-0.00003, 0.000233)	(-0.000322, 0.000484)	(-0.00018, 0.000103)
## 302	(-0.000072, 0.001711)	(-0.003432, 0.002034)	(-0.001612, 0.000306)
## 303	(-0.000139, 0.000212)	(-0.000469, 0.000608)	(0.000064, 0.000442)
## 304	(-0.000881, 0.000154)	(-0.003073, 0.000103)	(0.000088, 0.001203)
## 305	(-0.000159, 0.000136)	(-0.000853, 0.000051)	(-0.000365, -0.000048)
## 306	(-0.000606, 0.000523)	(-0.000382, 0.003079)	(-0.000077, 0.001137)
## 307	(-0.002248, 0.00166)	(-0.006931, 0.005051)	(-0.001632, 0.002572)
## 308	(-0.000671, 0.00308)	(-0.011884, -0.000381)	(-0.004278, -0.000241)
## 309	(-0.000252, 0.000046)	(-0.000426, 0.000485)	(-0.000119, 0.000201)
## 310	(-0.000186, 0.000093)	(-0.000448, 0.00041)	(-0.000209, 0.000092)
## 311	(-0.000236, 0.001997)	(-0.004326, 0.002523)	(-0.003161, -0.000757)
## 312	(-0.001221, 0.001788)	(-0.007547, 0.00168)	(-0.001373, 0.001864)
## 313	(-0.000622, 0.000733)	(-0.002412, 0.001743)	(-0.001082, 0.000376)
## 314	(-0.000625, -0.000126)	(-0.00175, -0.000218)	(0.000245, 0.000782)

## 315	(-0.000111, 0.000019)	(-0.000364, 0.000035)	(-0.000013, 0.000128)
## 316	(-0.000067, 0.00022)	(-0.000859, 0.000022)	(-0.000016, 0.000293)
## 317	(-0.000111, 0.000085)	(-0.000374, 0.000229)	(-0.0001, 0.000112)
## 318	(-0.000086, 0.000724)	(-0.002431, 0.000051)	(-0.000118, 0.000753)
## 319	(-0.000086, 0.000067)	(-0.00035, 0.000119)	(-0.000136, 0.000028)
## 320	(-0.000028, 0.000541)	(-0.000953, 0.000792)	(-0.000507, 0.000105)
## 321	(-0.005162, 0.001327)	(-0.019135, 0.000759)	(-0.005033, 0.001947)
## 322	(-0.001878, 0.001357)	(-0.006442, 0.003475)	(-0.002803, 0.000677)
## 323	(-0.000769, -0.000262)	(-0.00129, 0.000264)	(-0.0007, -0.000155)
## 324	(-0.001405, 0.00108)	(-0.003034, 0.004586)	(-0.000707, 0.001967)
## 325	(0.000004, 0.000902)	(-0.001017, 0.001738)	(-0.001226, -0.000259)
## 326	(-0.000229, 0.000031)	(-0.000509, 0.000287)	(-0.000102, 0.000177)
## 327	(-0.012928, 0.043541)	(-0.096666, 0.076473)	(-0.043617, 0.017138)
## 328	(-0.0004, 0.000233)	(-0.000477, 0.001465)	(-0.000653, 0.000028)
## 329	(-0.000696, 0.000274)	(-0.000073, 0.002901)	(0.000782, 0.001825)
## 330	(-0.001468, 0.001313)	(0.000044, 0.008573)	(0.00078, 0.003772)
## 331	(0.000756, 0.005855)	(-0.007775, 0.007859)	(-0.004346, 0.00114)
## 332	(-0.000062, 0.000106)	(-0.000587, -0.000071)	(-0.000011, 0.00017)
## 333	(-0.00003, 0.000039)	(-0.000067, 0.000147)	(-0.000059, 0.000017)
## 334	(-0.000132, 0.000161)	(0.000094, 0.000994)	(-0.000071, 0.000244)
## 335	(-0.000001, 0.000376)	(-0.000432, 0.000725)	(-0.000117, 0.000289)
## 336	(-0.000416, 0.000456)	(-0.001564, 0.001107)	(-0.0006, 0.000337)
## 337	(-0.000533, -0.00013)	(0.000297, 0.001531)	(-0.000096, 0.000337)
## 338	(-0.000975, 0.00516)	(-0.009399, 0.009412)	(-0.005077, 0.001523)
## 339	(-0.000983, 0.000211)	(-0.001417, 0.002246)	(0.000011, 0.001296)
## 340	(-0.000117, 0.000116)	(0.000062, 0.000776)	(-0.000076, 0.000175)
## 341	(-0.001404, 0.000577)	(-0.004689, 0.001382)	(-0.002174, -0.000043)
## 342	(-0.000154, 0.00037)	(-0.000177, 0.001431)	(-0.000237, 0.000327)
## 343	(-0.000275, 0.000021)	(-0.001159, -0.000251)	(-0.000281, 0.000038)
## 344	(-0.000652, 0.001562)	(-0.005114, 0.001673)	(-0.001423, 0.000959)
## 345	(-0.000367, 0.000512)	(-0.000917, 0.00178)	(-0.00129, -0.000343)
## 346	(-0.000094, -0.000006)	(0.000028, 0.000297)	(-0.000044, 0.000051)
## 347	(-0.000595, 0.000175)	(-0.001578, 0.000785)	(-0.000231, 0.000598)
## 348	(-0.003666, -0.000678)	(-0.003824, 0.005335)	(-0.000454, 0.00276)
## 349	(-0.000053, 0.000102)	(-0.000306, 0.000169)	(-0.000032, 0.000135)
## 350	(0.00004, 0.00022)	(0.000231, 0.000782)	(-0.000052, 0.000141)
## 351	(-0.00007, 0.000697)	(-0.001446, 0.000906)	(-0.000835, -0.00001)
## 352	(-0.000046, 0.000121)	(-0.000648, -0.000136)	(-0.000144, 0.000036)
## 353	(-0.000366, 0.000299)	(-0.001821, 0.000218)	(-0.000169, 0.000546)
## 354	(-0.000446, 0.000867)	(-0.004438, -0.000413)	(-0.001898, -0.000486)
## 355	(-0.000211, 0.000251)	(-0.000142, 0.001275)	(0.000063, 0.00056)
## 356	(0.07414, 0.495036)	(0.050407, 1.340906)	(0.453186, 0.906028)
## 357	(-0.001653, 0.001855)	(-0.009822, 0.000933)	(-0.005136, -0.001362)
## 358	(-0.000334, 0.000442)	(-0.00221, 0.000168)	(-0.001176, -0.000342)
## 359	(-0.001326, 0.002332)	(-0.008121, 0.003094)	(-0.002746, 0.001189)

## 360	(-0.000212, 0.000022)	(0.000427, 0.001144)	(0.000119, 0.00037)
## 361	(-0.00117, -0.000314)	(-0.001502, 0.001123)	(0.000389, 0.00131)
## 362	(-0.000105, 0.005403)	(-0.009201, 0.007689)	(-0.001979, 0.003948)
## 363	(-0.000886, 0.000653)	(0.001888, 0.006608)	(-0.000385, 0.001271)
## 364	(-0.000678, 0.00151)	(-0.004127, 0.002582)	(0.000228, 0.002582)
## 365	(-0.000056, 0.000134)	(-0.000395, 0.000189)	(0.000043, 0.000248)
## 366	(0.000022, 0.000744)	(-0.001168, 0.001048)	(-0.000531, 0.000247)
## 367	(-0.000084, 0.000238)	(-0.001258, -0.000269)	(-0.000369, -0.000022)
## 368	(-0.001498, 0.000063)	(-0.002889, 0.001897)	(-0.000597, 0.001083)
## 369	(-0.001642, -0.000374)	(-0.00308, 0.000806)	(0.000108, 0.001472)
## 370	(-0.000267, 0.00004)	(-0.00022, 0.00072)	(-0.000195, 0.000135)
## 371	(-0.000726, 0.000043)	(-0.002051, 0.000307)	(0.000518, 0.001346)
## 372	(-0.000872, 0.000213)	(-0.002005, 0.00132)	(-0.000418, 0.000749)
## 373	(-0.000005, 0.000224)	(-0.000486, 0.000217)	(-0.000013, 0.000234)
## 374	(-0.000621, 0.000717)	(-0.001129, 0.002973)	(-0.000816, 0.000624)
## 375	(-0.00031, 0.000263)	(-0.000334, 0.001422)	(-0.000382, 0.000235)
## 376	(0.000078, 0.00024)	(-0.000056, 0.000439)	(-0.000077, 0.000097)
## 377	(-0.002012, -0.000029)	(0.002673, 0.008754)	(-0.001244, 0.000889)
## 378	(-0.000287, 0.000319)	(-0.000535, 0.001323)	(-0.000398, 0.000254)
## 379	(-0.001622, 0.007724)	(-0.008585, 0.020072)	(-0.003474, 0.006582)
## 380	(-0.000144, 0.000031)	(-0.000609, -0.00007)	(-0.000172, 0.000017)
## 381	(-0.000356, 0.000083)	(-0.000378, 0.000969)	(-0.000468, 0.000005)
## 382	(0.000006, 0.000334)	(-0.000477, 0.00053)	(-0.000386, -0.000033)
## 383	(-0.000133, 0.00006)	(-0.000162, 0.00043)	(-0.000195, 0.000013)
## 384	(0.000031, 0.000185)	(-0.000269, 0.000202)	(0.000097, 0.000262)
## 385	(-0.00016, 0.000177)	(-0.000988, 0.000047)	(-0.000128, 0.000235)
## 386	(-0.00114, 0.000399)	(-0.00128, 0.00344)	(-0.000817, 0.000839)
## 387	(-0.000844, 0.002771)	(0.003555, 0.014638)	(-0.000354, 0.003535)
## 388	(-0.000569, 0.000178)	(-0.00007, 0.002222)	(-0.000219, 0.000585)
## 389	(-0.001842, 0.002047)	(0.004878, 0.016803)	(-0.003494, 0.000691)
## 390	(-0.000925, 0.000253)	(-0.003306, 0.000306)	(-0.000428, 0.00084)
## 391	(-0.001525, 0.000291)	(-0.00176, 0.00381)	(0.001104, 0.003058)
## 392	(0.000483, 0.002377)	(-0.005492, 0.000314)	(-0.002285, -0.000248)
## 393	(-0.000047, 0.000141)	(-0.000357, 0.000221)	(-0.000181, 0.000022)
## 394	(-0.000052, 0.000058)	(-0.000332, 0.000006)	(-0.000034, 0.000085)
## 395	(-0.001596, 0.000024)	(-0.00086, 0.004108)	(-0.001376, 0.000367)
## 396	(-0.000262, 0.000087)	(-0.000538, 0.000533)	(-0.000371, 0.000005)
## 397	(-0.003049, -0.000272)	(0.00201, 0.010527)	(-0.000214, 0.002775)
## 398	(-0.000194, 0.000179)	(-0.000703, 0.00044)	(-0.000602, -0.000201)
## 399	(-0.000445, 0.000754)	(-0.000736, 0.002941)	(-0.001846, -0.000556)
## 400	(-0.000164, 0.000406)	(-0.000161, 0.001587)	(-0.00036, 0.000253)
## 401	(-0.000144, 0.00006)	(0.000187, 0.000811)	(-0.000036, 0.000183)
## 402	(-0.002424, 0.001281)	(-0.004387, 0.006973)	(-0.000242, 0.003745)
## 403	(-0.000175, -0.000011)	(-0.000002, 0.000503)	(-0.000158, 0.00002)
## 404	(-0.000195, 0.000196)	(-0.000446, 0.000752)	(-0.000365, 0.000055)

## 405	(-0.000402, -0.000075)	(0.000126, 0.001129)	(-0.000435, -0.000083)
## 406	(-0.001265, 0.00059)	(-0.005669, 0.000019)	(-0.001748, 0.000248)
## 407	(0.000045, 0.000231)	(0.000183, 0.000753)	(-0.0001, 0.0001)
## 408	(-0.000235, 0.000117)	(-0.000075, 0.001004)	(-0.000677, -0.000298)
## 409	(-0.000185, 0.000133)	(-0.001503, -0.000529)	(-0.000261, 0.00008)
## 410	(-0.003286, -0.000246)	(-0.004742, 0.00458)	(-0.000862, 0.002409)
## 411	(-0.000897, 0.005161)	(0.000763, 0.019336)	(-0.00055, 0.005967)
## 412	(-0.00006, 0.000188)	(-0.000446, 0.000316)	(-0.000166, 0.000101)
## 413	(0.000024, 0.000279)	(-0.00028, 0.0005)	(-0.000114, 0.00016)
## 414	(-0.000003, 0.000162)	(-0.000316, 0.000189)	(0.000042, 0.000219)
## 415	(-0.00004, 0.000106)	(-0.000279, 0.000166)	(0.000148, 0.000304)
## 416	(-0.020796, 0.010743)	(-0.053601, 0.0431)	(-0.002045, 0.031888)
## 417	(-0.000298, 0.000232)	(-0.000549, 0.001074)	(-0.000437, 0.000132)
## 418	(-0.000562, 0.000024)	(-0.001933, -0.000138)	(-0.000392, 0.000238)
## 419	(-0.00051, 0.000963)	(-0.000311, 0.004205)	(0.000519, 0.002104)
## 420	(-0.000805, 0.002451)	(-0.010319, -0.000336)	(-0.003464, 0.000039)
## 421	(-0.000417, -0.000133)	(-0.000735, 0.000135)	(-0.000069, 0.000236)
## 422	(-0.000101, 0.000163)	(-0.000189, 0.000621)	(0.000133, 0.000417)
## 423	(-0.000319, 0.000218)	(-0.00121, 0.000434)	(-0.000009, 0.000568)
## 424	(-0.001238, 0.001223)	(-0.005673, 0.00187)	(0.00055, 0.003197)
## 425	(0.000005, 0.002894)	(0.002952, 0.011812)	(-0.003484, -0.000375)
## 426	(-0.001255, 0.000656)	(-0.003683, 0.002179)	(-0.001135, 0.000922)
## 427	(-0.00278, -0.000433)	(-0.004975, 0.002223)	(-0.001288, 0.001238)
## 428	(-0.0012, -0.000383)	(-0.000578, 0.001926)	(-0.000418, 0.00046)
## 429	(-0.001721, 0.001235)	(-0.006575, 0.002488)	(-0.001457, 0.001723)
## 430	(0.000252, 0.000623)	(-0.000002, 0.001134)	(0.00006, 0.000459)
## 431	(-0.000094, 0.00006)	(-0.000182, 0.000291)	(-0.000105, 0.000061)
## 432	(0.000011, 0.000145)	(-0.000301, 0.000109)	(-0.000169, -0.000025)
## 433	(-0.000426, 0.003183)	(0.000543, 0.011609)	(-0.002451, 0.001432)
## 434	(0.000022, 0.000597)	(-0.001607, 0.000154)	(-0.000371, 0.000247)
## 435	(0.000014, 0.000152)	(0.000075, 0.000498)	(-0.000118, 0.00003)
## 436	(-0.000216, 0.000082)	(-0.000532, 0.000381)	(-0.000079, 0.000241)
## 437	(0.000009, 0.000687)	(0.001021, 0.003101)	(-0.000665, 0.000065)
## 438	(0.000116, 0.001643)	(-0.002039, 0.002643)	(-0.001177, 0.000466)
## 439	(-0.068913, 0.059874)	(-0.482484, -0.087613)	(-0.0704, 0.068163)
## 440	(-0.000407, 0.001056)	(-0.000199, 0.004285)	(-0.00103, 0.000543)
## 441	(-0.003837, 0.000704)	(0.000137, 0.014062)	(-0.005367, -0.000481)
## 442	(-0.000752, 0.00087)	(-0.006504, -0.001532)	(-0.001253, 0.000492)
## 443	(0.000173, 0.001143)	(-0.001676, 0.001299)	(-0.001003, 0.000041)
## 444	(0.000007, 0.000157)	(-0.000342, 0.000117)	(-0.000068, 0.000093)
## 445	(-0.000988, 0.001138)	(-0.001239, 0.005278)	(-0.001807, 0.00048)
## 446	(-0.002305, 0.000618)	(-0.007552, 0.001412)	(-0.000678, 0.002467)
## 447	(-0.003149, 0.001652)	(-0.005921, 0.008801)	(-0.00274, 0.002426)
## 448	(-0.000077, 0.000203)	(-0.000282, 0.000576)	(-0.000293, 0.000008)
## 449	(-0.001005, 0.000259)	(-0.005068, -0.001191)	(-0.000516, 0.000845)

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## 450 (-0.00016, -0.000012) (-0.000192, 0.000263) (0.000023, 0.000182)
## 451 (-0.00006, 0.000109) (-0.000659, -0.000143) (0.000013, 0.000194)
## 452 (-0.000374, -0.000178) (-0.000457, 0.000145) (0.000007, 0.000282)
## 453 (-0.002112, 0.00086) (-0.00943, -0.000317) (-0.001876, 0.001322)
## 454 (-0.000226, 0.001307) (-0.005518, -0.000816) (-0.000745, 0.000905)
## 455 (-0.001001, -0.000176) (-0.001126, 0.001402) (0.000108, 0.000996)
## 456 (-0.000224, 0.000054) (-0.000273, 0.000581) (-0.000165, 0.000134)
## 457 (-0.123842, 0.019706) (-0.407352, 0.032778) (-0.099757, 0.054687)
## 458 (-0.000043, 0.000041) (-0.00014, 0.000118) (-0.000047, 0.000044)
## 459 (-0.000125, 0.001265) (-0.000679, 0.003584) (0.000029, 0.001786)
## 460 (-0.000378, -0.000221) (-0.000447, 0.000036) (-0.000149, 0.000021)
## 461 (-0.053709, -0.007961) (0.078765, 0.219031) (-0.008692, 0.040528)
## 462 (0.000131, 0.000626) (-0.001503, 0.000016) (-0.000807, -0.000274)
## 463 (-0.000478, 0.000204) (0.000197, 0.002286) (0.000055, 0.000788)
## 464 (-0.000231, 0.000025) (-0.000765, 0.000023) (-0.000136, 0.00014)
## 465 (-0.000085, 0.000377) (-0.001348, 0.000067) (-0.000097, 0.0004)
## 466 (-0.000507, -0.000019) (0.000366, 0.001861) (-0.000082, 0.000443)
## 467 (-0.001441, 0.000361) (-0.005403, 0.000122) (-0.001664, 0.000275)
## 468 (-0.001229, 0.001533) (-0.001812, 0.006658) (-0.003609, -0.000637)
## 469 (-0.254828, 0.022083) (-0.988357, -0.139324) (-0.397923, -0.099994)
## 470 (-0.000218, 0.000649) (-0.000973, 0.001685) (-0.00012, 0.000813)
## 471 (-0.000415, 0.000416) (-0.001285, 0.001262) (-0.000149, 0.000745)
## 472 (-0.002142, 0.000203) (-0.006007, 0.001184) (-0.002013, 0.000511)
## 473 (0.000054, 0.000273) (-0.000906, -0.000234) (-0.000012, 0.000224)
## 474 (-0.000153, 0.000083) (-0.001588, -0.000866) (-0.000085, 0.000169)
## 475 (-0.000208, 0.002785) (-0.004271, 0.004907) (-0.001885, 0.001336)
## 476 (-0.000154, 0.000013) (-0.000202, 0.00031) (-0.000125, 0.000055)
## 477 (-0.062, 0.006375) (-0.014183, 0.19546) (-0.101328, -0.027763)
## 478 (-0.000052, 0.000071) (-0.000023, 0.000356) (-0.000079, 0.000054)
## 479 (-0.001554, 0.000764) (-0.002294, 0.004813) (-0.00124, 0.001253)
## 480 (-0.000191, 0.000127) (-0.000293, 0.000683) (-0.000092, 0.000251)
## 481 (-0.000488, 0.00308) (-0.013386, -0.002444) (-0.002297, 0.001542)
## 482 (-0.000207, 0.000476) (0.000331, 0.002427) (-0.001101, -0.000365)
## 483 (-0.000785, 0.000988) (-0.004452, 0.000985) (-0.001623, 0.000285)
## 484 (-0.269879, 0.050386) (-0.272576, 0.709381) (-0.32251, 0.022063)
## 485 (-0.001388, 0.00283) (-0.007101, 0.005834) (-0.00436, 0.000179)
## 486 (0.000046, 0.000448) (-0.000124, 0.001109) (-0.000343, 0.00009)
## 487 (-0.000124, 0.000559) (-0.000116, 0.001978) (-0.000178, 0.000557)
## 488 (-0.0014, 0.000517) (-0.00383, 0.00205) (-0.002388, -0.000325)
## 489 (-0.000112, 0.00064) (-0.000212, 0.002096) (-0.000769, 0.000041)
## 490 (-0.000182, 0.000098) (-0.000689, 0.000169) (-0.000176, 0.000125)
## 491 (-0.000409, 0.000201) (-0.000408, 0.00146) (-0.000654, 0.000002)
## 492 (-0.000063, 0.000359) (-0.000913, 0.000378) (0.000066, 0.000519)
## 493 (-0.001082, 0.000068) (-0.000438, 0.003086) (-0.000053, 0.001183)
## 494 (-0.000028, 0.000102) (-0.000433, -0.000036) (-0.000088, 0.000051)

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## 495	(-0.000157, 0.000753)	(-0.001627, 0.001163)	(-0.000296, 0.000683)
## 496	(0.000019, 0.000441)	(-0.000296, 0.000997)	(-0.000346, 0.000107)
## 497	(-0.000108, 0.003231)	(-0.003157, 0.00708)	(-0.001863, 0.00173)
## 498	(-0.000124, 0.000213)	(-0.001315, -0.000284)	(-0.000092, 0.00027)
## 499	(-0.000846, 0.002605)	(0.003704, 0.014287)	(-0.001885, 0.001828)
## 500	(-0.000584, 0.00088)	(-0.004282, 0.000209)	(-0.00095, 0.000626)
## 501	(0.000044, 0.000477)	(0.000836, 0.002166)	(-0.000533, -0.000067)
## 502	(-0.048342, 0.123103)	(-0.482597, 0.043068)	(-0.218469, -0.034011)
## 503	(-0.000492, 0.000453)	(0.00034, 0.003237)	(-0.000371, 0.000645)
## 504	(-0.002188, 0.000513)	(-0.010646, -0.002367)	(0.000137, 0.003043)
## 505	(-0.000223, 0.000442)	(-0.000605, 0.001434)	(0.000488, 0.001203)
## 506	(0.000145, 0.001034)	(-0.001259, 0.001466)	(-0.000718, 0.000238)
## 507	(-0.000355, 0.000141)	(-0.000594, 0.000924)	(-0.000255, 0.000278)
## 508	(-0.004436, 0.003324)	(0.023221, 0.047012)	(-0.000181, 0.008168)
## 509	(0.000032, 0.002285)	(-0.005903, 0.001004)	(0.000268, 0.002692)
## 510	(-0.002978, 0.001249)	(-0.006983, 0.005979)	(0.000607, 0.005156)
## 511	(-0.00027, 0.000414)	(-0.001009, 0.001088)	(-0.000151, 0.000585)
## 512	(-0.000979, 0.001939)	(-0.003897, 0.005048)	(-0.002049, 0.00109)
## 513	(0.000014, 0.000269)	(-0.000689, 0.000092)	(-0.000137, 0.000137)
## 514	(-0.000097, 0.000012)	(0.000013, 0.000347)	(0.000003, 0.000147)
## 515	(-0.001122, 0.001269)	(-0.004781, 0.002551)	(-0.000872, 0.001701)
## 516	(-0.00212, 0.003301)	(-0.013063, 0.003558)	(-0.001565, 0.004267)
## 517	(-0.000655, 0.000736)	(-0.00204, 0.002225)	(-0.001314, 0.000182)
## 518	(0.000234, 0.00097)	(-0.000428, 0.001831)	(-0.000693, 0.0001)
## 519	(-0.004155, 0.00575)	(-0.017191, 0.013179)	(-0.003849, 0.006808)
## 520	(-0.000111, 0.000844)	(-0.00208, 0.00085)	(-0.001546, -0.000518)
## 521	(-0.000087, 0.000066)	(-0.000133, 0.000335)	(-0.000005, 0.000115)
## 522	(-0.046008, 0.052407)	(-0.004634, 0.297114)	(-0.108851, -0.002966)
## 523	(-0.000176, 0.001288)	(-0.001092, 0.003395)	(0.001258, 0.002833)
## 524	(-0.00099, 0.002636)	(-0.009534, 0.001583)	(-0.002958, 0.000943)
## 525	(-0.017287, 0.011644)	(-0.008765, 0.07994)	(-0.023966, 0.007161)
## 526	(-0.000395, 0.000349)	(0.000616, 0.002898)	(-0.000818, -0.000017)
## 527	(-0.000258, 0.000299)	(-0.000344, 0.001363)	(0.000143, 0.000741)
## 528	(-0.001176, 0.000406)	(0.001438, 0.006287)	(0.000181, 0.001882)
## 529	(0.000083, 0.0017)	(-0.00106, 0.003899)	(0.000109, 0.001848)
## 530	(-0.002136, 0.00053)	(-0.004861, 0.003313)	(-0.002098, 0.00077)
## 531	(-0.000262, -0.000013)	(-0.00092, -0.000159)	(-0.000119, 0.000149)
## 532	(-0.000085, 0.003173)	(-0.006507, 0.003481)	(-0.000551, 0.002954)
## 533	(0.000015, 0.001192)	(0.000224, 0.003832)	(-0.000844, 0.000422)
## 534	(-0.083376, -0.012102)	(-0.224496, -0.005964)	(-0.081365, -0.004681)
## 535	(-0.002274, 0.000173)	(-0.002038, 0.005462)	(-0.003566, -0.000934)
## 536	(-0.034454, 0.055896)	(-0.013075, 0.263944)	(-0.070107, 0.0271)
## 537	(-0.002841, 0.000691)	(-0.006632, 0.004197)	(-0.002803, 0.000997)
## 538	(-0.002548, -0.000013)	(0.001368, 0.009142)	(-0.000383, 0.002345)
## 539	(-0.001322, 0.001568)	(-0.006301, 0.002561)	(-0.002444, 0.000666)

## 540	(-0.001623, 0.001809)	(-0.000892, 0.00963)	(-0.003925, -0.000233)
## 541	(-0.001175, 0.000237)	(-0.003561, 0.00077)	(-0.001221, 0.000299)
## 542	(-0.003908, 0.045351)	(-0.211472, -0.060439)	(-0.007878, 0.04512)
## 543	(-0.001671, 0.000897)	(-0.008018, -0.000144)	(-0.002215, 0.000549)
## 544	(-0.001323, -0.000259)	(-0.002049, 0.001213)	(-0.000859, 0.000285)
## 545	(-0.000488, 0.002037)	(-0.002931, 0.00481)	(-0.000904, 0.001812)
## 546	(-0.001009, 0.001061)	(-0.000796, 0.005549)	(-0.001517, 0.000709)
## 547	(-0.000509, 0.001746)	(0.000989, 0.007902)	(-0.002056, 0.00037)
## 548	(-0.00092, 0.001349)	(-0.003006, 0.003951)	(-0.001207, 0.001234)
## 549	<NA>	<NA>	<NA>
## 550	<NA>	<NA>	<NA>
## 551	<NA>	<NA>	<NA>
## 552	<NA>	<NA>	<NA>
## 553	<NA>	<NA>	<NA>
## 554	<NA>	<NA>	<NA>
## 555	<NA>	<NA>	<NA>
## 556	<NA>	<NA>	<NA>
## 557	<NA>	<NA>	<NA>
## 558	<NA>	<NA>	<NA>
## 559	<NA>	<NA>	<NA>
## 560	<NA>	<NA>	<NA>
## 561	<NA>	<NA>	<NA>
## 562	<NA>	<NA>	<NA>
## 563	<NA>	<NA>	<NA>
##	model4_int	modellme	brmodel2
## 1	(-0.003674, 0.001276)	(-0.065693, 0.145525)	<NA>
## 2	(-0.000081, 0.000019)	(-0.039521, -0.002294)	(-0.000167, 0.000102)
## 3	(-0.000033, 0.000063)	(-0.042982, -0.000698)	(-0.000138, 0.00011)
## 4	(-0.000215, -0.000078)	(-0.089987, -0.032277)	(-0.000312, 0.000056)
## 5	(-0.000034, 0.000023)	(-0.046223, 0.034155)	(-0.000061, 0.000078)
## 6	(-0.000043, 0.000034)	(-0.023125, 0.015989)	(-0.000022, 0.000188)
## 7	(-0.000002, 0.000075)	(-0.010584, 0.042901)	(-0.00005, 0.000151)
## 8	(0.000002, 0.00019)	(-0.00358, 0.055815)	(0.000082, 0.000578)
## 9	(0.000026, 0.000173)	(0.003592, 0.036933)	(-0.000157, 0.000258)
## 10	(0, 0.000043)	(0.002111, 0.049384)	(-0.000026, 0.000083)
## 11	(0.000002, 0.000128)	(-0.080643, 0.227163)	(-0.000032, 0.000294)
## 12	(-0.000005, 0.000124)	(0.01131, 0.120187)	(-0.000144, 0.000203)
## 13	(-0.001658, 0.007282)	(0.00553, 0.041423)	(-0.00692, 0.014788)
## 14	(-0.000064, 0.000041)	(-0.046759, -0.005679)	(-0.000013, 0.000249)
## 15	(-0.000004, 0.000046)	(-0.033646, 0.017717)	(-0.000027, 0.000197)
## 16	(-0.000139, 0.000004)	(-0.122474, 0.040075)	(-0.000373, 0.00011)
## 17	(-0.000028, 0.000039)	(-0.00597, 0.041886)	(-0.000152, 0.000024)
## 18	(-0.000041, 0.000042)	(-0.079361, 0.069906)	(-0.000162, 0.000065)
## 19	(-0.000031, 0.000098)	(-0.027358, 0.024998)	(-0.000012, 0.000373)
## 20	(-0.000216, -0.000065)	(-0.049133, 0.024001)	(-0.000346, 0.000125)

## 21	(-0.000683, -0.000039)	(-0.060553, 0.017421)	(-0.001533, 0.000087)
## 22	(-0.000109, 0.000039)	(-0.244604, 0.016114)	(-0.000175, 0.000154)
## 23	(-0.000008, 0.00004)	(-0.006806, 0.055756)	(-0.000034, 0.000086)
## 24	(-0.000041, 0.000128)	(-0.005686, 0.033455)	(-0.000051, 0.000418)
## 25	(-0.000138, -0.000001)	(-0.151817, 0.017503)	(-0.000257, 0.000073)
## 26	(-0.000156, 0.000011)	(-0.09348, 0.055027)	(-0.0002, 0.000189)
## 27	(-0.00021, 0.00034)	(-0.012678, 0.014481)	(-0.001704, 0.000776)
## 28	(-0.000057, 0.000022)	(-0.072476, 0.01259)	(-0.000164, 0.000044)
## 29	(-0.02034, 0.01693)	(-0.006707, 0.025914)	(-0.04792, 0.048455)
## 30	(-0.000001, 0.000059)	(-0.009651, 0.028694)	(-0.000027, 0.000129)
## 31	(-0.000051, 0.000021)	(-0.040447, 0.015111)	(-0.000123, 0.000072)
## 32	(-0.000059, 0.00001)	(-0.03508, 0.009557)	(-0.000079, 0.000102)
## 33	(-0.000029, 0.000143)	(-0.034681, 0.024935)	(-0.000167, 0.000357)
## 34	(-0.000025, 0.000028)	(-0.036082, 0.006607)	(-0.000132, 0.000002)
## 35	(-0.000075, 0.000018)	(-0.04208, 0.018346)	(-0.000107, 0.000164)
## 36	(-0.000241, -0.00006)	(-0.352938, -0.022746)	(-0.000758, -0.000147)
## 37	(-0.000046, 0.000142)	(-0.085435, 0.031877)	(-0.000327, 0.00016)
## 38	(-0.00006, 0.000036)	(-0.088102, -0.023175)	(-0.000125, 0.000144)
## 39	(0.000868, 0.00218)	(0.023382, 0.049778)	(-0.001314, 0.002217)
## 40	(-0.000097, 0.000059)	(-0.04553, 0.041231)	(-0.000335, 0.000077)
## 41	(-0.000009, 0.000047)	(-0.011912, 0.043269)	(-0.000062, 0.000092)
## 42	(-0.003141, 0.020092)	(-0.002509, 0.030421)	(-0.002726, 0.051516)
## 43	(-0.000082, 0.000043)	(-0.048792, 0.01676)	(-0.000095, 0.00023)
## 44	(-0.000064, 0.000014)	(0.016276, 0.14934)	(-0.000124, 0.000073)
## 45	(-0.001173, 0.000206)	(-0.012371, 0.015133)	(-0.002075, 0.00142)
## 46	(-0.002586, -0.000745)	(-0.12527, -0.025416)	(-0.007155, 0.001986)
## 47	(-0.000058, 0.000083)	(-0.0161, 0.011772)	(-0.000216, 0.000261)
## 48	(0.000017, 0.000067)	(0.011184, 0.042166)	(-0.00001, 0.000111)
## 49	(-0.000047, 0.00001)	(-0.014329, 0.041723)	(-0.000094, 0.00006)
## 50	(-0.000074, -0.000001)	(-0.050893, 0.023251)	(-0.000111, 0.000073)
## 51	(-0.000002, 0.000134)	(-0.006397, 0.067055)	(-0.000126, 0.000309)
## 52	(-0.000035, 0.00008)	(0.017057, 0.065853)	(-0.000142, 0.000153)
## 53	(-0.000061, 0.000223)	(-0.087032, 0.131117)	(0.00001, 0.000813)
## 54	(0.000017, 0.000143)	(-0.002215, 0.1037)	(-0.000058, 0.000258)
## 55	(-0.000169, 0.000021)	(-0.03548, 0.003275)	(-0.000369, 0.000133)
## 56	(-0.000064, -0.00001)	(-0.300132, -0.010401)	(-0.000097, 0.000056)
## 57	(-0.000028, 0.000018)	(-0.016451, 0.078781)	(-0.000109, 0.000006)
## 58	(0.000023, 0.000099)	(-0.02703, 0.022095)	(-0.000053, 0.000143)
## 59	(-0.000059, 0.000014)	(-0.166387, 0.136117)	(-0.00016, 0.000015)
## 60	(-0.000039, 0.000042)	(-0.095564, 0.025798)	(-0.00006, 0.000143)
## 61	(-0.000058, 0.000004)	(-0.020632, 0.011036)	(-0.000089, 0.000078)
## 62	(-0.000049, 0.000059)	(-0.051674, 0.032747)	(-0.000012, 0.00026)
## 63	(-0.00001, 0.000043)	(-0.251447, 0.180051)	(-0.000056, 0.000085)
## 64	(-0.000039, 0.000019)	(-0.023594, 0.01752)	(-0.000111, 0.000044)
## 65	(-0.000106, -0.000013)	(-0.017848, 0.019607)	(-0.000211, 0.000024)

## 66	(-0.000091, 0.000028)	(-0.055784, 0.016665)	(-0.000134, 0.000189)
## 67	(-0.029879, 0.13187)	(0.012451, 0.043425)	(-0.065895, 0.375149)
## 68	(-0.000031, 0.000027)	(-0.054989, -0.008151)	(-0.000066, 0.000087)
## 69	(-0.000003, 0.000022)	(-0.071735, 0.043402)	(-0.000077, 0.000054)
## 70	(-0.000005, 0.000018)	(-0.195894, -0.02522)	(-0.000105, 0.00006)
## 71	(-0.000181, 0.000014)	(-0.019936, 0.00926)	(-0.000184, 0.00025)
## 72	(0.068373, 0.416682)	(0.022671, 0.055813)	(-0.55409, 0.34152)
## 73	(0.001578, 0.004801)	(0.049044, 0.090832)	(-0.00059, 0.00729)
## 74	(-0.000145, -0.000001)	(-0.110222, -0.04019)	(-0.000441, -0.000041)
## 75	(-0.000055, 0.000126)	(-0.121459, 0.085376)	(-0.000221, 0.000263)
## 76	(-0.000161, -0.000053)	(-0.044227, -0.006842)	(-0.000302, -0.000031)
## 77	(-0.000007, 0.000266)	(-0.01252, 0.02074)	(-0.000241, 0.000458)
## 78	(-0.001511, 0.001882)	(-0.014583, 0.010603)	(-0.005372, 0.003676)
## 79	(0, 0.000072)	(0.043534, 0.284385)	(-0.000067, 0.000117)
## 80	(-0.000088, 0.000065)	(-0.040249, 0.134435)	(-0.000224, 0.000193)
## 81	(-0.000038, 0.000044)	(-0.015107, 0.032169)	(-0.000146, 0.000072)
## 82	(-0.000044, 0.000009)	(-0.068117, 0.060047)	(-0.000087, 0.000064)
## 83	(-0.000005, 0.000006)	(-0.070941, -0.020323)	(-0.000122, 0.000039)
## 84	(-0.000163, 0.000041)	(-0.119575, 0.006066)	(-0.000365, 0.000233)
## 85	(-0.000085, 0.000014)	(-0.108276, 0.027816)	(-0.000197, 0.00008)
## 86	(-0.175044, 0.281265)	(-0.008097, 0.020078)	(0.013296, 1.208002)
## 87	(-0.000039, 0.000015)	(-0.069144, -0.007138)	(-0.000069, 0.000068)
## 88	(-0.000084, 0.000021)	(-0.039032, 0.043307)	(-0.000207, 0.000065)
## 89	(0.000021, 0.000087)	(-0.050772, 0.108974)	(-0.000035, 0.000129)
## 90	(-0.000026, 0.00006)	(0.000553, 0.053398)	(-0.000061, 0.000171)
## 91	(-0.000135, 0.000027)	(-0.060687, 0.036672)	(-0.000138, 0.000246)
## 92	(-0.000024, 0.000072)	(0.00335, 0.081166)	(-0.000168, 0.000085)
## 93	(0.013688, 0.030734)	(0.032642, 0.074079)	(0.003811, 0.049942)
## 94	(-0.00006, 0.000002)	(-0.022599, 0.021696)	(-0.000131, 0.00004)
## 95	(-0.000222, 0.000064)	(-0.044842, 0.048042)	(-0.000458, 0.000224)
## 96	(-0.000051, 0.000048)	(-0.034689, 0.04532)	(-0.000131, 0.000148)
## 97	(-0.003164, 0.002959)	(-0.014245, 0.016886)	(-0.007106, 0.009104)
## 98	(-0.000018, 0.000059)	(-0.025368, 0.011003)	(-0.000054, 0.000157)
## 99	(-0.000058, 0.000018)	(-0.076277, 0.008518)	(-0.000188, 0.000028)
## 100	(-0.000188, 0.000264)	(-0.008809, 0.031567)	(-0.000651, 0.000587)
## 101	(-0.000067, 0.000009)	(-0.040046, -0.0013)	(-0.000163, 0.000031)
## 102	(-0.000032, 0.000049)	(-0.025951, 0.010346)	(-0.000109, 0.000104)
## 103	(-0.000029, 0.000014)	(-0.056483, 0.01725)	(-0.000088, 0.000028)
## 104	(-0.000085, 0.000085)	(-0.087218, 0.042122)	(-0.000191, 0.000216)
## 105	(-0.000882, -0.000219)	(-0.020675, 0.033503)	(-0.001797, -0.000122)
## 106	(-0.000196, -0.000012)	(-0.054957, 0.012376)	(-0.000368, 0.000116)
## 107	(-0.000072, 0.000062)	(-0.048506, 0.181638)	(-0.000233, 0.000092)
## 108	(-0.000073, 0.000055)	(-0.105952, 0.009735)	(-0.00016, 0.000187)
## 109	(-0.000094, 0.000103)	(-0.006311, 0.103498)	(-0.000235, 0.000214)
## 110	(0.000006, 0.000143)	(0.006738, 0.033255)	(-0.000057, 0.000272)

## 111	(-0.000575, 0.000438)	(-0.03429, -0.005049)	(-0.000604, 0.002138)
## 112	(-0.000048, -0.000003)	(-0.055792, -0.006228)	(-0.000044, 0.000072)
## 113	(-0.000009, 0.000036)	(-0.009562, 0.023154)	(-0.000048, 0.000067)
## 114	(-0.0001, 0.000029)	(-0.109626, 0.034662)	(-0.000289, 0.00005)
## 115	(-0.000014, 0.000053)	(-0.02706, 0.040646)	(-0.000093, 0.000075)
## 116	(-0.001576, -0.000081)	(-0.035386, -0.00423)	(-0.003283, 0.000765)
## 117	(-0.000052, 0.000011)	(-0.064462, -0.010425)	(-0.000066, 0.000108)
## 118	(-0.000281, 0.000007)	(-0.050037, -0.002564)	(-0.000739, 0.000212)
## 119	(-0.000118, 0.000086)	(-0.025668, 0.023686)	(-0.000129, 0.000345)
## 120	(-0.000204, 0.000019)	(-0.026167, 0.009031)	(-0.000627, -0.000048)
## 121	(-0.000316, 0.00168)	(0.002876, 0.029487)	(-0.003853, 0.00142)
## 122	(-0.000012, 0.000066)	(-0.023058, 0.024995)	(-0.00003, 0.000187)
## 123	(-0.000016, 0.000082)	(-0.006638, 0.040115)	(-0.000164, 0.000146)
## 124	(-0.000022, 0.000069)	(-0.035143, 0.013222)	(-0.000036, 0.000208)
## 125	(-0.000174, 0.000193)	(-0.017383, 0.021383)	(-0.000519, 0.000437)
## 126	(0.000021, 0.000139)	(-0.131977, 0.052829)	(-0.000164, 0.000139)
## 127	(-0.000058, 0.000021)	(-0.011785, 0.028126)	(-0.000233, 0.000001)
## 128	(-0.000032, 0.000048)	(-0.064142, -0.016791)	(-0.000082, 0.000136)
## 129	(-0.000058, 0.000042)	(-0.03346, 0.012656)	(-0.000155, 0.000104)
## 130	(-0.000117, 0.000038)	(-0.140895, 0.062389)	(-0.000342, 0.000109)
## 131	(-0.002799, 0.000265)	(-0.028304, 0.003585)	(-0.006718, 0.00074)
## 132	(-0.000099, -0.000003)	(-0.061584, -0.008329)	(-0.000105, 0.000147)
## 133	(-0.000026, 0.000113)	(0.002285, 0.041111)	(-0.000173, 0.000225)
## 134	(-0.000061, 0.000145)	(-0.007535, 0.038197)	(-0.000251, 0.000286)
## 135	(-0.000186, 0.000016)	(-0.060701, -0.020465)	(-0.000199, 0.000312)
## 136	(-0.006529, 0.00364)	(-0.008305, 0.029291)	(-0.037789, -0.008788)
## 137	(0.00006, 0.000184)	(0.011472, 0.068547)	(-0.00013, 0.000178)
## 138	(-0.00007, -0.000004)	(-0.147923, -0.039)	(-0.000106, 0.000069)
## 139	(-0.000055, 0.000096)	(0.02003, 0.14329)	(-0.000052, 0.000339)
## 140	(-0.000164, 0.001475)	(-0.010121, 0.072765)	(-0.001497, 0.003336)
## 141	(-0.000038, 0.000031)	(-0.065396, -0.008736)	(-0.000079, 0.0001)
## 142	(-0.000064, -0.000006)	(-0.020679, 0.037127)	(-0.000093, 0.000063)
## 143	(-0.000009, 0.000079)	(-0.019405, 0.052356)	(-0.000112, 0.000117)
## 144	(0.000095, 0.000256)	(0.033488, 0.181056)	(-0.000122, 0.000271)
## 145	(-0.000134, -0.000014)	(-0.086469, 0.130315)	(-0.000168, 0.000167)
## 146	(0.000042, 0.000431)	(-0.005209, 0.058976)	(-0.000276, 0.000731)
## 147	(-0.000199, 0.000102)	(-0.090591, 0.011777)	(-0.000426, 0.000361)
## 148	(-0.000413, 0.000307)	(0.000161, 0.036242)	(-0.001315, 0.000923)
## 149	(-0.00006, 0.000116)	(-0.076559, 0.063957)	(-0.000812, -0.000106)
## 150	(-0.00007, 0.000007)	(0.009157, 0.065254)	(-0.000101, 0.000103)
## 151	(-0.000011, 0.000049)	(-0.009427, 0.022267)	(-0.000114, 0.000021)
## 152	(-0.000055, 0.000014)	(-0.027585, 0.008865)	(-0.000125, 0.000051)
## 153	(-0.000128, 0.000211)	(-0.007891, 0.089017)	(-0.000453, 0.000436)
## 154	(-0.000073, 0.000039)	(-0.036753, 0.008847)	(-0.000104, 0.000172)
## 155	(-0.000073, 0.000053)	(-0.026518, 0.010675)	(-0.00031, -0.000011)

## 156	(-0.000017, 0.000047)	(0.0114, 0.063918)	(-0.000071, 0.000102)
## 157	(-0.000069, 0.000002)	(-0.000501, 0.114676)	(-0.000082, 0.000104)
## 158	(-0.000057, 0.000054)	(-0.047001, 0.086097)	(-0.000144, 0.000157)
## 159	(-0.008889, 0.062716)	(0.001498, 0.033575)	(-0.067318, 0.122312)
## 160	(-0.000061, 0.000084)	(-0.052598, 0.06355)	(-0.000071, 0.000299)
## 161	(-0.000697, 0.00016)	(-0.053141, -0.008885)	(-0.001918, 0.000212)
## 162	(-0.000063, 0.000123)	(-0.076708, -0.000033)	(-0.000321, 0.000155)
## 163	(-0.000022, 0.000118)	(-0.015043, 0.051071)	(0.000044, 0.00039)
## 164	(-0.000064, 0.000086)	(-0.020369, 0.009125)	(0.000002, 0.000356)
## 165	(0.000009, 0.000184)	(-0.01168, 0.09693)	(-0.000067, 0.000355)
## 166	(-0.000153, 0.000067)	(-0.097261, 0.019539)	(-0.000364, 0.000306)
## 167	(0.000036, 0.000526)	(-0.032026, 0.045132)	(-0.000284, 0.000665)
## 168	(-0.000009, 0.000075)	(0.001131, 0.047173)	(-0.000088, 0.000124)
## 169	(0.000017, 0.000163)	(-0.059138, 0.02617)	(-0.000047, 0.000294)
## 170	(-0.000033, 0.000051)	(-0.01472, 0.021785)	(-0.000018, 0.000207)
## 171	(-0.000032, 0.000024)	(-0.031998, 0.011597)	(-0.00011, 0.000047)
## 172	(0.000031, 0.000133)	(0.034428, 0.078575)	(-0.000048, 0.000241)
## 173	(-0.000263, 0.00008)	(-0.100446, 0.003143)	(-0.000517, 0.000453)
## 174	(-0.00022, 0.000233)	(-0.010402, 0.030683)	(-0.000392, 0.000561)
## 175	(-0.000049, 0.000124)	(-0.009551, 0.040626)	(-0.000518, -0.000039)
## 176	(-0.000047, 0.000018)	(-0.041505, 0.008942)	(-0.000048, 0.000123)
## 177	(-0.001695, 0.006949)	(0.001993, 0.026676)	(-0.016451, 0.006161)
## 178	(-0.02361, 0.068101)	(-0.008355, 0.017837)	(-0.062344, 0.175809)
## 179	(-0.000019, 0.000031)	(-0.017569, 0.013092)	(-0.000061, 0.000073)
## 180	(-0.019711, 0.009839)	(-0.021104, 0.009004)	(-0.079355, 0.002491)
## 181	(-0.000027, 0.000035)	(0.06689, 0.3302)	(-0.000123, 0.000049)
## 182	(-0.010358, 0.029834)	(0.019035, 0.048054)	(-0.041304, 0.066307)
## 183	(-0.000026, 0.000062)	(0.003792, 0.038942)	(-0.000201, 0.000058)
## 184	(-0.008746, 0.013144)	(0.016188, 0.048709)	(-0.033895, 0.020296)
## 185	(-0.001122, 0.001276)	(-0.011433, 0.019959)	(-0.001811, 0.004501)
## 186	(0.009061, 0.022603)	(0.057686, 0.089617)	(-0.003486, 0.028483)
## 187	(0.00262, 0.014802)	(0.011996, 0.044154)	(-0.003525, 0.025118)
## 188	(-0.000105, 0.00003)	(-0.07016, -0.008076)	(-0.000351, -0.000018)
## 189	(-0.000099, 0.00004)	(-0.006437, 0.051558)	(-0.000206, 0.000146)
## 190	(-0.000028, 0.000171)	(-0.01232, 0.035014)	(-0.000204, 0.000492)
## 191	(-0.000415, 0.000134)	(-0.052287, -0.00093)	(-0.000674, 0.000557)
## 192	(-0.00004, 0.000074)	(-0.001413, 0.052495)	(-0.000184, 0.000101)
## 193	(-0.000169, 0.000349)	(-0.033491, 0.017636)	(-0.000814, 0.000473)
## 194	(0.002239, 0.008414)	(0.020456, 0.050548)	(-0.000378, 0.015394)
## 195	(0.00003, 0.000222)	(-0.084145, 0.056845)	(-0.00022, 0.000265)
## 196	(-0.039788, 0.103619)	(-0.01905, 0.012561)	(-0.15883, 0.236621)
## 197	(-0.037964, 0.1058)	(0.008494, 0.035438)	(-0.107936, 0.274911)
## 198	(-0.000032, 0.000136)	(0.000316, 0.057154)	(0.000028, 0.000442)
## 199	(-0.000092, 0.000027)	(-0.046596, 0.033235)	(-0.000152, 0.000155)
## 200	(-0.000161, 0.000027)	(-0.097353, 0.003007)	(-0.000269, 0.000235)

## 201	(-0.000013, 0.000047)	(-0.02055, 0.028033)	(-0.000059, 0.000109)
## 202	(0.000011, 0.000136)	(-0.011726, 0.018384)	(0.000026, 0.000333)
## 203	(-0.000074, 0.000036)	(-0.064601, 0.175756)	(-0.000146, 0.000158)
## 204	(-0.000016, 0.000064)	(-0.021942, 0.01067)	(-0.000025, 0.000199)
## 205	(-0.007993, 0.024423)	(-0.005799, 0.025226)	(-0.039844, 0.042074)
## 206	(-0.000015, 0.000052)	(0.003616, 0.042511)	(-0.000108, 0.000062)
## 207	(-0.000312, 0.000377)	(-0.01915, 0.023379)	(-0.000507, 0.001233)
## 208	(0.000022, 0.000104)	(0.022012, 0.068265)	(0.000013, 0.000224)
## 209	(0.021946, 0.139953)	(0.022343, 0.050613)	(-0.036142, 0.268278)
## 210	(-0.001362, 0.00086)	(-0.020485, 0.007736)	(-0.001352, 0.004326)
## 211	(-0.000191, 0.000035)	(-0.060724, -0.012157)	(-0.00015, 0.000532)
## 212	(-0.000083, 0.000013)	(-0.152683, 0.012628)	(-0.000266, -0.000021)
## 213	(-0.041318, 0.214163)	(-0.004262, 0.024733)	(-0.206687, 0.419391)
## 214	(-0.000196, 0.001269)	(-0.012449, 0.022685)	(-0.001464, 0.002375)
## 215	(-0.00011, 0.000031)	(-0.023321, 0.007712)	(-0.000222, 0.000171)
## 216	(-0.006434, 0.03096)	(0.00727, 0.038775)	(-0.035062, 0.059604)
## 217	(-0.010028, 0.021683)	(-0.0168, 0.014433)	(-0.02999, 0.05704)
## 218	(-0.000016, 0.000071)	(0.022206, 0.11258)	(-0.000083, 0.000133)
## 219	(-0.000013, 0.000063)	(-0.216213, 0.012841)	(-0.000141, 0.000054)
## 220	(0.173196, 0.553146)	(0.036179, 0.065824)	(-0.058822, 0.907357)
## 221	(-0.000013, 0.000069)	(-0.000788, 0.093407)	(-0.000032, 0.000185)
## 222	(-0.000036, 0.000023)	(-0.03739, 0.013367)	(-0.00009, 0.000065)
## 223	(-0.000042, 0.000038)	(-0.032944, 0.055603)	(-0.000055, 0.000167)
## 224	(-0.000155, -0.000025)	(-0.06392, -0.010122)	(-0.000234, 0.000123)
## 225	(-0.000042, 0.000095)	(-0.022632, 0.068743)	(-0.000189, 0.000151)
## 226	(-0.000004, 0.000077)	(-0.003867, 0.022824)	(-0.000098, 0.000091)
## 227	(-0.000072, 0.00016)	(-0.009857, 0.039453)	(-0.000242, 0.000338)
## 228	(-0.000045, 0.000049)	(-0.027583, 0.030141)	(-0.000067, 0.000176)
## 229	(-0.001755, 0.000173)	(-0.017888, 0.009026)	(-0.001682, 0.003486)
## 230	(0.000106, 0.000466)	(0.025982, 0.064576)	(-0.000241, 0.001381)
## 231	(-0.000321, 0.000048)	(-0.03017, 0.045911)	(-0.000482, 0.000247)
## 232	(-0.000006, 0.000088)	(-0.091786, 0.067142)	(-0.000071, 0.000165)
## 233	(-0.000015, 0.000089)	(0.008687, 0.05017)	(-0.000133, 0.00012)
## 234	(-0.030556, 0.066711)	(-0.012055, 0.018044)	(-0.055432, 0.17652)
## 235	(-0.004137, 0.002098)	(-0.007606, 0.021563)	(-0.007927, 0.006427)
## 236	(-0.000142, -0.000041)	(-0.03913, 0.087659)	(-0.000197, 0.000078)
## 237	(-0.00038, 0.000907)	(-0.014453, 0.014391)	(-0.00076, 0.002526)
## 238	(0.000001, 0.000102)	(0.026292, 0.101895)	(-0.000192, 0.000077)
## 239	(0.000022, 0.00016)	(0.017876, 0.085455)	(-0.000101, 0.000261)
## 240	(-0.000257, 0.000106)	(-0.03629, -0.003537)	(-0.000209, 0.00077)
## 241	(-0.019189, -0.004099)	(-0.070353, -0.018762)	(-0.031281, 0.009779)
## 242	(-0.000032, 0.000089)	(-0.034467, 0.168583)	(-0.000054, 0.000281)
## 243	(-0.000486, 0.000855)	(-0.006893, 0.036376)	(-0.001583, 0.001633)
## 244	(-0.000121, -0.000024)	(-0.049421, 0.019827)	(-0.000188, 0.000076)
## 245	(0.000069, 0.000627)	(-0.010249, 0.055393)	(-0.000655, 0.000764)

## 246	(0.000158, 0.000397)	(0.018053, 0.052181)	(-0.000278, 0.000449)
## 247	(-0.000052, 0.000055)	(-0.00691, 0.025255)	(-0.000309, -0.000033)
## 248	(-0.006764, 0.00526)	(-0.003098, 0.024641)	(-0.015076, 0.015962)
## 249	(-0.005811, 0.047235)	(0.002794, 0.028182)	(-0.026652, 0.119971)
## 250	(-0.256842, 0.184082)	(-0.014646, 0.012609)	(-0.155448, 1.003143)
## 251	(-0.033588, 0.028939)	(-0.016418, 0.013212)	(-0.079042, 0.095569)
## 252	(-0.000017, 0.000198)	(-0.04141, 0.056446)	(-0.00049, 0.000107)
## 253	(-0.008965, 0.00405)	(-0.031157, -0.003517)	(-0.019445, 0.018657)
## 254	(-0.002099, -0.000347)	(-0.033199, 0.003023)	(-0.004705, 0.000445)
## 255	(-0.000086, -0.000002)	(-0.146422, 0.020444)	(-0.000243, -0.000011)
## 256	(-0.000085, 0.000108)	(-0.00711, 0.035952)	(-0.000217, 0.000238)
## 257	(-0.005043, 0.00089)	(-0.018111, 0.008621)	(-0.012339, 0.003204)
## 258	(-0.000054, 0.000027)	(-0.124595, 0.001067)	(-0.000163, 0.000046)
## 259	(-0.000041, 0.000078)	(0.001224, 0.032147)	(-0.000196, 0.000109)
## 260	(-0.018301, 0.036726)	(0.01489, 0.047109)	(-0.127374, 0.022191)
## 261	(-0.000209, -0.000004)	(0.005232, 0.084823)	(-0.000542, 0.000053)
## 262	(-0.000324, 0.000219)	(-0.046922, 0.013212)	(-0.001056, 0.000556)
## 263	(-0.003581, 0.005722)	(-0.011697, 0.020856)	(-0.003387, 0.022591)
## 264	(-0.000109, -0.000009)	(-0.060334, 0.014822)	(-0.000193, 0.00007)
## 265	(-0.000186, 0.000003)	(-0.021998, 0.015366)	(-0.000368, 0.00015)
## 266	(-0.00005, 0.000142)	(0.022974, 0.116831)	(-0.000214, 0.000266)
## 267	(-0.008916, 0.013493)	(-0.011179, 0.016135)	(-0.022538, 0.032419)
## 268	(0.003342, 0.026308)	(0.022484, 0.047894)	(-0.003401, 0.054865)
## 269	(-0.000078, 0.000002)	(0.016797, 0.169186)	(-0.000148, 0.000093)
## 270	(0.000019, 0.000633)	(-0.005792, 0.033904)	(-0.000397, 0.001128)
## 271	(-0.000025, 0.000143)	(-0.002346, 0.024218)	(-0.000181, 0.000284)
## 272	(-0.000303, 0.000323)	(-0.020514, 0.033566)	(-0.000297, 0.00137)
## 273	(-0.000006, 0.000182)	(-0.004173, 0.055462)	(0.000008, 0.000473)
## 274	(0.028182, 0.067761)	(0.018686, 0.047911)	(0.04234, 0.127343)
## 275	(-0.000265, 0.000486)	(-0.01931, 0.015646)	(-0.000568, 0.001099)
## 276	(-0.025694, 0.042737)	(-0.010581, 0.025646)	(-0.149559, 0.065008)
## 277	(-0.001422, 0.000593)	(-0.009159, 0.016591)	(-0.001193, 0.004219)
## 278	(-0.000099, 0.000071)	(-0.018301, 0.125387)	(-0.000345, 0.000071)
## 279	(-0.00055, 0.000488)	(-0.021762, 0.021171)	(-0.00196, 0.000666)
## 280	(-0.00078, 0.001304)	(-0.004355, 0.023005)	(-0.002164, 0.003302)
## 281	(-0.000137, 0.000002)	(-0.120064, -0.016124)	(-0.000152, 0.000326)
## 282	(-0.004786, 0.003449)	(-0.018333, 0.007151)	(-0.023441, -0.001257)
## 283	(0.007638, 0.036532)	(0.008878, 0.039267)	(-0.008432, 0.067099)
## 284	(-0.000783, 0.000629)	(-0.020237, 0.006975)	(-0.002747, 0.00096)
## 285	(-0.000001, 0.000183)	(-0.038944, 0.023945)	(-0.000047, 0.000424)
## 286	(-0.002446, -0.000325)	(-0.033656, -0.003734)	(-0.00173, 0.002884)
## 287	(-0.000424, -0.000021)	(-0.04606, 0.013615)	(-0.000453, 0.000691)
## 288	(-0.000766, -0.000189)	(-0.060466, -0.011027)	(-0.001295, 0.000035)
## 289	(-0.000046, 0.000022)	(-0.142734, 0.14989)	(-0.000052, 0.000113)
## 290	(0.000046, 0.000311)	(0.013942, 0.051497)	(0.001403, 0.005457)

## 291	(-0.000469, 0.000057)	(-0.058046, 0.012497)	(-0.00215, 0.006614)
## 292	(-0.000879, 0.000582)	(-0.106453, 0.015895)	(-0.009265, 0.004786)
## 293	(-0.000388, 0.00049)	(-0.085218, 0.095881)	(-0.014888, -0.002509)
## 294	(-0.000182, 0.000149)	(-0.028047, 0.028002)	(-0.003529, 0.000927)
## 295	(-0.000423, 0.000134)	(-0.074771, 0.026073)	(-0.015349, 0.000232)
## 296	(-0.000846, 0.000473)	(-0.074244, 0.040789)	(0.000895, 0.014951)
## 297	(-0.000179, 0.000292)	(-0.028427, 0.072059)	(-0.002154, 0.008901)
## 298	(-0.00495, 0.003317)	(-0.237928, 0.499535)	(-0.057141, 0.039846)
## 299	(-0.000918, 0.00111)	(-0.069764, 0.175986)	(-0.017435, 0.001948)
## 300	(-0.0003, 0.000312)	(-0.040391, 0.030134)	(0.002378, 0.012809)
## 301	(-0.000233, 0.000329)	(-0.031325, 0.061364)	(-0.001897, 0.006522)
## 302	(-0.002085, 0.00173)	(-0.2, 0.177792)	(-0.016612, 0.044778)
## 303	(-0.000256, 0.000496)	(-0.011437, 0.090949)	(-0.014727, -0.001267)
## 304	(-0.001509, 0.000707)	(-0.1995, 0.150915)	(0.006238, 0.039297)
## 305	(-0.000522, 0.000109)	(-0.08817, -0.006167)	(-0.001838, 0.009984)
## 306	(-0.000595, 0.00182)	(-0.022696, 0.142933)	(0.002621, 0.061782)
## 307	(-0.004436, 0.003927)	(-0.084627, 0.087273)	(-0.062633, 0.046483)
## 308	(-0.00641, 0.001619)	(-0.557393, 0.066366)	(-0.018523, 0.090865)
## 309	(-0.000329, 0.000307)	(-0.078333, 0.05995)	(-0.012061, 0.004642)
## 310	(-0.000341, 0.000258)	(-0.035412, 0.014437)	(-0.00539, 0.002878)
## 311	(-0.00305, 0.00173)	(-0.320876, 0.081174)	(0.0036, 0.066806)
## 312	(-0.004022, 0.002419)	(-0.19598, 0.157415)	(-0.043582, 0.022959)
## 313	(-0.001661, 0.001239)	(-0.02814, 0.016211)	(-0.072903, 0.020443)
## 314	(-0.000816, 0.000252)	(-0.119422, 0.069428)	(-0.010576, 0.008148)
## 315	(-0.00019, 0.000088)	(-0.03813, 0.02009)	(-0.000871, 0.004697)
## 316	(-0.000375, 0.00024)	(-0.046203, 0.064735)	(-0.002302, 0.005955)
## 317	(-0.000237, 0.000184)	(-0.04405, 0.034168)	(-0.006301, -0.000845)
## 318	(-0.001051, 0.000682)	(-0.044496, 0.080607)	(-0.004416, 0.017329)
## 319	(-0.000224, 0.000104)	(-0.062636, 0.012982)	(-0.003208, 0.002083)
## 320	(-0.000618, 0.0006)	(-0.062463, 0.069895)	(-0.014941, -0.000931)
## 321	(-0.01116, 0.002727)	(-0.765633, 0.02616)	(0.001865, 0.17151)
## 322	(-0.004397, 0.002525)	(-0.212036, 0.066848)	(-0.113624, 0.008847)
## 323	(-0.001028, 0.000057)	(-0.208291, -0.068963)	(-0.001223, 0.012148)
## 324	(-0.002245, 0.003074)	(-0.014193, 0.022632)	(-0.032372, 0.026468)
## 325	(-0.000938, 0.000985)	(-0.113045, 0.076322)	(0.003066, 0.025313)
## 326	(-0.000335, 0.000221)	(-0.078602, 0.036724)	(-0.004896, 0.001553)
## 327	(-0.063105, 0.057752)	(-0.02605, 0.027408)	(-0.859098, 0.214991)
## 328	(-0.000645, 0.00071)	(-0.091675, 0.047823)	(-0.002592, 0.010588)
## 329	(-0.000203, 0.001873)	(0.080534, 0.393791)	(-0.003716, 0.025547)
## 330	(-0.000807, 0.005146)	(0.003599, 0.043578)	(-0.034959, 0.052914)
## 331	(-0.004875, 0.006038)	(-0.053027, 0.120559)	(-0.126671, -0.003687)
## 332	(-0.000256, 0.000105)	(-0.014837, 0.012621)	(0.001663, 0.007214)
## 333	(-0.000067, 0.000082)	(-0.018561, 0.015806)	(-0.001376, 0.000338)
## 334	(-0.000099, 0.000529)	(-0.00179, 0.118094)	(-0.012306, -0.001456)
## 335	(-0.000264, 0.000544)	(-0.011502, 0.146871)	(-0.013111, -0.000093)

## 336	(-0.001046, 0.000819)	(-0.100394, 0.059923)	(-0.016027, 0.033228)
## 337	(-0.000197, 0.000665)	(-0.033879, 0.059108)	(-0.009017, 0.000714)
## 338	(-0.006458, 0.006673)	(-0.248867, 0.271811)	(-0.187309, 0.011347)
## 339	(-0.001051, 0.001506)	(-0.085396, 0.159008)	(-0.007509, 0.031908)
## 340	(-0.000094, 0.000405)	(-0.00473, 0.03427)	(-0.00143, 0.011462)
## 341	(-0.003178, 0.00106)	(-0.664001, 0.031062)	(-0.027102, 0.010611)
## 342	(-0.000301, 0.000821)	(-0.042571, 0.180326)	(-0.023082, 0.009912)
## 343	(-0.000635, -0.000001)	(-0.122461, -0.024913)	(-0.003587, 0.005057)
## 344	(-0.002868, 0.00187)	(-0.414438, 0.310401)	(-0.00026, 0.068966)
## 345	(-0.001045, 0.000837)	(-0.231033, 0.052124)	(-0.046085, 0.00984)
## 346	(-0.000055, 0.000133)	(-0.015271, 0.020449)	(-0.001593, 0.000748)
## 347	(-0.000965, 0.000684)	(-0.120162, 0.069796)	(-0.007914, 0.006056)
## 348	(-0.003285, 0.003109)	(-0.649643, 0.386847)	(-0.074741, -0.008645)
## 349	(-0.000163, 0.000168)	(-0.021832, 0.047913)	(-0.002276, 0.001842)
## 350	(0.000035, 0.000419)	(0.020193, 0.072965)	(-0.002653, 0.002004)
## 351	(-0.000947, 0.000694)	(-0.100686, 0.060079)	(-0.024287, 0.011396)
## 352	(-0.000315, 0.000042)	(-0.077891, 0.004968)	(-0.003749, -0.000104)
## 353	(-0.000927, 0.000496)	(-0.156175, 0.115104)	(-0.024528, 0.005832)
## 354	(-0.00254, 0.000269)	(-0.474564, -0.067691)	(-0.069431, -0.008998)
## 355	(-0.000195, 0.000794)	(0.001735, 0.048449)	(-0.016657, 0.002441)
## 356	(0.102879, 1.003689)	(0.026093, 0.077582)	(-1.605395, 6.886842)
## 357	(-0.006284, 0.001222)	(-0.043972, -0.005882)	(-0.019053, 0.077532)
## 358	(-0.001405, 0.000255)	(-0.168024, -0.016863)	(-0.010252, 0.026756)
## 359	(-0.004844, 0.002984)	(-0.317734, 0.174736)	(-0.015569, 0.047907)
## 360	(0.000062, 0.000562)	(0.018511, 0.075202)	(0.004461, 0.013233)
## 361	(-0.000944, 0.000889)	(-0.028171, 0.032475)	(-0.006044, 0.018626)
## 362	(-0.004936, 0.006854)	(-0.007568, 0.023987)	(-0.127119, 0.04862)
## 363	(-0.000122, 0.003172)	(0.025182, 0.600366)	(-0.037745, 0.030531)
## 364	(-0.001992, 0.002691)	(-0.06963, 0.343445)	(-0.031888, 0.047231)
## 365	(-0.000177, 0.000231)	(-0.012309, 0.062909)	(-0.002572, 0.001635)
## 366	(-0.000713, 0.000834)	(-0.110552, 0.192157)	(-0.015197, 0.007481)
## 367	(-0.000639, 0.000051)	(-0.116658, -0.006283)	(0.000909, 0.023061)
## 368	(-0.001994, 0.001347)	(-0.187409, 0.098856)	(-0.014266, 0.032174)
## 369	(-0.001808, 0.000905)	(-0.226165, 0.093838)	(-0.021129, 0.037711)
## 370	(-0.000293, 0.000364)	(-0.077351, 0.057555)	(-0.003088, 0.006835)
## 371	(-0.000917, 0.000729)	(-0.054737, 0.131339)	(-0.008702, 0.032926)
## 372	(-0.001329, 0.000992)	(-0.23482, 0.144447)	(-0.018467, 0.014012)
## 373	(-0.000217, 0.000274)	(-0.013195, 0.076535)	(-0.003286, 0.003103)
## 374	(-0.00114, 0.001723)	(-0.089801, 0.13255)	(-0.014303, 0.025691)
## 375	(-0.000464, 0.000762)	(-0.075177, 0.095792)	(-0.005029, 0.017562)
## 376	(-0.000052, 0.000293)	(0.012054, 0.086202)	(-0.001775, 0.003104)
## 377	(-0.000617, 0.003627)	(-0.057627, 0.151392)	(-0.021332, 0.081272)
## 378	(-0.000536, 0.000761)	(-0.081771, 0.0949)	(-0.008379, 0.004673)
## 379	(-0.006552, 0.013451)	(-0.006479, 0.030695)	(-0.173359, 0.00311)
## 380	(-0.000346, 0.000003)	(-0.065715, -0.008555)	(-0.003577, 0.002413)

## 381	(-0.000494, 0.000446)	(-0.147542, 0.03805)	(-0.009389, 0.008558)
## 382	(-0.000356, 0.000347)	(-0.01425, 0.011277)	(-0.007333, -0.002663)
## 383	(-0.000204, 0.000209)	(-0.045714, 0.018832)	(-0.003014, 0.002759)
## 384	(-0.00008, 0.000249)	(0.017306, 0.067717)	(-0.000076, 0.003765)
## 385	(-0.000497, 0.000225)	(-0.110691, 0.057377)	(-0.012557, 0.011265)
## 386	(-0.001407, 0.001887)	(-0.148741, 0.150248)	(-0.032997, 0.009524)
## 387	(0.000016, 0.007752)	(0.023501, 0.140367)	(-0.045394, 0.058705)
## 388	(-0.000446, 0.001154)	(-0.035932, 0.099651)	(-0.051864, -0.019419)
## 389	(-0.000982, 0.007342)	(-0.122699, 0.428475)	(-0.046806, 0.079656)
## 390	(-0.001804, 0.000717)	(-0.201892, 0.065382)	(-0.014501, 0.018592)
## 391	(-0.001115, 0.002773)	(-0.018182, 0.235072)	(-0.008698, 0.031679)
## 392	(-0.002835, 0.001218)	(-0.027895, 0.012489)	(0.000796, 0.045452)
## 393	(-0.000235, 0.000168)	(-0.064401, 0.036025)	(-0.001512, 0.002362)
## 394	(-0.000163, 0.000073)	(-0.031946, 0.021541)	(0.000331, 0.003047)
## 395	(-0.001623, 0.001845)	(-0.24219, 0.097395)	(-0.018293, 0.023187)
## 396	(-0.000465, 0.000283)	(-0.119686, 0.0257)	(-0.001214, 0.010375)
## 397	(-0.00101, 0.004935)	(-0.009253, 0.033665)	(-0.056004, -0.006971)
## 398	(-0.000579, 0.000219)	(-0.125889, -0.007661)	(-0.009999, -0.000126)
## 399	(-0.001265, 0.001302)	(-0.075061, 0.021338)	(-0.019272, 0.007855)
## 400	(-0.00035, 0.00087)	(-0.024554, 0.061656)	(-0.002844, 0.010842)
## 401	(-0.000041, 0.000395)	(-0.002258, 0.025303)	(-0.002988, 0.002699)
## 402	(-0.003141, 0.004789)	(-0.011551, 0.025133)	(-0.060666, 0.024122)
## 403	(-0.000147, 0.000206)	(-0.051181, 0.020159)	(-0.009131, 0.00265)
## 404	(-0.000419, 0.000418)	(-0.060649, 0.033126)	(-0.009373, 0.00263)
## 405	(-0.000307, 0.000394)	(-0.080433, 0.012738)	(-0.000616, 0.008081)
## 406	(-0.003289, 0.000681)	(-0.445429, -0.007071)	(0.010792, 0.082259)
## 407	(0.000003, 0.000401)	(0.014417, 0.077085)	(-0.001957, 0.002843)
## 408	(-0.000404, 0.000349)	(-0.099501, -0.002159)	(-0.000567, 0.011922)
## 409	(-0.000717, -0.000037)	(-0.098441, -0.012939)	(-0.005555, 0.004698)
## 410	(-0.003611, 0.002896)	(-0.304507, 0.17968)	(-0.016571, 0.032214)
## 411	(-0.001519, 0.011446)	(0.005715, 0.051706)	(-0.008326, 0.102125)
## 412	(-0.000277, 0.000255)	(-0.042342, 0.046908)	(-0.019319, -0.000513)
## 413	(-0.000178, 0.000367)	(-0.006009, 0.046689)	(-0.008577, 0.000393)
## 414	(-0.000127, 0.000225)	(0.001888, 0.033428)	(-0.001244, 0.004902)
## 415	(-0.000088, 0.000223)	(0.006914, 0.03205)	(-0.001947, 0.002427)
## 416	(-0.032202, 0.035298)	(-0.02539, 0.043109)	(-0.747216, 0.196204)
## 417	(-0.000541, 0.000592)	(-0.073045, 0.043496)	(-0.005748, 0.016248)
## 418	(-0.001087, 0.000166)	(-0.258861, -0.008635)	(-0.001182, 0.014376)
## 419	(-0.000415, 0.002738)	(0.047543, 0.333612)	(-0.012414, 0.033358)
## 420	(-0.005557, 0.001412)	(-0.101798, 0.017451)	(-0.018648, 0.039804)
## 421	(-0.000467, 0.00014)	(-0.112671, 0.001664)	(-0.004766, 0.002661)
## 422	(-0.000109, 0.000456)	(0.018774, 0.138583)	(-0.004786, -0.000694)
## 423	(-0.000627, 0.000521)	(-0.063267, 0.094738)	(-0.050678, 0.011323)
## 424	(-0.002645, 0.002621)	(-0.094414, 0.255531)	(-0.047331, 0.005523)
## 425	(-0.000792, 0.005393)	(-0.092125, 0.425915)	(-0.038232, 0.024778)

## 426	(-0.002432, 0.00166)	(-0.089364, 0.046546)	(-0.03002, 0.017745)
## 427	(-0.003514, 0.00151)	(-0.20865, 0.024412)	(-0.018542, 0.053574)
## 428	(-0.000906, 0.000842)	(-0.027199, 0.006856)	(-0.023734, 0.004891)
## 429	(-0.003881, 0.002445)	(-0.199995, 0.133718)	(-0.095028, 0.016981)
## 430	(0.000024, 0.000818)	(0.073038, 0.191176)	(-0.007, 0.0075)
## 431	(-0.00016, 0.00017)	(-0.03033, 0.022716)	(-0.001183, 0.002171)
## 432	(-0.000182, 0.000105)	(-0.03553, 0.016095)	(-0.002915, 0.001932)
## 433	(-0.001547, 0.006177)	(-0.029693, 0.196359)	(-0.072439, 0.030492)
## 434	(-0.000775, 0.000455)	(-0.049706, 0.047272)	(0.004161, 0.0346)
## 435	(-0.000039, 0.000256)	(-0.002228, 0.031758)	(-0.002135, 0.000546)
## 436	(-0.000339, 0.000298)	(-0.058097, 0.051366)	(-0.008104, 0.000257)
## 437	(-0.000023, 0.001429)	(0.004866, 0.272066)	(-0.006084, 0.008679)
## 438	(-0.001359, 0.001909)	(-0.099833, 0.214749)	(-0.013428, 0.08665)
## 439	(-0.234712, 0.040921)	(-0.034859, 0.008615)	(-4.632542, 0.633315)
## 440	(-0.000857, 0.002273)	(-0.078789, 0.191791)	(0.001423, 0.063266)
## 441	(-0.00399, 0.00573)	(-0.068273, 0.026829)	(-0.114116, 0.021667)
## 442	(-0.003182, 0.000289)	(-0.170327, 0.006045)	(-0.047968, 0.013359)
## 443	(-0.001042, 0.001035)	(-0.065022, 0.083085)	(-0.019067, 0.001999)
## 444	(-0.000166, 0.000154)	(-0.008095, 0.016799)	(-0.004477, -0.000243)
## 445	(-0.001797, 0.002752)	(-0.126755, 0.128631)	(-0.015492, 0.029397)
## 446	(-0.004135, 0.002122)	(-0.180917, 0.094895)	(-0.005737, 0.064039)
## 447	(-0.00496, 0.005316)	(-0.094836, 0.08423)	(0.006543, 0.116883)
## 448	(-0.000277, 0.000322)	(-0.051352, 0.03496)	(-0.004291, 0.003115)
## 449	(-0.002466, 0.00024)	(-0.183531, 0.01268)	(-0.032506, 0.019672)
## 450	(-0.000141, 0.000176)	(-0.018665, 0.030159)	(-0.000002, 0.003916)
## 451	(-0.000271, 0.000089)	(-0.037909, 0.040419)	(-0.004883, -0.001029)
## 452	(-0.000295, 0.000125)	(-0.051942, 0.010339)	(-0.007911, 0.001667)
## 453	(-0.005106, 0.001255)	(-0.206358, 0.031734)	(-0.073334, -0.012779)
## 454	(-0.00249, 0.000792)	(-0.050912, 0.026777)	(0.008875, 0.056181)
## 455	(-0.000848, 0.000916)	(-0.053454, 0.051791)	(-0.03867, 0.010308)
## 456	(-0.00028, 0.000316)	(-0.054702, 0.049154)	(-0.009085, -0.00015)
## 457	(-0.24091, 0.066316)	(-0.029237, 0.002674)	(-1.897978, 1.012233)
## 458	(-0.000094, 0.000085)	(-0.019413, 0.019638)	(-0.003415, -0.000738)
## 459	(-0.000468, 0.002508)	(0.120105, 0.748783)	(-0.008308, 0.022576)
## 460	(-0.000358, -0.000021)	(-0.08189, -0.034969)	(-0.001929, 0.000636)
## 461	(-0.004295, 0.093616)	(-0.004756, 0.043823)	(-0.800535, 0.933133)
## 462	(-0.000832, 0.000228)	(-0.109723, 0.011236)	(-0.006028, 0.011307)
## 463	(-0.00022, 0.001238)	(-0.000663, 0.121905)	(-0.007449, 0.0101)
## 464	(-0.000432, 0.000117)	(-0.044448, 0.005125)	(-0.016482, 0.001495)
## 465	(-0.000608, 0.00038)	(-0.016791, 0.020603)	(-0.005722, 0.017236)
## 466	(-0.000178, 0.000865)	(-0.025108, 0.084713)	(-0.008729, 0.006572)
## 467	(-0.00322, 0.000637)	(-0.100853, -0.00327)	(-0.082733, -0.02789)
## 468	(-0.002805, 0.003107)	(-0.24114, 0.092354)	(-0.050789, 0.007902)
## 469	(-0.60605, -0.013398)	(-0.058596, -0.012793)	(-2.290509, 0.406643)
## 470	(-0.000622, 0.001234)	(-0.015273, 0.105941)	(-0.005607, 0.005596)

## 471	(-0.000793, 0.000985)	(-0.060869, 0.118958)	(-0.048879, -0.003406)
## 472	(-0.003887, 0.001133)	(-0.231559, 0.004555)	(-0.049394, 0.038141)
## 473	(-0.000335, 0.000134)	(-0.032526, 0.062518)	(-0.001502, 0.003834)
## 474	(-0.000659, -0.000155)	(-0.051821, -0.009519)	(-0.00074, 0.004606)
## 475	(-0.002759, 0.003647)	(-0.182176, 0.393125)	(-0.03736, 0.062361)
## 476	(-0.000196, 0.000161)	(-0.036765, 0.011452)	(-0.005898, -0.000679)
## 477	(-0.073742, 0.072595)	(-0.041259, 0.005182)	(-0.709193, 1.471713)
## 478	(-0.000078, 0.000187)	(-0.014994, 0.038433)	(-0.005467, -0.000116)
## 479	(-0.00219, 0.00277)	(-0.040618, 0.039545)	(-0.050574, 0.017798)
## 480	(-0.00026, 0.000421)	(-0.03012, 0.06174)	(-0.003351, 0.006508)
## 481	(-0.006151, 0.001487)	(-0.02461, 0.010516)	(-0.006331, 0.055115)
## 482	(-0.000471, 0.000992)	(-0.052631, 0.035065)	(-0.012811, 0.005489)
## 483	(-0.002664, 0.001131)	(-0.30897, 0.084767)	(-0.030685, -0.000313)
## 484	(-0.356574, 0.328863)	(-0.019943, 0.010648)	(-2.16253, 0.722914)
## 485	(-0.005182, 0.003847)	(-0.0408, 0.021323)	(-0.098483, -0.015967)
## 486	(-0.000226, 0.000635)	(-0.01131, 0.043243)	(-0.004477, 0.016646)
## 487	(-0.000285, 0.001177)	(-0.015263, 0.190794)	(-0.007911, 0.019825)
## 488	(-0.002948, 0.001156)	(-0.555727, -0.007697)	(-0.053542, -0.01022)
## 489	(-0.000525, 0.001086)	(-0.083461, 0.133885)	(-0.017232, 0.021689)
## 490	(-0.000409, 0.00019)	(-0.080131, 0.02597)	(-0.002931, 0.00494)
## 491	(-0.00062, 0.000684)	(-0.148373, 0.052938)	(0.002234, 0.036771)
## 492	(-0.000393, 0.000509)	(-0.007205, 0.09306)	(-0.008449, -0.000457)
## 493	(-0.000769, 0.00169)	(-0.064252, 0.144897)	(-0.026807, 0.002227)
## 494	(-0.000211, 0.000067)	(-0.023826, 0.009176)	(-0.002073, -0.000209)
## 495	(-0.000887, 0.00106)	(-0.029368, 0.069038)	(-0.045189, 0.020824)
## 496	(-0.000297, 0.000605)	(-0.033394, 0.085767)	(-0.017603, 0.002082)
## 497	(-0.002421, 0.004725)	(-0.007773, 0.027585)	(-0.040473, 0.054529)
## 498	(-0.000582, 0.000138)	(-0.020032, 0.010817)	(-0.007069, 0.003349)
## 499	(-0.000411, 0.006976)	(-0.001443, 0.174627)	(-0.04612, 0.010825)
## 500	(-0.002251, 0.000884)	(-0.275974, 0.101209)	(-0.026391, 0.035645)
## 501	(0.000023, 0.000951)	(0.00091, 0.084872)	(-0.005495, 0.007616)
## 502	(-0.28634, 0.080591)	(-0.034314, 0.006341)	(-2.396894, 0.48094)
## 503	(-0.000376, 0.001646)	(-0.034689, 0.258759)	(-0.013957, 0.015448)
## 504	(-0.004808, 0.000972)	(-0.030439, 0.012633)	(-0.079507, 0.007506)
## 505	(-0.000255, 0.001168)	(0.045833, 0.214422)	(0.006906, 0.032587)
## 506	(-0.0008, 0.001102)	(-0.048124, 0.101847)	(-0.031978, 0.009421)
## 507	(-0.000507, 0.000553)	(-0.018577, 0.017312)	(-0.010551, -0.002244)
## 508	(0.004548, 0.021155)	(0.00987, 0.03341)	(-0.105331, 0.094431)
## 509	(-0.002348, 0.002474)	(-0.071472, 0.412097)	(-0.04615, 0.012762)
## 510	(-0.004019, 0.005029)	(-0.023195, 0.056375)	(-0.053345, 0.042387)
## 511	(-0.000622, 0.000842)	(-0.046329, 0.109666)	(-0.002317, 0.032744)
## 512	(-0.00293, 0.003314)	(-0.067538, 0.073102)	(-0.007373, 0.072919)
## 513	(-0.000325, 0.00022)	(-0.012725, 0.017803)	(-0.001063, 0.003578)
## 514	(-0.000041, 0.000192)	(-0.001374, 0.026347)	(-0.002295, 0.001138)
## 515	(-0.002768, 0.00235)	(-0.113268, 0.115731)	(-0.016829, 0.047393)

## 516	(-0.006738, 0.004864)	(-0.030331, 0.035756)	(-0.038048, 0.078098)
## 517	(-0.001633, 0.001344)	(-0.267027, 0.129162)	(-0.001269, 0.048833)
## 518	(-0.000453, 0.001124)	(-0.016487, 0.068607)	(-0.00982, 0.013195)
## 519	(-0.010509, 0.01069)	(-0.014319, 0.019857)	(-0.076355, 0.262662)
## 520	(-0.001449, 0.000596)	(-0.275493, 0.019639)	(-0.007753, 0.018723)
## 521	(-0.000122, 0.000204)	(-0.010214, 0.021937)	(-0.00542, 0.022579)
## 522	(-0.074138, 0.136492)	(-0.020844, 0.022507)	(-1.96334, -0.117011)
## 523	(-0.000315, 0.002817)	(0.084429, 0.26382)	(-0.046835, -0.014145)
## 524	(-0.005267, 0.002493)	(-0.089124, 0.04547)	(-0.018601, 0.074361)
## 525	(-0.022838, 0.039081)	(-0.027567, 0.031921)	(-0.134815, 0.993314)
## 526	(-0.000357, 0.001235)	(-0.069925, 0.100364)	(-0.000105, 0.023069)
## 527	(-0.000272, 0.00092)	(0.001131, 0.067632)	(-0.01125, 0.023216)
## 528	(-0.000189, 0.003195)	(0.016681, 0.23165)	(-0.003468, 0.063244)
## 529	(-0.000634, 0.002827)	(0.059272, 0.42123)	(-0.013022, 0.018282)
## 530	(-0.0036, 0.002106)	(-0.066026, 0.015353)	(-0.040154, 0.028881)
## 531	(-0.000487, 0.000045)	(-0.032383, -0.001326)	(-0.000363, 0.004691)
## 532	(-0.003075, 0.003897)	(-0.023316, 0.09245)	(-0.075658, 0.038247)
## 533	(-0.000452, 0.002066)	(-0.01203, 0.12004)	(-0.030016, 0.000656)
## 534	(-0.144935, 0.007607)	(-0.051147, -0.011778)	(-2.827893, -1.806458)
## 535	(-0.003147, 0.002088)	(-0.06607, -0.000999)	(-0.038292, 0.029972)
## 536	(-0.058467, 0.134902)	(-0.011849, 0.026989)	(-1.394658, 0.854026)
## 537	(-0.004845, 0.002714)	(-0.027365, 0.006704)	(-0.066685, 0.052731)
## 538	(-0.001061, 0.004365)	(-0.083466, 0.256489)	(-0.038824, 0.048139)
## 539	(-0.003972, 0.002214)	(-0.054484, 0.020146)	(-0.114328, 0.00875)
## 540	(-0.002878, 0.004467)	(-0.020193, 0.014389)	(-0.060018, 0.012785)
## 541	(-0.002287, 0.000737)	(-0.231374, 0.008093)	(-0.053305, 0.017695)
## 542	(-0.084918, 0.020509)	(-0.021231, 0.018223)	(-0.42498, 0.792546)
## 543	(-0.004515, 0.000981)	(-0.035688, 0.001206)	(-0.020919, 0.063132)
## 544	(-0.001637, 0.00064)	(-0.140663, -0.003419)	(-0.047397, -0.001037)
## 545	(-0.001979, 0.003425)	(-0.007344, 0.022157)	(-0.03848, 0.013908)
## 546	(-0.001548, 0.002881)	(-0.052014, 0.075453)	(-0.034377, 0.012455)
## 547	(-0.001006, 0.00382)	(-0.02394, 0.074197)	(-0.005064, 0.064264)
## 548	(-0.002195, 0.002662)	(-0.294998, 0.406813)	(-0.049389, 0.041104)
## 549	<NA>	(-0.215901, 0.423368)	(-0.010781, 0.108533)
## 550	<NA>	<NA>	(-0.012315, 0.000253)
## 551	<NA>	<NA>	(0.000072, 0.003812)
## 552	<NA>	<NA>	(0.000291, 0.052837)
## 553	<NA>	<NA>	(-0.546953, 0.957498)
## 554	<NA>	<NA>	(0.000072, 0.000078)
## 555	<NA>	<NA>	(0.002644, 0.007728)
## 556	<NA>	<NA>	(-0.00015, 0.002167)
## 557	<NA>	<NA>	(0.000044, 0.002361)
## 558	<NA>	<NA>	(0.000056, 0.002375)
## 559	<NA>	<NA>	(-0.01908, 0.005007)
## 560	<NA>	<NA>	(-0.019113, 0.004952)

## 561	<NA>	<NA>	(-0.019066, 0.005024)
## 562	<NA>	<NA>	(-6.282727, -6.282427)
## 563	<NA>	<NA>	(16079.83684, 16175.951294)

##	Parameter	Rhat
## 1	Intercept	1.012724
## 2	Neuron_3	1.001526
## 3	Neuron_4	1.001907
## 4	Neuron_5	1.001037
## 5	Neuron_6	0.999662
## 6	Neuron_7	1.003088
## 7	Neuron_8	1.001508
## 8	Neuron_9	1.000785
## 9	Neuron_11	1.007624
## 10	Neuron_12	1.001597
## 11	Neuron_13	1.002114
## 12	Neuron_14	1.002347
## 13	Neuron_15	1.008048
## 14	Neuron_16	0.999875
## 15	Neuron_17	1.000421
## 16	Neuron_18	1.003997
## 17	Neuron_19	1.003504
## 18	Neuron_20	1.000871
## 19	Neuron_21	1.001423
## 20	Neuron_22	1.001490
## 21	Neuron_23	1.002702
## 22	Neuron_24	1.002111
## 23	Neuron_25	1.001530
## 24	Neuron_26	1.004694
## 25	Neuron_27	1.000253
## 26	Neuron_28	1.002758
## 27	Neuron_29	1.001974
## 28	Neuron_30	1.003352
## 29	Neuron_31	1.009536
## 30	Neuron_32	1.000012
## 31	Neuron_33	1.001288
## 32	Neuron_34	1.005041
## 33	Neuron_35	1.003098
## 34	Neuron_36	1.002750
## 35	Neuron_37	1.005540
## 36	Neuron_38	1.001177
## 37	Neuron_39	1.001284
## 38	Neuron_40	1.000575
## 39	Neuron_41	1.004506
## 40	Neuron_42	1.002018

## 41	Neuron_43	1.012392
## 42	Neuron_44	1.002165
## 43	Neuron_45	1.001375
## 44	Neuron_46	1.000418
## 45	Neuron_47	1.000190
## 46	Neuron_48	1.009474
## 47	Neuron_50	1.000075
## 48	Neuron_51	1.000135
## 49	Neuron_52	1.003508
## 50	Neuron_53	1.005457
## 51	Neuron_54	0.999705
## 52	Neuron_55	1.009322
## 53	Neuron_56	1.000523
## 54	Neuron_57	1.001724
## 55	Neuron_58	1.001552
## 56	Neuron_59	1.001006
## 57	Neuron_60	1.004699
## 58	Neuron_61	1.004281
## 59	Neuron_62	1.000791
## 60	Neuron_63	1.004615
## 61	Neuron_64	1.000211
## 62	Neuron_65	1.008853
## 63	Neuron_66	1.000006
## 64	Neuron_67	1.000285
## 65	Neuron_68	1.008065
## 66	Neuron_69	1.004352
## 67	Neuron_70	1.036860
## 68	Neuron_71	1.005539
## 69	Neuron_72	1.003675
## 70	Neuron_73	1.000334
## 71	Neuron_74	0.999746
## 72	Neuron_75	1.092189
## 73	Neuron_76	1.002349
## 74	Neuron_77	1.008532
## 75	Neuron_78	1.001258
## 76	Neuron_79	1.004178
## 77	Neuron_80	1.006017
## 78	Neuron_81	1.002295
## 79	Neuron_82	1.000585
## 80	Neuron_83	1.001174
## 81	Neuron_84	1.000629
## 82	Neuron_85	1.001529
## 83	Neuron_86	1.000587
## 84	Neuron_87	1.003787
## 85	Neuron_88	1.000865

## 86	Neuron_89	1.123057
## 87	Neuron_90	1.006580
## 88	Neuron_91	1.000545
## 89	Neuron_93	1.001646
## 90	Neuron_94	1.007372
## 91	Neuron_95	1.004094
## 92	Neuron_96	1.003962
## 93	Neuron_97	1.043027
## 94	Neuron_98	1.005337
## 95	Neuron_99	0.999786
## 96	Neuron_100	1.001801
## 97	Neuron_101	1.015876
## 98	Neuron_102	0.999865
## 99	Neuron_103	1.005219
## 100	Neuron_104	1.002728
## 101	Neuron_105	1.000754
## 102	Neuron_106	1.002353
## 103	Neuron_107	1.006170
## 104	Neuron_108	1.001109
## 105	Neuron_109	1.003202
## 106	Neuron_110	1.002830
## 107	Neuron_111	1.001981
## 108	Neuron_112	1.001563
## 109	Neuron_113	1.005860
## 110	Neuron_114	1.002740
## 111	Neuron_115	1.001761
## 112	Neuron_116	1.000501
## 113	Neuron_117	1.004864
## 114	Neuron_118	1.004025
## 115	Neuron_119	1.001970
## 116	Neuron_120	1.006422
## 117	Neuron_121	1.000268
## 118	Neuron_122	1.002235
## 119	Neuron_123	1.004101
## 120	Neuron_124	1.005931
## 121	Neuron_125	1.007251
## 122	Neuron_127	1.001162
## 123	Neuron_128	1.002220
## 124	Neuron_129	1.000415
## 125	Neuron_130	1.001800
## 126	Neuron_131	1.007468
## 127	Neuron_132	1.002674
## 128	Neuron_133	1.006537
## 129	Neuron_134	1.001440
## 130	Neuron_135	1.006053

## 131	Neuron_136	1.010384
## 132	Neuron_137	1.001824
## 133	Neuron_138	0.999617
## 134	Neuron_139	1.003278
## 135	Neuron_140	1.002859
## 136	Neuron_141	1.019049
## 137	Neuron_142	0.999767
## 138	Neuron_143	1.005673
## 139	Neuron_144	1.000422
## 140	Neuron_145	1.000759
## 141	Neuron_146	1.001968
## 142	Neuron_147	1.002517
## 143	Neuron_148	1.002833
## 144	Neuron_149	1.002533
## 145	Neuron_150	1.000738
## 146	Neuron_151	1.003787
## 147	Neuron_152	1.011384
## 148	Neuron_153	1.000387
## 149	Neuron_154	1.000223
## 150	Neuron_155	1.002047
## 151	Neuron_156	1.000086
## 152	Neuron_157	1.003446
## 153	Neuron_158	1.003944
## 154	Neuron_159	1.000348
## 155	Neuron_160	1.002097
## 156	Neuron_161	0.999913
## 157	Neuron_162	1.001122
## 158	Neuron_163	1.004139
## 159	Neuron_164	1.042922
## 160	Neuron_165	1.000432
## 161	Neuron_166	1.000561
## 162	Neuron_167	1.000286
## 163	Neuron_169	1.001400
## 164	Neuron_170	1.002002
## 165	Neuron_171	1.002624
## 166	Neuron_172	1.003478
## 167	Neuron_173	1.001860
## 168	Neuron_174	1.003765
## 169	Neuron_175	1.006601
## 170	Neuron_176	1.003140
## 171	Neuron_177	1.003779
## 172	Neuron_180	1.004493
## 173	Neuron_181	1.000453
## 174	Neuron_182	1.003874
## 175	Neuron_183	1.007070

## 176	Neuron_184	1.000783
## 177	Neuron_185	1.017315
## 178	Neuron_186	1.023808
## 179	Neuron_187	1.002203
## 180	Neuron_189	1.040473
## 181	Neuron_190	1.003089
## 182	Neuron_191	1.014301
## 183	Neuron_192	1.001267
## 184	Neuron_193	1.047951
## 185	Neuron_194	1.026273
## 186	Neuron_196	1.050667
## 187	Neuron_197	1.040594
## 188	Neuron_198	1.000411
## 189	Neuron_199	1.004865
## 190	Neuron_200	1.000192
## 191	Neuron_202	1.002499
## 192	Neuron_204	1.003175
## 193	Neuron_205	1.004622
## 194	Neuron_207	1.007259
## 195	Neuron_208	1.000505
## 196	Neuron_209	1.028031
## 197	Neuron_210	1.027697
## 198	Neuron_211	1.001202
## 199	Neuron_212	1.000628
## 200	Neuron_213	1.001343
## 201	Neuron_214	1.003643
## 202	Neuron_215	1.001231
## 203	Neuron_216	1.006209
## 204	Neuron_218	1.002351
## 205	Neuron_219	1.028906
## 206	Neuron_220	1.000704
## 207	Neuron_221	1.003615
## 208	Neuron_222	1.007131
## 209	Neuron_223	1.034648
## 210	Neuron_224	1.005372
## 211	Neuron_225	1.003199
## 212	Neuron_227	1.001334
## 213	Neuron_228	1.019758
## 214	Neuron_229	1.004641
## 215	Neuron_230	1.000141
## 216	Neuron_231	1.023796
## 217	Neuron_232	1.025001
## 218	Neuron_233	1.001725
## 219	Neuron_235	1.008084
## 220	Neuron_236	1.024364

## 221	Neuron_237	1.006973
## 222	Neuron_238	1.004133
## 223	Neuron_239	1.000442
## 224	Neuron_240	1.001079
## 225	Neuron_241	1.000484
## 226	Neuron_242	1.000349
## 227	Neuron_244	1.003294
## 228	Neuron_246	1.000877
## 229	Neuron_247	1.001873
## 230	Neuron_248	1.001207
## 231	Neuron_249	1.000156
## 232	Neuron_250	1.001175
## 233	Neuron_251	1.001721
## 234	Neuron_253	1.019297
## 235	Neuron_254	1.008482
## 236	Neuron_255	1.002499
## 237	Neuron_256	1.002450
## 238	Neuron_257	1.000666
## 239	Neuron_258	1.006214
## 240	Neuron_259	1.000816
## 241	Neuron_260	1.025785
## 242	Neuron_261	1.001127
## 243	Neuron_262	1.003008
## 244	Neuron_264	1.004978
## 245	Neuron_265	1.006831
## 246	Neuron_266	1.002296
## 247	Neuron_267	1.002558
## 248	Neuron_268	1.036808
## 249	Neuron_269	1.019394
## 250	Neuron_270	1.010984
## 251	Neuron_271	1.011471
## 252	Neuron_273	1.004453
## 253	Neuron_274	1.011788
## 254	Neuron_275	1.008584
## 255	Neuron_276	1.002758
## 256	Neuron_277	1.001248
## 257	Neuron_278	1.016832
## 258	Neuron_279	1.000996
## 259	Neuron_280	1.001174
## 260	Neuron_282	1.004527
## 261	Neuron_283	1.010018
## 262	Neuron_284	1.000364
## 263	Neuron_285	1.026253
## 264	Neuron_286	1.001013
## 265	Neuron_287	1.000226

## 266	Neuron_288	1.001545
## 267	Neuron_289	1.023493
## 268	Neuron_290	1.034307
## 269	Neuron_291	1.004358
## 270	Neuron_292	1.001563
## 271	Neuron_293	1.000198
## 272	Neuron_294	1.000912
## 273	Neuron_295	1.001641
## 274	Neuron_296	1.033225
## 275	Neuron_297	1.001599
## 276	Neuron_298	1.037058
## 277	Neuron_299	1.015180
## 278	Neuron_300	0.999727
## 279	Neuron_301	1.007847
## 280	Neuron_302	1.000256
## 281	Neuron_303	1.001697
## 282	Neuron_304	1.017732
## 283	Neuron_306	1.031356
## 284	Neuron_308	1.004190
## 285	Neuron_309	1.000513
## 286	Neuron_311	1.003036
## 287	Neuron_312	1.003036
## 288	Neuron_314	1.002040
## 289	Neuron_315	1.000010
## 290	reach	1.861013
## 291	Neuron_3:reach	1.810027
## 292	Neuron_4:reach	1.585849
## 293	Neuron_5:reach	1.274722
## 294	Neuron_6:reach	1.523542
## 295	Neuron_7:reach	1.384130
## 296	Neuron_8:reach	2.189141
## 297	Neuron_9:reach	1.332224
## 298	Neuron_12:reach	1.719404
## 299	Neuron_13:reach	1.350488
## 300	Neuron_14:reach	1.260879
## 301	Neuron_16:reach	1.899279
## 302	Neuron_17:reach	1.813801
## 303	Neuron_18:reach	1.596550
## 304	Neuron_19:reach	1.477561
## 305	Neuron_20:reach	1.709362
## 306	Neuron_21:reach	1.812095
## 307	Neuron_22:reach	2.285873
## 308	Neuron_23:reach	1.967391
## 309	Neuron_24:reach	2.094183
## 310	Neuron_25:reach	1.617153

```

## 311      Neuron_26:reach 1.353875
## 312      Neuron_27:reach 1.457150
## 313      Neuron_28:reach 1.217639
## 314      Neuron_29:reach 2.208310
## 315      Neuron_30:reach 1.908639
## 316      Neuron_32:reach 2.052224
## 317      Neuron_33:reach 1.453770
## 318      Neuron_34:reach 1.137418
## 319      Neuron_35:reach 1.874775
## 320      Neuron_36:reach 1.724659
## 321      Neuron_37:reach 1.434475
## 322      Neuron_38:reach 1.466666
## 323      Neuron_39:reach 2.232128
## 324      Neuron_40:reach 1.643701
## 325      Neuron_41:reach 1.362715
## 326      Neuron_42:reach 1.414106
## 327      Neuron_43:reach 1.591131
## 328      Neuron_44:reach 1.181290
## 329      Neuron_45:reach 1.243380
## 330      Neuron_46:reach 1.839556
## 331      Neuron_47:reach 1.826700
## 332      Neuron_48:reach 1.683818
## 333      Neuron_50:reach 1.144886
## 334      Neuron_51:reach 1.298581
## 335      Neuron_52:reach 1.814378
## 336      Neuron_53:reach 1.403657
## 337      Neuron_54:reach 1.744303
## 338      Neuron_55:reach 1.137787
## 339      Neuron_56:reach 2.043181
## 340      Neuron_57:reach 1.836658
## 341      Neuron_58:reach 1.809030
## 342      Neuron_59:reach 1.133724
## 343      Neuron_60:reach 2.139508
## 344      Neuron_61:reach 1.771717
## 345      Neuron_62:reach 1.333827
## 346      Neuron_63:reach 1.662484
## 347      Neuron_64:reach 1.367313
## 348      Neuron_65:reach 1.152594
## 349      Neuron_66:reach 1.465553
## 350      Neuron_67:reach 1.835129
## 351      Neuron_68:reach 1.373990
## 352      Neuron_69:reach 1.463997
## 353      Neuron_71:reach 1.113800
## 354      Neuron_72:reach 1.162054
## 355      Neuron_73:reach 1.682855

```

356 Neuron_74:reach 1.329219
357 Neuron_75:reach 2.314838
358 Neuron_76:reach 1.954513
359 Neuron_77:reach 1.516919
360 Neuron_78:reach 1.511346
361 Neuron_79:reach 1.657756
362 Neuron_80:reach 1.826488
363 Neuron_81:reach 2.379398
364 Neuron_82:reach 2.002199
365 Neuron_83:reach 2.099116
366 Neuron_84:reach 1.287203
367 Neuron_85:reach 1.839807
368 Neuron_86:reach 1.356243
369 Neuron_87:reach 1.485067
370 Neuron_88:reach 1.804938
371 Neuron_90:reach 1.415017
372 Neuron_91:reach 1.821010
373 Neuron_93:reach 1.748690
374 Neuron_94:reach 1.349703
375 Neuron_95:reach 1.862953
376 Neuron_96:reach 1.209402
377 Neuron_98:reach 1.353608
378 Neuron_99:reach 1.740995
379 Neuron_100:reach 1.285982
380 Neuron_101:reach 1.089557
381 Neuron_102:reach 1.648293
382 Neuron_103:reach 1.933064
383 Neuron_104:reach 1.060199
384 Neuron_105:reach 1.891725
385 Neuron_106:reach 1.061176
386 Neuron_107:reach 1.583558
387 Neuron_108:reach 1.217737
388 Neuron_109:reach 1.767787
389 Neuron_110:reach 1.307640
390 Neuron_111:reach 1.885718
391 Neuron_112:reach 1.462825
392 Neuron_113:reach 1.825860
393 Neuron_115:reach 1.093391
394 Neuron_116:reach 1.179611
395 Neuron_117:reach 1.327058
396 Neuron_118:reach 1.131766
397 Neuron_119:reach 1.497660
398 Neuron_120:reach 1.363517
399 Neuron_121:reach 1.695285
400 Neuron_122:reach 1.086660

401 Neuron_123:reach 1.462845
402 Neuron_124:reach 1.545337
403 Neuron_125:reach 1.754769
404 Neuron_127:reach 2.186683
405 Neuron_128:reach 1.857606
406 Neuron_129:reach 1.195656
407 Neuron_131:reach 1.765657
408 Neuron_132:reach 1.774563
409 Neuron_133:reach 2.026787
410 Neuron_134:reach 1.159424
411 Neuron_135:reach 1.125019
412 Neuron_136:reach 1.285059
413 Neuron_137:reach 1.335999
414 Neuron_138:reach 1.517290
415 Neuron_139:reach 1.889002
416 Neuron_140:reach 2.190181
417 Neuron_141:reach 1.284897
418 Neuron_142:reach 2.171949
419 Neuron_143:reach 1.588779
420 Neuron_144:reach 1.803218
421 Neuron_145:reach 1.119920
422 Neuron_146:reach 1.441890
423 Neuron_147:reach 1.043629
424 Neuron_148:reach 1.649627
425 Neuron_149:reach 1.309599
426 Neuron_150:reach 1.352201
427 Neuron_151:reach 1.477320
428 Neuron_152:reach 1.496166
429 Neuron_153:reach 1.264512
430 Neuron_154:reach 1.514729
431 Neuron_155:reach 2.266144
432 Neuron_156:reach 1.213108
433 Neuron_157:reach 1.846228
434 Neuron_158:reach 1.777564
435 Neuron_159:reach 1.906378
436 Neuron_160:reach 1.185356
437 Neuron_161:reach 1.380925
438 Neuron_162:reach 1.029260
439 Neuron_163:reach 2.114794
440 Neuron_164:reach 2.917383
441 Neuron_165:reach 2.039824
442 Neuron_166:reach 1.354319
443 Neuron_167:reach 1.856123
444 Neuron_169:reach 1.177942
445 Neuron_170:reach 1.656235

446 Neuron_171:reach 1.266074
447 Neuron_172:reach 1.345043
448 Neuron_173:reach 1.389643
449 Neuron_174:reach 1.475691
450 Neuron_175:reach 1.604033
451 Neuron_176:reach 1.383574
452 Neuron_177:reach 1.104276
453 Neuron_180:reach 1.764510
454 Neuron_181:reach 1.190922
455 Neuron_182:reach 1.451104
456 Neuron_183:reach 2.005050
457 Neuron_184:reach 1.678215
458 Neuron_186:reach 2.012910
459 Neuron_187:reach 1.340545
460 Neuron_190:reach 1.193593
461 Neuron_192:reach 1.344120
462 Neuron_193:reach 1.998449
463 Neuron_198:reach 1.692679
464 Neuron_199:reach 1.850648
465 Neuron_200:reach 1.761763
466 Neuron_202:reach 1.627370
467 Neuron_204:reach 1.192106
468 Neuron_205:reach 1.541626
469 Neuron_208:reach 1.485818
470 Neuron_209:reach 1.348561
471 Neuron_211:reach 1.122495
472 Neuron_212:reach 1.687540
473 Neuron_213:reach 1.847291
474 Neuron_214:reach 1.234669
475 Neuron_215:reach 1.135820
476 Neuron_216:reach 1.566646
477 Neuron_218:reach 1.566360
478 Neuron_219:reach 2.183267
479 Neuron_220:reach 1.894829
480 Neuron_221:reach 1.338432
481 Neuron_222:reach 1.385450
482 Neuron_224:reach 1.419921
483 Neuron_225:reach 1.881604
484 Neuron_227:reach 1.058649
485 Neuron_228:reach 2.024469
486 Neuron_229:reach 1.252515
487 Neuron_230:reach 2.082965
488 Neuron_233:reach 1.672421
489 Neuron_235:reach 1.266700
490 Neuron_237:reach 1.616914

491 Neuron_238:reach 1.687256
492 Neuron_239:reach 2.618166
493 Neuron_240:reach 1.543394
494 Neuron_241:reach 1.495580
495 Neuron_242:reach 1.145219
496 Neuron_244:reach 1.842679
497 Neuron_246:reach 1.903968
498 Neuron_247:reach 1.987295
499 Neuron_248:reach 1.493469
500 Neuron_249:reach 1.292953
501 Neuron_250:reach 2.017765
502 Neuron_251:reach 1.424569
503 Neuron_253:reach 1.463961
504 Neuron_255:reach 1.704327
505 Neuron_256:reach 1.383484
506 Neuron_257:reach 1.511643
507 Neuron_258:reach 1.912023
508 Neuron_259:reach 1.172722
509 Neuron_260:reach 1.917849
510 Neuron_261:reach 1.227602
511 Neuron_262:reach 1.696263
512 Neuron_264:reach 1.964603
513 Neuron_265:reach 1.410555
514 Neuron_266:reach 1.464254
515 Neuron_267:reach 1.385636
516 Neuron_273:reach 1.585746
517 Neuron_275:reach 1.761418
518 Neuron_276:reach 1.588014
519 Neuron_277:reach 2.609039
520 Neuron_278:reach 1.550908
521 Neuron_279:reach 1.150412
522 Neuron_280:reach 1.574541
523 Neuron_282:reach 1.040724
524 Neuron_283:reach 1.558547
525 Neuron_284:reach 1.673228
526 Neuron_285:reach 2.230535
527 Neuron_286:reach 1.181919
528 Neuron_287:reach 1.421917
529 Neuron_288:reach 2.045229
530 Neuron_291:reach 1.088172
531 Neuron_292:reach 1.409381
532 Neuron_293:reach 1.442139
533 Neuron_294:reach 1.799846
534 Neuron_295:reach 1.441385
535 Neuron_296:reach 2.038837

```

## 536      Neuron_297:reach 1.448801
## 537      Neuron_298:reach 2.150239
## 538      Neuron_299:reach 1.558214
## 539      Neuron_300:reach 2.198206
## 540      Neuron_301:reach 2.329415
## 541      Neuron_302:reach 1.081953
## 542      Neuron_303:reach 1.766752
## 543      Neuron_306:reach 1.664313
## 544      Neuron_308:reach 2.350866
## 545      Neuron_309:reach 2.154345
## 546      Neuron_311:reach 1.171534
## 547      Neuron_312:reach 1.349600
## 548      Neuron_314:reach 1.926785
## 549      Neuron_315:reach 1.960932
## 550          sd(Intercept) 1.888333
## 551          sd(reach) 2.233853
## 552 cor(Intercept,reach) 1.940348

```

##	Parameter	Bulk_ESS	Tail_ESS
## 1	Intercept	55.34027	158.1211
## 2	Neuron_3	915.58965	2610.0920
## 3	Neuron_4	787.74067	3103.6310
## 4	Neuron_5	949.80104	2392.8325
## 5	Neuron_6	1483.85056	2858.8309
## 6	Neuron_7	587.35661	1551.7617
## 7	Neuron_8	1456.58661	2908.6702
## 8	Neuron_9	1045.33657	2529.3460
## 9	Neuron_11	506.47671	1181.5387
## 10	Neuron_12	1805.83082	2961.5222
## 11	Neuron_13	372.80055	1238.9288
## 12	Neuron_14	942.56972	2322.6926
## 13	Neuron_15	131.89741	253.1900
## 14	Neuron_16	491.42628	1811.2017
## 15	Neuron_17	842.01467	2614.6392
## 16	Neuron_18	646.78119	1793.0692
## 17	Neuron_19	1252.85879	2979.0380
## 18	Neuron_20	812.00916	2604.6031
## 19	Neuron_21	1510.36902	2753.1038
## 20	Neuron_22	2209.22678	2959.6542
## 21	Neuron_23	1028.20855	2072.7902
## 22	Neuron_24	706.15075	2563.7791
## 23	Neuron_25	1160.08911	2570.9256
## 24	Neuron_26	712.74470	3130.7730
## 25	Neuron_27	1645.13317	2953.7979
## 26	Neuron_28	715.64118	2486.7953

## 27	Neuron_29	1381.17782	2463.0213
## 28	Neuron_30	876.93088	2511.8559
## 29	Neuron_31	102.54438	258.7171
## 30	Neuron_32	1023.60398	1854.0400
## 31	Neuron_33	968.67870	2512.6144
## 32	Neuron_34	459.73119	2043.1926
## 33	Neuron_35	860.21802	2284.8313
## 34	Neuron_36	682.20613	2060.6636
## 35	Neuron_37	421.67316	1575.0295
## 36	Neuron_38	1038.75527	2245.3687
## 37	Neuron_39	2826.90601	2627.6436
## 38	Neuron_40	712.10793	1979.5054
## 39	Neuron_41	845.63134	1257.8865
## 40	Neuron_42	748.83430	2566.0063
## 41	Neuron_43	412.89868	1792.6810
## 42	Neuron_44	101.98106	221.3254
## 43	Neuron_45	757.03210	2734.2259
## 44	Neuron_46	1088.47286	2965.3527
## 45	Neuron_47	739.35694	1414.5987
## 46	Neuron_48	199.37849	420.6911
## 47	Neuron_50	1486.11252	3188.2259
## 48	Neuron_51	1436.99773	3299.9957
## 49	Neuron_52	1049.97340	2194.2704
## 50	Neuron_53	995.21026	1613.7867
## 51	Neuron_54	531.56871	1766.4869
## 52	Neuron_55	504.88636	2245.6238
## 53	Neuron_56	1378.52017	2679.0130
## 54	Neuron_57	763.82109	1805.2826
## 55	Neuron_58	1511.15076	3134.4199
## 56	Neuron_59	918.70762	2531.8680
## 57	Neuron_60	607.58158	3185.5445
## 58	Neuron_61	694.94305	2179.2167
## 59	Neuron_62	723.09242	2593.0125
## 60	Neuron_63	662.15905	2052.9563
## 61	Neuron_64	1022.31090	1953.8343
## 62	Neuron_65	382.44934	1862.7715
## 63	Neuron_66	2017.92608	3111.5966
## 64	Neuron_67	904.09768	2653.9180
## 65	Neuron_68	839.09427	3326.0145
## 66	Neuron_69	881.84739	2095.5000
## 67	Neuron_70	78.24974	131.2249
## 68	Neuron_71	705.02559	1744.2547
## 69	Neuron_72	1399.56743	2990.0704
## 70	Neuron_73	725.44001	2765.2856
## 71	Neuron_74	1258.66336	2448.3839

## 72	Neuron_75	24.13562	116.1549
## 73	Neuron_76	223.73172	499.5651
## 74	Neuron_77	685.05606	2753.0399
## 75	Neuron_78	826.62541	2049.5155
## 76	Neuron_79	747.12453	2156.8415
## 77	Neuron_80	870.07499	2629.8081
## 78	Neuron_81	176.51273	339.5515
## 79	Neuron_82	531.51909	2483.8799
## 80	Neuron_83	499.24668	2365.5413
## 81	Neuron_84	747.72181	2520.3854
## 82	Neuron_85	610.87040	2357.6446
## 83	Neuron_86	1112.85206	3512.9030
## 84	Neuron_87	849.59484	1899.3927
## 85	Neuron_88	399.28152	1832.8683
## 86	Neuron_89	11.22737	10.9670
## 87	Neuron_90	601.46554	2063.5027
## 88	Neuron_91	1530.89308	3054.5864
## 89	Neuron_93	832.18716	2517.8365
## 90	Neuron_94	749.10812	2801.1465
## 91	Neuron_95	591.56862	2077.1882
## 92	Neuron_96	893.57874	2264.2869
## 93	Neuron_97	55.22579	123.7273
## 94	Neuron_98	541.88488	1741.2212
## 95	Neuron_99	710.28507	2249.8675
## 96	Neuron_100	1242.05342	3021.4594
## 97	Neuron_101	105.82477	110.8202
## 98	Neuron_102	1496.06987	3294.6719
## 99	Neuron_103	616.62449	2599.7749
## 100	Neuron_104	851.58582	2493.2521
## 101	Neuron_105	1140.28221	2760.7304
## 102	Neuron_106	829.29081	2344.6569
## 103	Neuron_107	593.43418	1899.5256
## 104	Neuron_108	714.32058	1873.7539
## 105	Neuron_109	888.35955	1911.6609
## 106	Neuron_110	680.11725	2103.9425
## 107	Neuron_111	1007.09015	2391.8283
## 108	Neuron_112	631.29400	2493.3831
## 109	Neuron_113	541.54855	2179.7620
## 110	Neuron_114	682.66445	2611.7075
## 111	Neuron_115	680.44236	2165.7281
## 112	Neuron_116	1129.51344	2509.5107
## 113	Neuron_117	1209.82954	1913.2201
## 114	Neuron_118	949.50534	2412.9235
## 115	Neuron_119	1045.57395	2444.1674
## 116	Neuron_120	582.93811	1059.6336

## 117	Neuron_121	1650.93298	3520.7857
## 118	Neuron_122	1688.79003	2698.0319
## 119	Neuron_123	1867.85514	3267.5673
## 120	Neuron_124	957.06810	2511.1637
## 121	Neuron_125	241.22544	955.9189
## 122	Neuron_127	662.07950	2230.5776
## 123	Neuron_128	1021.20554	2427.7075
## 124	Neuron_129	787.72211	2721.7204
## 125	Neuron_130	591.83610	1860.6114
## 126	Neuron_131	286.56554	1904.9702
## 127	Neuron_132	679.28727	2248.3819
## 128	Neuron_133	400.20402	952.1865
## 129	Neuron_134	1064.04537	2714.8037
## 130	Neuron_135	384.75185	1382.4977
## 131	Neuron_136	215.37394	655.4342
## 132	Neuron_137	762.58464	2104.8648
## 133	Neuron_138	817.80784	1627.8851
## 134	Neuron_139	917.16536	2034.4367
## 135	Neuron_140	971.78674	2268.6875
## 136	Neuron_141	93.07821	165.6909
## 137	Neuron_142	888.74367	2590.3973
## 138	Neuron_143	628.16537	2598.6884
## 139	Neuron_144	1024.12817	2173.5481
## 140	Neuron_145	523.99936	1105.8650
## 141	Neuron_146	578.79811	2481.1303
## 142	Neuron_147	658.40142	2046.5379
## 143	Neuron_148	427.45125	1549.4764
## 144	Neuron_149	499.04266	1701.7711
## 145	Neuron_150	946.45277	2722.6868
## 146	Neuron_151	1260.86105	2476.8995
## 147	Neuron_152	570.89758	2329.2976
## 148	Neuron_153	878.31218	2133.2759
## 149	Neuron_154	2143.39891	3385.8813
## 150	Neuron_155	457.79039	1516.0164
## 151	Neuron_156	2116.48826	3635.0017
## 152	Neuron_157	875.30860	2249.6803
## 153	Neuron_158	1076.71877	2794.2613
## 154	Neuron_159	2951.18513	3408.9581
## 155	Neuron_160	688.47285	2280.6118
## 156	Neuron_161	2228.25896	2983.0926
## 157	Neuron_162	1587.74743	3088.0623
## 158	Neuron_163	660.05966	2228.3819
## 159	Neuron_164	60.98012	119.4182
## 160	Neuron_165	644.78461	1910.6461
## 161	Neuron_166	1710.34709	2912.4247

## 162	Neuron_167	2182.42360	3320.3631
## 163	Neuron_169	817.05272	2044.3515
## 164	Neuron_170	662.17101	2120.3695
## 165	Neuron_171	908.84607	2754.5446
## 166	Neuron_172	669.26324	1822.8492
## 167	Neuron_173	1098.09377	2369.3471
## 168	Neuron_174	1104.34226	2599.2348
## 169	Neuron_175	843.00356	2449.2747
## 170	Neuron_176	834.07065	1933.4969
## 171	Neuron_177	466.20532	2058.0535
## 172	Neuron_180	913.51471	2649.6713
## 173	Neuron_181	1889.69576	2642.4909
## 174	Neuron_182	1240.24989	2222.5688
## 175	Neuron_183	541.07699	1440.0118
## 176	Neuron_184	1404.57487	3243.6834
## 177	Neuron_185	73.01273	164.0845
## 178	Neuron_186	87.08846	126.8842
## 179	Neuron_187	1018.31765	3059.2608
## 180	Neuron_189	71.07511	170.7617
## 181	Neuron_190	758.54775	2779.7654
## 182	Neuron_191	93.63917	195.8598
## 183	Neuron_192	471.88562	1715.8290
## 184	Neuron_193	52.69430	199.1462
## 185	Neuron_194	181.37362	661.5466
## 186	Neuron_196	68.31379	154.7982
## 187	Neuron_197	75.34788	168.3156
## 188	Neuron_198	1487.07304	3268.4248
## 189	Neuron_199	604.59687	2217.1251
## 190	Neuron_200	1046.90380	2085.6218
## 191	Neuron_202	881.03960	2998.9687
## 192	Neuron_204	889.41633	2082.8265
## 193	Neuron_205	572.41340	1619.5159
## 194	Neuron_207	122.32903	326.7026
## 195	Neuron_208	1532.38132	3353.4895
## 196	Neuron_209	48.81866	139.1159
## 197	Neuron_210	104.60465	141.9329
## 198	Neuron_211	649.43733	2357.0451
## 199	Neuron_212	1207.46189	3230.7731
## 200	Neuron_213	1441.92618	3099.8131
## 201	Neuron_214	521.77861	1626.5186
## 202	Neuron_215	1460.48504	2604.8050
## 203	Neuron_216	988.05224	2630.6523
## 204	Neuron_218	784.70269	1773.7822
## 205	Neuron_219	58.15984	80.6476
## 206	Neuron_220	784.26616	2847.3683

## 207	Neuron_221	1255.64362	2488.8123
## 208	Neuron_222	645.43708	1966.3941
## 209	Neuron_223	100.86779	226.8002
## 210	Neuron_224	397.99389	721.3589
## 211	Neuron_225	1274.71618	2623.1945
## 212	Neuron_227	918.09540	2601.7706
## 213	Neuron_228	89.91989	214.7883
## 214	Neuron_229	503.47852	1035.5243
## 215	Neuron_230	1554.81531	2778.5590
## 216	Neuron_231	70.33455	170.1652
## 217	Neuron_232	85.85547	187.8607
## 218	Neuron_233	934.06250	2183.9550
## 219	Neuron_235	512.09564	2722.4844
## 220	Neuron_236	51.95018	106.8230
## 221	Neuron_237	674.37515	2789.5840
## 222	Neuron_238	648.71823	2095.3474
## 223	Neuron_239	939.03658	2528.1144
## 224	Neuron_240	932.06253	2570.2173
## 225	Neuron_241	780.19649	2193.0929
## 226	Neuron_242	2042.31476	3463.1953
## 227	Neuron_244	600.76229	1910.3888
## 228	Neuron_246	1090.04825	2000.2382
## 229	Neuron_247	323.07033	1003.3551
## 230	Neuron_248	1065.63256	2129.2105
## 231	Neuron_249	820.61400	1715.7802
## 232	Neuron_250	1292.67467	2886.4442
## 233	Neuron_251	1042.28181	3189.0999
## 234	Neuron_253	44.50874	85.9584
## 235	Neuron_254	96.77008	315.4009
## 236	Neuron_255	1105.42210	3113.2742
## 237	Neuron_256	1101.03084	1659.6609
## 238	Neuron_257	782.73384	2917.6056
## 239	Neuron_258	549.80984	2282.0244
## 240	Neuron_259	1415.64665	2273.3743
## 241	Neuron_260	58.81269	161.5389
## 242	Neuron_261	869.06202	1915.7305
## 243	Neuron_262	633.09929	1631.8861
## 244	Neuron_264	730.33894	2712.2896
## 245	Neuron_265	927.43004	1741.2369
## 246	Neuron_266	983.75522	2798.9477
## 247	Neuron_267	771.29958	2530.9647
## 248	Neuron_268	56.78418	178.4092
## 249	Neuron_269	75.86562	84.5283
## 250	Neuron_270	80.88139	175.2749
## 251	Neuron_271	114.91734	248.3284

## 252	Neuron_273	650.97169	2247.8224
## 253	Neuron_274	104.52846	227.8597
## 254	Neuron_275	408.93164	1091.5537
## 255	Neuron_276	816.89345	2775.9752
## 256	Neuron_277	1222.40490	2196.5670
## 257	Neuron_278	103.60956	201.8771
## 258	Neuron_279	1051.05476	2386.1053
## 259	Neuron_280	1786.82161	3327.9350
## 260	Neuron_282	60.98312	118.9962
## 261	Neuron_283	695.42174	1979.2727
## 262	Neuron_284	1525.26455	2113.2763
## 263	Neuron_285	62.77570	182.7691
## 264	Neuron_286	1151.72238	2982.6634
## 265	Neuron_287	2156.83397	3078.8559
## 266	Neuron_288	1521.56215	2988.9132
## 267	Neuron_289	67.05430	79.9549
## 268	Neuron_290	98.76909	258.6150
## 269	Neuron_291	639.82092	2610.0030
## 270	Neuron_292	1143.06440	2467.4304
## 271	Neuron_293	1324.09824	3274.0530
## 272	Neuron_294	1358.54285	2675.8826
## 273	Neuron_295	1670.79431	3051.6116
## 274	Neuron_296	91.17417	221.5711
## 275	Neuron_297	1657.96128	2665.4461
## 276	Neuron_298	54.71038	101.0347
## 277	Neuron_299	335.20357	464.8190
## 278	Neuron_300	845.47544	2432.7415
## 279	Neuron_301	677.69708	1693.3132
## 280	Neuron_302	400.75355	735.8071
## 281	Neuron_303	1467.87351	3292.1876
## 282	Neuron_304	146.65010	321.9064
## 283	Neuron_306	72.53142	131.7370
## 284	Neuron_308	529.13577	1290.2177
## 285	Neuron_309	1937.90838	2904.3402
## 286	Neuron_311	365.30550	929.0057
## 287	Neuron_312	1011.71594	1717.2283
## 288	Neuron_314	1372.23891	2449.0418
## 289	Neuron_315	1132.19256	2655.5232
## 290	reach	2.88519	24.7902
## 291	Neuron_3:reach	2.97665	56.5553
## 292	Neuron_4:reach	3.56952	15.4630
## 293	Neuron_5:reach	6.76491	69.9027
## 294	Neuron_6:reach	3.64073	19.9980
## 295	Neuron_7:reach	4.75935	13.2108
## 296	Neuron_8:reach	2.55527	11.6991

## 297	Neuron_9:reach	5.21977	18.5377
## 298	Neuron_12:reach	3.19452	14.4589
## 299	Neuron_13:reach	4.73266	12.4067
## 300	Neuron_14:reach	6.18541	40.8284
## 301	Neuron_16:reach	2.85389	15.2213
## 302	Neuron_17:reach	3.04424	52.8296
## 303	Neuron_18:reach	3.48647	26.6437
## 304	Neuron_19:reach	3.92462	21.0849
## 305	Neuron_20:reach	3.12757	20.1770
## 306	Neuron_21:reach	2.98975	11.3879
## 307	Neuron_22:reach	2.54363	27.4887
## 308	Neuron_23:reach	2.78308	14.2085
## 309	Neuron_24:reach	2.66615	23.3544
## 310	Neuron_25:reach	3.42240	14.7766
## 311	Neuron_26:reach	5.09707	42.3525
## 312	Neuron_27:reach	4.22610	45.7529
## 313	Neuron_28:reach	6.84197	35.3990
## 314	Neuron_29:reach	2.57594	28.6822
## 315	Neuron_30:reach	2.84106	12.9654
## 316	Neuron_32:reach	2.66426	11.9595
## 317	Neuron_33:reach	4.03498	17.3835
## 318	Neuron_34:reach	11.96532	132.1209
## 319	Neuron_35:reach	2.91421	26.3296
## 320	Neuron_36:reach	3.23459	36.0737
## 321	Neuron_37:reach	4.15891	32.1900
## 322	Neuron_38:reach	3.84087	17.8812
## 323	Neuron_39:reach	2.55156	17.1277
## 324	Neuron_40:reach	3.36425	12.6461
## 325	Neuron_41:reach	4.49046	11.8790
## 326	Neuron_42:reach	4.60106	31.6727
## 327	Neuron_43:reach	3.55421	42.5581
## 328	Neuron_44:reach	8.21754	13.4689
## 329	Neuron_45:reach	6.41663	22.4672
## 330	Neuron_46:reach	2.95140	31.4934
## 331	Neuron_47:reach	2.96326	12.4220
## 332	Neuron_48:reach	3.32312	22.7581
## 333	Neuron_50:reach	11.95574	41.6779
## 334	Neuron_51:reach	5.52517	12.7804
## 335	Neuron_52:reach	2.96340	11.5897
## 336	Neuron_53:reach	4.23149	11.2433
## 337	Neuron_54:reach	3.14066	13.2971
## 338	Neuron_55:reach	10.39068	32.2560
## 339	Neuron_56:reach	2.68249	15.0807
## 340	Neuron_57:reach	2.95141	12.8420
## 341	Neuron_58:reach	3.01042	22.7458

## 342	Neuron_59:reach	22.00482	21.4019
## 343	Neuron_60:reach	2.58795	12.4572
## 344	Neuron_61:reach	3.08101	14.6997
## 345	Neuron_62:reach	5.09531	13.4892
## 346	Neuron_63:reach	3.18518	15.9677
## 347	Neuron_64:reach	4.71029	12.3904
## 348	Neuron_65:reach	11.13871	57.9363
## 349	Neuron_66:reach	3.89878	14.3531
## 350	Neuron_67:reach	2.93011	14.2033
## 351	Neuron_68:reach	4.70650	16.0175
## 352	Neuron_69:reach	3.85901	27.1744
## 353	Neuron_71:reach	13.24433	16.6180
## 354	Neuron_72:reach	13.05732	12.0815
## 355	Neuron_73:reach	3.19264	15.3811
## 356	Neuron_74:reach	4.90756	39.6526
## 357	Neuron_75:reach	2.50136	13.9997
## 358	Neuron_76:reach	2.80386	22.1353
## 359	Neuron_77:reach	3.66300	12.2520
## 360	Neuron_78:reach	3.79347	11.4606
## 361	Neuron_79:reach	3.18850	16.4145
## 362	Neuron_80:reach	2.93622	10.9882
## 363	Neuron_81:reach	2.49890	15.5032
## 364	Neuron_82:reach	2.79372	25.0601
## 365	Neuron_83:reach	2.67512	19.9496
## 366	Neuron_84:reach	5.55533	46.2315
## 367	Neuron_85:reach	2.91965	23.9497
## 368	Neuron_86:reach	4.81912	27.7852
## 369	Neuron_87:reach	3.74042	10.9730
## 370	Neuron_88:reach	3.01333	16.5692
## 371	Neuron_90:reach	4.22871	53.9179
## 372	Neuron_91:reach	2.91270	12.6037
## 373	Neuron_93:reach	3.03136	21.9400
## 374	Neuron_94:reach	5.46502	20.5463
## 375	Neuron_95:reach	2.84099	11.5970
## 376	Neuron_96:reach	7.85645	36.8869
## 377	Neuron_98:reach	4.93560	14.2129
## 378	Neuron_99:reach	3.15404	11.7234
## 379	Neuron_100:reach	6.05635	47.9419
## 380	Neuron_101:reach	33.32512	23.2894
## 381	Neuron_102:reach	3.26778	21.2655
## 382	Neuron_103:reach	2.80058	20.7939
## 383	Neuron_104:reach	27.18433	100.0131
## 384	Neuron_105:reach	2.85150	18.0000
## 385	Neuron_106:reach	24.98054	30.8737
## 386	Neuron_107:reach	3.39735	19.6501

## 387	Neuron_108:reach	7.60376	30.0323
## 388	Neuron_109:reach	3.03321	11.1668
## 389	Neuron_110:reach	5.41485	27.9643
## 390	Neuron_111:reach	2.85349	21.3328
## 391	Neuron_112:reach	4.06636	15.2212
## 392	Neuron_113:reach	2.92381	25.2819
## 393	Neuron_115:reach	18.35588	41.8017
## 394	Neuron_116:reach	7.86680	34.4513
## 395	Neuron_117:reach	4.81981	17.2490
## 396	Neuron_118:reach	12.78666	14.0182
## 397	Neuron_119:reach	4.02764	57.3903
## 398	Neuron_120:reach	4.69435	53.2899
## 399	Neuron_121:reach	3.18955	19.7123
## 400	Neuron_122:reach	17.08736	17.7372
## 401	Neuron_123:reach	3.91634	13.7013
## 402	Neuron_124:reach	3.60576	18.9932
## 403	Neuron_125:reach	3.16386	14.3888
## 404	Neuron_127:reach	2.56751	12.6517
## 405	Neuron_128:reach	2.91602	40.0046
## 406	Neuron_129:reach	7.50532	34.6424
## 407	Neuron_131:reach	3.01254	17.9480
## 408	Neuron_132:reach	3.06114	23.8278
## 409	Neuron_133:reach	2.66989	14.2991
## 410	Neuron_134:reach	11.44017	24.1350
## 411	Neuron_135:reach	15.27182	32.8469
## 412	Neuron_136:reach	5.56024	15.2213
## 413	Neuron_137:reach	4.74319	13.9197
## 414	Neuron_138:reach	3.87953	24.0201
## 415	Neuron_139:reach	2.82195	10.8668
## 416	Neuron_140:reach	2.58332	11.4594
## 417	Neuron_141:reach	6.22135	43.5589
## 418	Neuron_142:reach	2.58457	16.9701
## 419	Neuron_143:reach	3.45008	14.7631
## 420	Neuron_144:reach	3.00513	19.6492
## 421	Neuron_145:reach	14.19201	107.0926
## 422	Neuron_146:reach	4.22810	11.4514
## 423	Neuron_147:reach	23.71850	45.5856
## 424	Neuron_148:reach	3.34877	15.4387
## 425	Neuron_149:reach	5.12976	82.6452
## 426	Neuron_150:reach	4.77413	48.2817
## 427	Neuron_151:reach	3.92690	13.4481
## 428	Neuron_152:reach	3.70023	14.7422
## 429	Neuron_153:reach	5.84113	12.7687
## 430	Neuron_154:reach	3.82534	11.4793
## 431	Neuron_155:reach	2.56678	21.7445

## 432	Neuron_156:reach	7.33716	20.3305
## 433	Neuron_157:reach	2.98458	18.1081
## 434	Neuron_158:reach	3.07098	28.3552
## 435	Neuron_159:reach	2.84068	12.6516
## 436	Neuron_160:reach	8.85058	50.5068
## 437	Neuron_161:reach	4.64859	26.2106
## 438	Neuron_162:reach	28.98115	60.7595
## 439	Neuron_163:reach	2.62899	12.9606
## 440	Neuron_164:reach	2.29434	10.9153
## 441	Neuron_165:reach	2.69918	11.1587
## 442	Neuron_166:reach	4.85150	25.8813
## 443	Neuron_167:reach	2.90842	19.3784
## 444	Neuron_169:reach	8.80600	91.9028
## 445	Neuron_170:reach	3.33789	11.9473
## 446	Neuron_171:reach	6.47231	20.8256
## 447	Neuron_172:reach	5.02014	48.5238
## 448	Neuron_173:reach	4.43659	51.9493
## 449	Neuron_174:reach	4.17058	36.3267
## 450	Neuron_175:reach	3.37269	16.2832
## 451	Neuron_176:reach	4.66775	19.6328
## 452	Neuron_177:reach	24.59568	71.8362
## 453	Neuron_180:reach	3.08772	15.4064
## 454	Neuron_181:reach	8.27220	21.0159
## 455	Neuron_182:reach	4.12294	16.5734
## 456	Neuron_183:reach	2.74921	13.3650
## 457	Neuron_184:reach	3.27366	39.7185
## 458	Neuron_186:reach	2.69116	10.8479
## 459	Neuron_187:reach	4.89539	43.3772
## 460	Neuron_190:reach	8.02906	38.9589
## 461	Neuron_192:reach	5.08471	32.0564
## 462	Neuron_193:reach	2.77615	15.3144
## 463	Neuron_198:reach	3.16797	13.2330
## 464	Neuron_199:reach	2.93419	38.4428
## 465	Neuron_200:reach	3.06466	25.6284
## 466	Neuron_202:reach	3.34530	20.5664
## 467	Neuron_204:reach	7.71746	31.2786
## 468	Neuron_205:reach	3.58495	12.2379
## 469	Neuron_208:reach	3.79598	47.1954
## 470	Neuron_209:reach	5.47092	10.8939
## 471	Neuron_211:reach	11.72069	86.0234
## 472	Neuron_212:reach	3.15548	10.9184
## 473	Neuron_213:reach	2.90739	12.7346
## 474	Neuron_214:reach	7.35497	17.4315
## 475	Neuron_215:reach	13.12202	19.9384
## 476	Neuron_216:reach	3.62242	11.4255

## 477	Neuron_218:reach	3.54871	12.9848
## 478	Neuron_219:reach	2.63018	11.3319
## 479	Neuron_220:reach	2.86657	24.4955
## 480	Neuron_221:reach	4.93530	53.7127
## 481	Neuron_222:reach	4.29842	16.7618
## 482	Neuron_224:reach	4.28906	64.6594
## 483	Neuron_225:reach	2.91022	17.4835
## 484	Neuron_227:reach	34.60363	81.8459
## 485	Neuron_228:reach	2.73320	21.0630
## 486	Neuron_229:reach	6.43210	60.3133
## 487	Neuron_230:reach	2.68175	11.3564
## 488	Neuron_233:reach	3.16549	15.6083
## 489	Neuron_235:reach	5.62520	13.2357
## 490	Neuron_237:reach	3.56883	19.0953
## 491	Neuron_238:reach	3.31682	26.6067
## 492	Neuron_239:reach	2.38260	12.5602
## 493	Neuron_240:reach	3.65906	62.3110
## 494	Neuron_241:reach	3.73683	21.2437
## 495	Neuron_242:reach	9.90445	30.2735
## 496	Neuron_244:reach	2.91759	10.9614
## 497	Neuron_246:reach	2.82891	20.0696
## 498	Neuron_247:reach	2.81006	37.2251
## 499	Neuron_248:reach	3.84726	15.5610
## 500	Neuron_249:reach	5.62116	33.7968
## 501	Neuron_250:reach	2.76113	13.3719
## 502	Neuron_251:reach	4.52047	34.2870
## 503	Neuron_253:reach	4.16088	16.9621
## 504	Neuron_255:reach	3.17354	136.7155
## 505	Neuron_256:reach	4.55174	34.5207
## 506	Neuron_257:reach	3.65874	18.0000
## 507	Neuron_258:reach	2.83867	12.3830
## 508	Neuron_259:reach	8.49701	25.5326
## 509	Neuron_260:reach	2.79536	19.4599
## 510	Neuron_261:reach	6.70049	12.4773
## 511	Neuron_262:reach	3.21683	21.0524
## 512	Neuron_264:reach	2.84328	32.6348
## 513	Neuron_265:reach	4.49759	19.9432
## 514	Neuron_266:reach	3.86789	19.2374
## 515	Neuron_267:reach	4.42946	13.8038
## 516	Neuron_273:reach	3.43023	15.0919
## 517	Neuron_275:reach	3.03715	30.0687
## 518	Neuron_276:reach	3.61504	14.5571
## 519	Neuron_277:reach	2.39337	12.2628
## 520	Neuron_278:reach	3.70767	15.9438
## 521	Neuron_279:reach	11.02738	25.7046

## 522	Neuron_280:reach	3.61579	24.4308
## 523	Neuron_282:reach	27.14789	71.3184
## 524	Neuron_283:reach	3.57077	31.8567
## 525	Neuron_284:reach	3.30202	22.8555
## 526	Neuron_285:reach	2.53021	15.2294
## 527	Neuron_286:reach	10.95552	11.5908
## 528	Neuron_287:reach	4.26851	11.2006
## 529	Neuron_288:reach	2.68289	12.7826
## 530	Neuron_291:reach	30.50419	27.9520
## 531	Neuron_292:reach	4.22567	19.0289
## 532	Neuron_293:reach	4.09822	42.2236
## 533	Neuron_294:reach	2.95470	13.4798
## 534	Neuron_295:reach	4.10994	17.2396
## 535	Neuron_296:reach	2.66642	10.9829
## 536	Neuron_297:reach	4.32962	46.9852
## 537	Neuron_298:reach	2.58942	11.3562
## 538	Neuron_299:reach	3.64145	10.8831
## 539	Neuron_300:reach	2.60560	24.0994
## 540	Neuron_301:reach	2.50774	31.9687
## 541	Neuron_302:reach	15.66288	26.5275
## 542	Neuron_303:reach	3.02582	15.4043
## 543	Neuron_306:reach	3.23282	36.3391
## 544	Neuron_308:reach	2.50124	20.6703
## 545	Neuron_309:reach	2.60245	22.0208
## 546	Neuron_311:reach	7.98454	17.6572
## 547	Neuron_312:reach	5.08044	29.6940
## 548	Neuron_314:reach	2.81693	27.2523
## 549	Neuron_315:reach	2.88416	14.4892
## 550	sd(Intercept)	2.92513	11.5300
## 551	sd(reach)	2.54701	11.5360
## 552	cor(Intercept,reach)	2.84850	15.9299